GEK-96874

DRIVE SYSTEM INSTRUCTIONS

EMERGENCY GENERATOR CONTROL PANEL

(WITH AUTOMATIC THROWOVER AND BACKFEEDING PROVISIONS)

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GE POWER SYSTEM
SONAT OFFSHORE DRILLING, INC
PRATT RATHER - DAEWOO HULL 3010
GEORGE RICHARDSON - DAEWOO HULL 3011

EMERGENCY GENERATOR CONTROL PANEL

(WITH AUTOMATIC THROWOVER AND BACKFEEDING PROVISIONS)

GEK-96874

ISSUED: AUGUST 1987

GE REQUISITION: F47-DC-841990
F47-DC-841991

The information contained herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise that are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, Drive Systems Operations, Salem, Virginia, U.S.A.

WARNING

THIS EQUIPMENT CONTAINS A POTENTIAL HAZARD OF ELECTRICAL SHOCK OR BURN. ONLY PERSONNEL WHO ARE ADEQUATELY TRAINED AND THOROUGHLY FAMILIAR WITH THE EQUIPMENT AND THESE INSTRUCTIONS SHOULD INSTALL, OPERATE OR MAINTAIN THIS EQUIPMENT.

ISOLATION OF TEST EQUIPMENT FROM THE EQUIPMENT UNDER TEST PRESENTS POTENTIAL ELECTRICAL HAZARDS. IF THE TEST EQUIPMENT CANNOT BE GROUNDED TO THE EQUIPMENT UNDER TEST, THE TEST EQUIPMENT'S CASE MUST BE SHIELDED TO PREVENT CONTACT BY PERSONNEL.

TO MINIMIZE HAZARD OF ELECTRICAL SHOCK OR BURN, APPROVED GROUNDING PRACTICES AND PROCEDURES MUST BE FOLLOWED.

CAUTION

THIS EQUIPMENT IS DESIGNED TO WORK IN A NORMAL INDUSTRIAL ATMOSPHERE. OPERATING RADIO TRANSMITTERS NEARBY MAY CAUSE THE CONTROL EQUIPMENT TO MALFUNCTION. IT IS RECOMMENDED THAT RADIO TRANSMITTERS NOT BE OPERATED WITHIN THE VICINITY OF THE CONTROL EQUIPMENT WHEN THE PANEL DOORS ARE OPEN, NOR AT ANY TIME WITHIN THE CONFINES OF A POWER CONTROL ROOM.

* NOTICE *

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Safety Symbol Legend

WARNING	Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in personal injury or death.
CAUTION	Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in damage to or destruction of equipment.
NOTE	Commands attention to an essential operating or maintenance procedure, condition, or statement that must be highlighted.

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CHAPTER 1
EMERGENCY GENERATOR CONTROL PANEL WITH AUTOMATIC THROWOVER/THROWBACK AND BACKFEED PROVISIONS

1-1. FUNCTION

The function of the emergency generator control panel is to provide rated power to critical loads on the emergency panel bus in the event of failure or shutdown (blackout) of the main supply generators.

1-2. OPERATION

Automatic and manual modes of operation are incorporated into the control, utilizing the features of electrically operated "Emergency Generator" and "Tie" circuit breakers. An automatic or manual mode operational selector switch is provided, in addition to a key lockable "Normal - Backfeed" selector switch. The positions of these switches will determine and govern the operational mode.

1-3. OPERATIONAL MODES

- Normal Mode (Manual, **not** in Backfeed)
- Automatic Throwover Mode (Auto)
- Automatic Throwback Mode (Auto)
- Manual Mode
- Backfeed Mode (Manual, Backfeed selected)

1-3.1. Normal Mode

In the Normal mode of operation, the ship's service motor control center (MCC) provides power to the emergency generator power distribution panel via the "Tie" circuit breaker, which is closed. Power flows from the main ship's service MCC through the "tie" circuit breaker to critical loads on the emergency generator power distribution bus. The "Tie" circuit breaker is in "cascade" with a circuit breaker from the main ship's service panel.

1-3.2. Automatic Throwover Mode

Should the normal flow of power via the "tie" circuit breaker be interrupted, the "Tie " Circuit breaker will trip on undervoltage (27VSR Relay).

If the "Manual - Automatic" selector switch is in the Automatic mode, a combination of this switch position coupled with tripping the "Tie" circuit breaker is permission to send a signal to the customer's emergency generator diesel engine, requesting engine start. Two items should be noted here, however, for they are critical to the automatic throwover mode.

- "27VSR - A Basler Electric Co. solid stare under/over voltage relay (BEI-27/59 - A3N C5L-AOSIP) must be adjusted for proper % under voltage ($\approx$ 85%) and inverse time delay ($\geq$ 5 sec.) to prevent false auto transfers.
- Emer. Gen. CB charging relay should be adjusted so EGCB properly charges prior to attempting to close ($\geq$ 10 sec.).

As emergency generator voltage builds up, emer. gen. voltage sensing relay "27GVSR" detects sufficient voltage (95%), permits circuit breaker undervoltage energizing, CB charging and CB closing. Power is now supplied to the main thru bus by the emergency generator.

1-3.3. Automatic Throwback Mode

The "Automatic Throwback" mode of operation is used to return power to the normal mode of operation; that is, the ship's service motor control center supply.

In going back to the Normal mode of operation via the Automatic Throwback method, the following conditions are required:

- LVIB Bus (main ship's service MCC) re-energizes, as detected by "Tie" CB voltage sensing relay "27VSR".
- "Manual - Off - Auto" switch in AUTO position
- "Normal - Backfeed" switch in NORMAL position
- Tie CB open (not backfeeding)

When the above conditions exist, the "Tie" breaker receives control and under voltage power from the main ship's service bus. The "Tie" breaker charge relay "ARTCB" is energized via a N.C. (B2) auxiliary contact on the Tie CB.

The Tie CB's undervoltage device picks up (UVTD2) and the tie CB charges (5-10 sec.) and closes. Both sources are now feeding the thru bus.

The period of time both Emerg. Gen. and normal power feed are supplying power to the thru bus is adjustable from zero(0) to thirty(30) seconds, per customer's request.

After about ten (10) seconds, the "overvoltage" portion of "Tie" CB under/over voltage relay "27VSR" energizes (sh. 241G05D). Auxiliary relay "27OVR" is energized, which energizes relay "CRNPR" (Control Relay Normal Power Restore) via NORMAL position of Backfeed-Off-Normal selector switch.

Relay "CRNPR" has a normally closed contact (N.C.) in series with emergency generator undervoltage time delay unit "UVTD1", which opens, tripping the emer-

gency Gen. CB off line, and initiating an adjustable diesel engine "cool down cycle" by energizing relay "ESDR" (sh. 241G line 19U).

As control is returned to the Normal mode via the "Automatic Throwback" scheme, provisions are made to energize diesel engine "Cool Down" circuitry. The diesel engine "Cool Down" control circuit permits the diesel to operate unloaded (EGEN circuit breaker open) for a sufficient period of time to cool down prior to shutdown.

1-3.4. Manual Mode (Local Operation)

In the Manual mode of operation, the "Manual - Off - Auto" selector switch is in the MANUAL position and all functions are operated manually. For example:

- Diesel engine is started manually, and
- Circuit breakers are charged and closed using manual pushbuttons

In the Manual mode, all operations are controlled manually by the operator locally at the emergency gen. panel.

1-3.5. Backfeed Mode

The "Backfeed" mode is a manual operational mode which permits the emergency generator distribution bus to "Backfeed" the main ship's service motor control center bus via the "Tie" circuit breaker in "cascade" with a circuit breaker off of the ship's service distribution bus.

The normal interlocking between the emergency generator circuit breaker and the "Tie" circuit breaker, which normally inhibits their simultaneous closing, is bypassed.

The "Normal - Backfeed" lockable control switch is placed in the "BACKFEED" position, which provides control and undervoltage power to the "Tie" circuit breaker. The "Tie" circuit breaker's "Automatic Throwback" mode is disabled at this time. This requires the operator to manually charge and close the "Tie" circuit breaker. Upon closing the "Tie" circuit breaker, the ship's service circuit breaker that is in "cascade" with the "Tie" circuit breaker may be closed. Moving the "Normal - Backfeed" switch back to NORMAL removes control and UV power from the "Tie" circuit breaker causing it to trip on undervoltage.

While in "backfeed" mode, power (KW) is monitored by a circuit located on preferential trip device AT1. Should the load in "backfeed" mode exceed 2000KW, circuit "KIB" on device AT1 opens in "Tie" CB undervoltage circuit, tripping the "Tie" CB off line and preserving the basic emergency generator load, preventing a total rig blackout.

1-4. EMERGENCY GENERATOR MANUAL/ AUTOMATIC VOLTAGE CONTROL

1-4.1. Manual (sh. 241F) Voltage Field Cont.

In Manual Mode, emergency generator field power is stepped down (480V to 120V), applied to a voltage varying "variac" auto transformer (0-140VAC output), rectified via a full wave diode bridge, and supplied to this emergency generator field exciter when the "Manual/ Auto" voltage control switch is in the manual mode.

1-4.2. Automatic Voltage Control

Automatic voltage control is via Stromberg (Finland) provided automatic voltage regulator. Three(3) phase, three(3) wire 120VAC power (60VA combined load max.) is supplied from "TEGCP1.2" CPTs.

The unit's controls are mounted upon the unit proper, with some portion possibly located at the emergency generator.

The device is field mounted, after the panel ships from DSO Salem, Va. The proposed mounting location was the lower front portion of Feeder Panel (EO2) no.2, on "square rods".

See elem. sh. 241AR for wiring and Stromberg instructions for operation information.

The output of the device is wired to the "Automatic" position of the Manual/Automatic Voltage Mode selector switch, such that when the switch is in "Auto" power is supplied to the emergency generator field exciter from the Stromberg voltage regulator.

1-5. BLACKOUT TEST SIMULATION

Provisions are made for simulating power loss from main ships service switch board bus "LVIB".

The "Tie" circuit breaker voltage sensing relay "27VSR" voltage sensing input is wired thru an SB1 type switch "Blackout Test Sw.". The switch is a "pull and hold, turn to test position" with spring return to normal.

Actuating the switch in the Auto Mode will initiate and complete the auto throwover sequence. Auto throwback will also occur if Mode Select switch remains in Auto Mode.

The interruption of sensing voltage to "27VSR" is at the heart of this feature. See elem. sh. 241E line 11S.

1-6. PREFERENTIAL TRIP CIRCUITS

Three(3) levels of preferential trips are provided, with three(3) independent normally closed (N.C.) contacts for shipyard use to alarm and trip non-essential loads.

Device nomenclature, device circuit, trip point and trip % are tabulated below.

DEVICE	CIRCUIT	TRIP POINT	TRIP %
AT1	K1A	1	100%
AT2	K1A	2	105%
AT2	K1B	3	110%

The preferential trip devices AT1 and AT2 are Rochester Instrument Systems (RIS) TYPE ET-1219 dual current/voltage alarms. See RIS Publication No. 1035-143 ET-1218 Instruction Manual for more information.

1-7. METERING

Switchboard type (1%) AB-40 meters are provided to monitor emergency generator volts, amps and frequency. KW metering uses a 0-1MA DB-40 switchboard meter. The emergency gen. temperature meter is a transducer driven (0-1MA) DB-40 switchboard type meter, scaled 0-180°C, with test calibration resistors (±80°C).

1-8. GROUND DETECTION (Sh. 241L)

A ground detector with visual and audible ground indication is provided, with silence and reset features.

Phase grounds are indicated via indication lights, with relay action setting off an audible alarm.

The audible alarm can be silenced while searching for the ground, with a reset provided after the ground is removed.

** NOTE **

The system cannot be reset and the alarm finally silenced until ground is removed.

1-9. ELV1 480VAC, 3ϕ, 3W, 60HZ MOTOR CONTROL CIRCUITS

Combination (circuit breaker and starter units) Motor Controls are provided. See MCC Publications for more details. MCC were inspected, approved and stamped by DNV (DET NORSKE VERITAS) in Salem, Va after mods to meet various DNV regulations. Each MCC unit has it's individual elem. control circuit.

The ELV1 MCC'S bus is direct connected to the main 480VAC, 3ϕ, 3W, 60HZ emergency generator control bus.

All MCC sections have space heaters and thermostat spaceheater controls (sh. 241P).

1-10. ELV2 120VAC, 3ϕ, 60HZ MOTOR CONTROL CIRCUITS (SHS. 241BA-241BJ)

Four(4) vertical sections of 120VAC, 3ϕ, 60HZ Motor Control feeders are provided, and are referred to as ELV2.

ELV2's incoming 120V 3ϕ power originates with the ELV1 480V 3ϕ feeders "EL RM/EM GEN.RM" (ET1) and "EL RM/EM GEN.RM" (ET2). These two 400 amp frame 300 amp trip CBs feed transformers ET1 and ET2 rated 225KVA 480V pri./120V, 3ϕ sec., where the voltage is stepped down to 120V, 3ϕ, 3W, 60HZ (Δ-Δ conn.)

From the step down transformer's 120V secondary, 3ϕ power enters ELV2 MCC via two key interlocked TKMA, 1200 amp circuit breakers, arranged such that only one CB at a time can be closed. The incoming power is then distributed to various 120V loads via selected CBs.

ELV2 MCC has a 0-150 volt scale voltmeter, and is equipped with ground detection as previously described.

1-11. DRAWINGS AND DIAGRAMS

The remainder of this section contains drawings and diagrams that apply to the equipment.

- Total System Diagram (one line diagram)
- Emergency Distribution Control Circuits
- ELV1 MCC Diagram
- ELV2 MCC diagram
- Assembly Diagrams

480VAC, 3 Ø, 3W, 60HZ, EMERGENCY GENERATOR CONTROL ELEM. SECTION

EMER. A-C GEN.
EMER. A-C SWBD.
EMER. A-C MCC

EMER. A-C GEN.
EMER. A-C SWBD.
EMER. A-C MCC

480VAC, 3 Ø, 3W, 60HZ.,-EMERGENCY

GENERATOR SWITCHBOARD; DISTRIBUTION AND MOTOR CONTROL CENTER ELEMENTARY SECTIONS - ELV1.

SPECIFICATION ITEMS
3A & 3B
REON: F47-DC-841991-C01
ML: 0504X0629W60,W61

- EMERGENCY GENERATOR CONT. SYSTEM ONE LINE DIAGRAM
- EMERGENCY GENERATOR SYSTEM DATA SHEET
- EMERGENCY GENERATOR BREAKER AND CONTROL SEQUENCING.
- TIE CIRCUIT BREAKER CONTROL AND SEQ.
- EMERG. GEN. MODE SELECTION
- EMERG. GEN. MCC FEEDERS & STARTERS

ML# 0504X0629W60,W61,W6C,W6D,W6E

E G C

ELV1

MAY 1987

E G C

NOTES: (SEE DWG NO. 237B9615)

PANEL WAS MODIFIED IN SALEM, VA TO MEET CUSTOMER CABLEING, AND CABLEING INSPECTION REQUIREMENTS. ADDITIONAL MODIFICATION WERE MADE TO MEET DET NORSKE VERITAS(DNV). SOME OF THE CHANGES ARE AS FOLLOWS:

1. INCOMINE E.GEN. & TIE CB CABLE TABS TO Ø.5" T X 3.0" HIGH TWO(2) PER PHASE.
2. RELOCATED BASE MOUNTED DEVICES TO SIDE WALL AND DOOR; CUT LOWER PORTION OF BASE TO PERMIT ACCESS TO INCOMING CABLE TABS.
3. CHANGED PANEL TYPE KW METER TO SWITCHBOARD TYPE.
4. ATTACHED ALL NAMEPLATES WITH SCREWS.
5. ADDED SPACE HEATERS TO E01-E06.
6. REPLACED SMALL FRAME CB CARD-BOARD LINE SHIELDS WITH GLASTIC.
7. CONNECTED GROUND CONNECTIONS USING SINGLE WIRE AND SCREW PER CONNECTIONS.

NOTES (CONT.):

8. ADDED WIRE LABELS TO MCC UNITS.
9. ADDED "FOLLOWERS" TO SOME MCC TB'S (≤14AWG).
10. ADDED TB'S TO AUTO TRANSFORMER CONNECTIONS UNITS A0 & A1.
11. ADD DNV NAMEPLATE TO ELV1, ELV1 MCC & ELV2 MCC.
12.
13

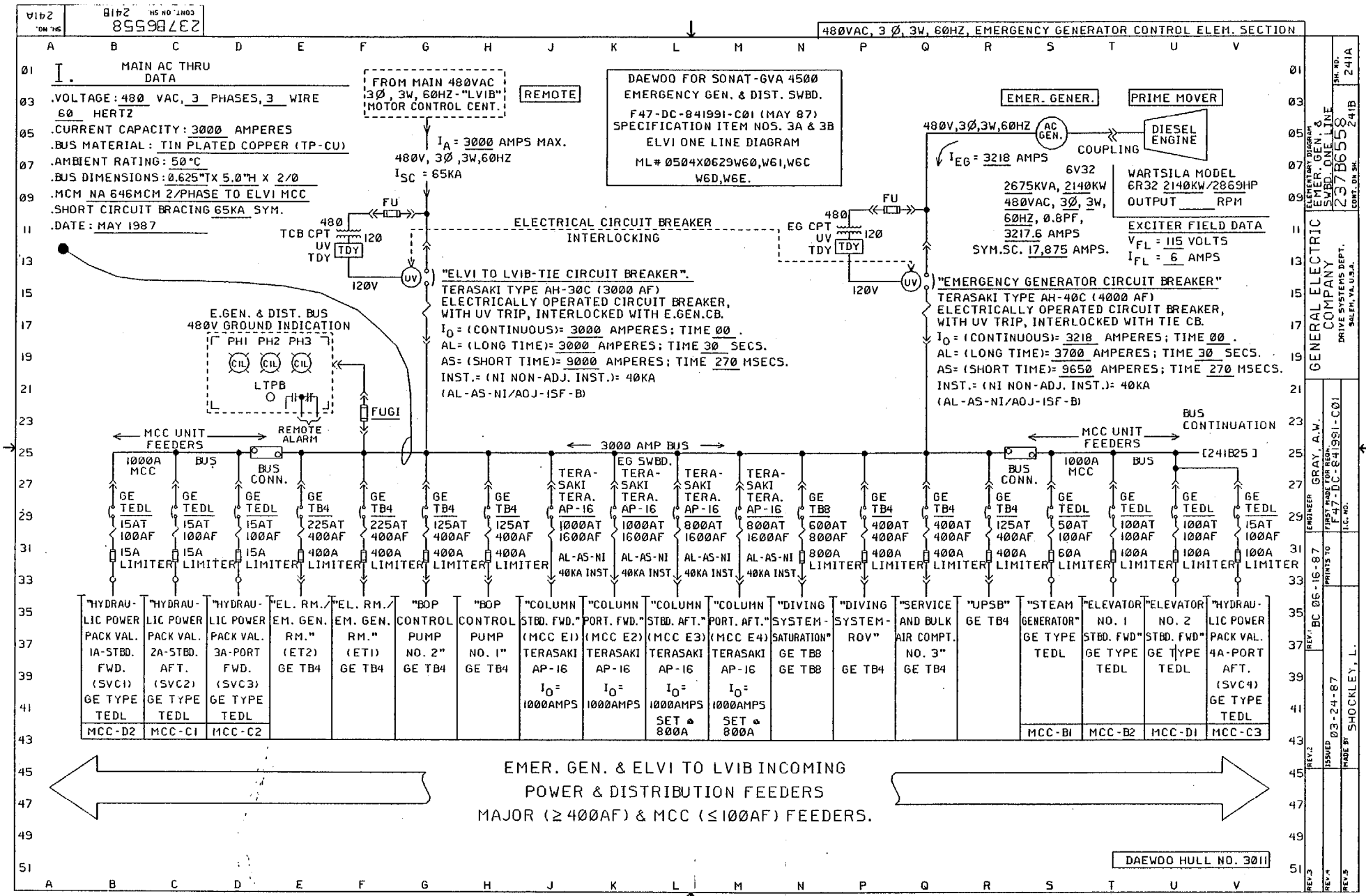
DAEWOO HULL NO. 3011

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
EMERG. GEN. & DIST. SWBD. ELEM.
237B6558 CONT. ON SH. 241A
SH. NO. 241

ENGINEER GRAY, A.W.
FIRST MADE FOR REON F47-DC-841991-C01
J.C. NO.

REV.1 BC 06-16-87 PRINTS TO
REV.2
REV.3
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.4
REV.5


480VAC, 3 Ø, 3W, 60HZ, EMERGENCY GENERATOR CONTROL ELEM. SECTION
I. MAIN AC THRU DATA
.VOLTAGE: 480 VAC, 3 PHASES, 3 WIRE
60 HERTZ
.CURRENT CAPACITY: 3000 AMPERES
.BUS MATERIAL: TIN PLATED COPPER (TP-CU)
.AMBIENT RATING: 50°C
.BUS DIMENSIONS: 0.625"TX 5.0"H X 2/Ø
.MCM NA 646MCM 2/PHASE TO ELVI MCC
.SHORT CIRCUIT BRACING 65KA SYM.
.DATE: MAY 1987
FROM MAIN 480VAC 3Ø, 3W, 60HZ-"LVIB" MOTOR CONTROL CENT.
REMOTE
I_A = 3000 AMPS MAX.
480V, 3Ø, 3W, 60HZ
I_{SC} = 65KA
DAEWOO FOR SONAT-GVA 4500 EMERGENCY GEN. & DIST. SWBD.
F47-DC-841991-C01 (MAY 87) SPECIFICATION ITEM NOS. 3A & 3B
ELVI ONE LINE DIAGRAM
ML# 0504X0629W60,W61,W6C W6D,W6E.
EMER. GENER.
PRIME MOVER
480V,3Ø,3W,60HZ
AC GEN.
COUPLING
DIESEL ENGINE
I_{EG} = 3218 AMPS
6V32
2675KVA, 2140KW
480VAC, 3Ø, 3W,
60HZ, 0.8PF,
3217.6 AMPS
SYM.SC. 17,875 AMPS.
WARTSILA MODEL
6R32 2140KW/2869HP
OUTPUT ____ RPM
EXCITER FIELD DATA
V_{FL} = 115 VOLTS
I_{FL} = 6 AMPS
FU
480
TCB CPT
120
UV
TDY
120V
ELECTRICAL CIRCUIT BREAKER INTERLOCKING
EG CPT
"ELVI TO LVIB-TIE CIRCUIT BREAKER".
TERASAKI TYPE AH-30C (3000 AF)
ELECTRICALLY OPERATED CIRCUIT BREAKER, WITH UV TRIP, INTERLOCKED WITH E.GEN.CB.
I_O = (CONTINUOUS)= 3000 AMPERES; TIME 00 .
AL= (LONG TIME)= 3000 AMPERES; TIME 30 SECS.
AS= (SHORT TIME)= 9000 AMPERES; TIME 270 MSECS.
INST.= (NI NON-ADJ. INST.)= 40KA
(AL-AS-NI/AOJ-ISF-B)
"EMERGENCY GENERATOR CIRCUIT BREAKER"
TERASAKI TYPE AH-40C (4000 AF)
ELECTRICALLY OPERATED CIRCUIT BREAKER, WITH UV TRIP, INTERLOCKED WITH TIE CB.
I_O = (CONTINUOUS)= 3218 AMPERES; TIME 00 .
AL= (LONG TIME)= 3700 AMPERES; TIME 30 SECS.
AS= (SHORT TIME)= 9650 AMPERES; TIME 270 MSECS.
INST.= (NI NON-ADJ. INST.)= 40KA
(AL-AS-NI/AOJ-ISF-B)
E.GEN. & DIST. BUS 480V GROUND INDICATION
PH1 PH2 PH3
CIL
LTPB
REMOTE ALARM
FUGI
MCC UNIT FEEDERS
3000 AMP BUS
BUS CONTINUATION
[241B25]
1000A MCC BUS
BUS CONN.
EG SWBD.
GE TEDL 15AT 100AF 15A LIMITER
GE TB4 225AT 400AF 400A LIMITER
GE TB4 125AT 400AF 400A LIMITER
TERA-SAKI TERA. AP-16 1000AT 1600AF AL-AS-NI 40KA INST
TERA-SAKI TERA. AP-16 800AT 1600AF AL-AS-NI 40KA INST
GE TB8 600AT 800AF 800A LIMITER
GE TB4 400AT 400AF 400A LIMITER
GE TB4 125AT 400AF 400A LIMITER
GE TEDL 50AT 100AF 60A LIMITER
GE TEDL 100AT 100AF 100A LIMITER
GE TEDL 15AT 100AF 100A LIMITER
"HYDRAULIC POWER PACK VAL. 1A-STBD. FWD. (SVC1) GE TYPE TEDL" MCC-D2
"HYDRAULIC POWER PACK VAL. 2A-STBD. AFT. (SVC2) GE TYPE TEDL" MCC-C1
"HYDRAULIC POWER PACK VAL. 3A-PORT FWD. (SVC3) GE TYPE TEDL" MCC-C2
"EL. RM./EM. GEN. RM." (ET2) GE TB4
"EL. RM./EM. GEN. RM." (ET1) GE TB4
"BOP CONTROL PUMP NO. 2" GE TB4
"BOP CONTROL PUMP NO. 1" GE TB4
"COLUMN STBD. FWD." (MCC E1) TERASAKI AP-16 I_O= 1000AMPS
"COLUMN PORT. FWD." (MCC E2) TERASAKI AP-16 I_O= 1000AMPS
"COLUMN STBD. AFT." (MCC E3) TERASAKI AP-16 I_O= 1000AMPS SET @ 800A
"COLUMN PORT. AFT." (MCC E4) TERASAKI AP-16 I_O= 1000AMPS SET @ 800A
"DIVING SYSTEM-SATURATION" GE TB8 GE TB8
"DIVING SYSTEM-ROV" GE TB4
"SERVICE AND BULK AIR COMPT. NO. 3" GE TB4
"UPSB" GE TB4
"STEAM GENERATOR" GE TYPE TEDL MCC-B1
"ELEVATOR NO. 1 STBD. FWD" GE TYPE TEDL MCC-B2
"ELEVATOR NO. 2 STBD. FWD" GE TYPE TEDL MCC-D1
"HYDRAULIC POWER PACK VAL. 4A-PORT AFT. (SVC4) GE TYPE TEDL" MCC-C3
EMER. GEN. & ELVI TO LVIB INCOMING POWER & DISTRIBUTION FEEDERS MAJOR (≥ 400AF) & MCC (≤ 100AF) FEEDERS.
DAEWOO HULL NO. 3011
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM EMER. GEN. & SWBD. ONE LINE
237B6558
SH. NO. 241A
241B
CONT. ON SH.
ENGINEER GRAY, A.W.
FIRST MADE FOR REQN. F47-DC-841991-C01
I.C. NO.
REV.1 BC 06-16-87
PRINTS TO
REV.2
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5

480VAC, 3 Ø, 3W, 60HZ, EMERGENCY GENERATOR CONTROL ELEM. SECTION

DAEWOD FOR SONAT GVA 4500
EMERGENCY GEN. & DIST. SWBD.
F47 - DC - 841991-CØ1 (MAY 1987)
SPECIFICATION ITEM NOS. 3A & 3B
ELVI ONE LINE DIAGRAM

II.

U.S. COAST GUARD REQUIRED INFORMATION
DOT - COAST GUARD - 46 CFR FED. REGISTER
.GE REQN. NO.: F47 - DC - 841991 - CØ1
.GE MTL. LIST : Ø504X0629W60,W61
.SHIPYARD NAME : DAEWOO OPKO KOREA
.VESSEL NAME : GEORGE RICHARDSON
.VESSEL OWNER : SONAT OFFSHORE DRILLING INC.
.VESSEL CLASS : GVA - 4500 SEMI - SUBMER.
.MODU - MOBILE OFFSHORE DRILLING UNIT.
.DATE : MAY 87
.HULL NO : 3011

III.

MCC MAIN AC THRU BUS DATA
.VOLTAGE : 480 VAC, 3 PHASES, 3 WIRES
60 HERTZ
.CURRENT CAPACITY : 1000 AMPERES
.BUS MATERIAL : TIN PLATED COPPER (TP - CU)
.AMBIENT RATING : 50 C
BUS DIMENSIONS : Ø.25"T X 2.0"H X 2 PER PHASE
.MCM 2/0 646MCM MAIN TO MCC
.SHORT CIRCUIT BRACING 65KA
.DATE : MAY '87

MCC BUS CONTINUATION [241A25]

1000 A BUS MCC.

MCC - E1: "RADAR X-BAND TRANS-FORMERS" TEDL — GE TEDL 15AT 100AF
MCC - E2: "RADAR ARPR TRANS-FORMERS" TEDL — GE TEDL 15AT 100AF
MCC - E3: "ACOUSTIC POS. IND. RELAY PORT" TEDL — GE TEDL 15AT 100AF
MCC - E4: "ACOUSTIC POS. IND. RELAY STDB." TEDL — GE TEDL 15AT 100AF
MCC - E5: "SPARE FEEDER NO. 3" TEDL — GE TEDL 15AT 100AF
MCC - E7: "SPARE FEEDER NO. 2" TEDL — GE TEDL 30AT 100AF
MCC - E6: "SPARE FEEDER NO. 1" TEDL — GE TEDL 50AT 100AF

MCC FEEDER UNITS

MCC STARTER UNITS

SEA WATER COOLING PUMP #3
MCC A0/A1
CB
R-RUN
IS-START
AT
2S 2S 2S
CLOSED TRANSITION ST. / RUN
AM Ø-400 AMPS
CT'S 600/5
OL'S
GND.
T1 T2 T3
SEA WATER COOLING PUMP #3 - RVNR ; NEMA 6-ATS

NEMA SIZE 6 RVNR AT START 227 KW / 295 HP.
MCC UNIT 48" WIDE

MCC D3
CB
M
CT 1300/15
OL
CT 200/5
AM Ø-200 AMPS
3 AC HOT
"MUD MIX PUMP NO.2" NEMA 5 FVNR 75KW/100HP

MCC C4
CB
M
CT 100/5
OL
AM Ø-100 AMPS
3 AC HOT
"LOW TEMP. FRESH WATER COOLING PUMP NO.1"-NEMA4, FVNR,49KW 65 HP.

MCC B5
CB
M
CT 100/5
OL
AM Ø-100 AMPS
3 AC HOT
"EMER. GEN. SEA COOLING PUMP NO.1" NEMA 3, FVNR 31KW/42HP

MCC B4
CB
M
OL
3 AC HOT
"SPARE STARTER NO. 1" NEMA 1, FVNR 7.5KW/10HP

MCC E8
CB
CT 80/5
AM
TWO PRIMARY TURNS.
Ø-40AMPS
SS FS
ØL SS
ØL FS
3 3
AC HOT
T-SS T-FS
(T1-T2-T3) (T5-T4-T6)
"EMERGENCY GENERATOR ROOM SUPPLY FAN." NEMA 2 TWO (2) SPEED, 30, 1W, VARIABLE TORQUE.

DAEWOO HULL 3011

ELEMENTARY DIAGRAM
EMER. GEN. & SWBD. ONE LINE
237B6558
SH. NO. 241B
241CA
CONT. ON SH.

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.

ENGINEER GRAY, A.W.
FIRST MADE FOR REQN. F47-DC-841991-CØ1
I.C. NO.
PRINTS TO
REV.1 BC 06-16-87
REV.2 LP 7-27-87
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5

SH. NO. 241CA

CONT. ON SH. 241C 237B6558

STANDARD NOMENCLATURE

A - AMPERE OR ACCELERATOR	GND - GROUND	REA - REACTOR	( ) - LOCATED ON OPERATORS STATION
AC - ALTERNATING CURRENT	HOA - HAND-OFF-AUTOMATIC SWITCH	REC - RECTIFIER	(___PNL) - PARENTHESIS AND "___PNL" INDICATES EQUIPMENT NOT ON OPER. STATION, BUT WIRED BY G.E.
AL - ALARM RELAY	HP - HORSEPOWER	RH - RHEOSTAT	
AM - AMMETER	HTR - HEATER	RS - RESISTOR	
AMP - AMPERE	IL - INDICATING LIGHT	RV - REDUCED VOLTAGE	
AMS - AMMETER TRANSFER SWITCH	INC - INCOMING	S - START	
AN - ANNUNCIATOR	INST - INSTANTANEOUS	SCP - SOLENOID CONTROL PANEL	CAT. NO. - CATALOG NUMBER OF ITEM
AT - AUTO TRANSFORMER	INTL - INTERLOCK	SEL - SELECTOR SWITCH	
AV - ADJUSTABLE VOLTAGE	J - JOG	SH - SHUNT	
BR - BRAKE	L.O. - LOCKOUT	SOL - SOLENOID	XXX - UNIT LOCATION OF ANOTHER STARTER
C - CONTACTOR	LR - LATCH RELAY	SP - SPARE	
(C) - CUSTOMER SUPPLIES	LS - LIMIT SWITCH	SR - SPRING RETURN	
CAP - CAPACITOR	LSW - LINE SWITCH	SS - SLOW SPEED	
CB - CIRCUIT BREAKER	LT - LIGHT	SS2P - SINGLE SOLENOID TWO POSITION	
CC - CLOSING COIL	M - MAIN CONTACTOR	SU - SPACE UNIT	
CLF - CURRENT LIMITING FUSE	MC - MAINTAINED CONTACT	SW - MANUAL SWITCH	
CLU - CLUTCH	MCC - MOTOR CONTROL CENTER	T - TRANSFORMER OR TRANSMITTER	
CP - CONSTANT POTENTIAL	MD - MAIN DRIVE	TB - TEMINAL BOARD	
CR - CONTROL RELAY	MN - MAIN	TC - TIME CLOCK OR TRIP COIL	
CSW - CONROL SWITCH	MP - MAINTAIN POSTION	TM - TIME METER	
CT - CURRENT TRANSFORMER	MTR - METER	TR - TIMING RELAY	
CTL - CONTROL	N - NEUTRAL	TSW - TRANSFER SWITCH	
D - DOWN	NA - NON-AUTOMATIC	TT - THERMOSTAT	
DB - DYNAMIC BRAKING	NC - NORMALLY CLOSED	U - UP	
DC - DIRECT CURRENT	NO - NORMALLY OPEN	UV - UNDERVOLTAGE	
DLS - DELIVERY SIDE	NR - NON-REVERSING	V - VOLT	
DM - DEMAND METER	OC - OVERCURRENT	VA - VOLT AMPERES	
DO - DRAWOUT	OL - OVERLOAD	VM - VOLTMETER	
DRS - DRIVE SIDE	OPS - OIL PRESSURE SWITCH	VMS - VOLTMETER TRANSFER SWITCH	
DS2P - DOUCLE SOLENOID TWO POSITION	OS - OPERATOR'S STATION	VS - VERTICAL SECTION	
DS3P - DOUBLE SOLENOID THREE POSITION	OSW - OPERATOR'S SWITCH	WHM - WATTHOUR METER	
DSW - DOOR SWITCH	OTT - OVERTEMPERATURE THERMOSTAT	WKS - WORK SIDE	
ENS - ENTRY SIDE	OV - OVERVOLTAGE	WM - WATTMETER	
ES - EMERGENCY STOP	PB - PUSHBUTTON	WPS - WATER PRESSURE SWITCH	
ETM - ELAPSED TIME METER	PF - POWER FACTOR	WR - WOUND ROTOR	
F - FORWARD	PFM - POWER FACTOR METER	2S1W - TWO SPEED, ONE WINDING	
FC - FIELD CONTACTOR	PH - PANEL HEATER	2S2W - TWO SPEED, TWO WINDING	
FDR - FEEDER	PNL - PANEL	-X - AUXILIARY DEVICE (MX,,FX,ETC.)	
FLSW - FLOAT SWITCH	PS - PRESSURE SWITCH	☐ - INDICATING LIGHT (LETTER BELOW	
FS - FAST SPEED	PT - POTENTIAL TRANSFORMER	R - DENOTES COLOR: A=AMBER, B=BLUE, C=CLEAR, G=GREEN, N=NEON(RED), R=RED W=WHITE)	
FSW - FIELD SWITCH	PW - PART WINDING		
FTSW - FOOT SWITCH	PWNR - PART WINDING NON-REVERSING		
FU - FUSE	PWR - PART WINDING REVERSING		
FV - FULL VOLTAGE	R - REVERSE OR RECEIVER	* - REMOTE MOUNTED, WIRED BY OTHERS	
GD - GROUND DETECTOR	(R) - REMOTE MOUNTED	① - REFER TO NOTE NO. 1	

DAEWOO HULL NO. 3011

ELEMENTARY DIAGRAM MCC
STANDARD NOMENCLATURE
SH. NO. 241CA
237B6558
CONT. ON SH. 241C

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.

ENGINEER A.W. GRAY
FIRST MADE FOR REQN. F47-DC-84-1991-C01
I.C. NO. DS4875
PRINTS TO
REV.1
REV.2 ISSUED 06-17-87
REV.3
REV.4
REV.5 MADE BY JOHNSON, RC

ORIG

DAEWOO FOR SONAT GVA 4500 CLASS SEMI. (MODU)
HULL NUMBER 3011
EMERGENCY GENERATOR CONTROL
DATA SHEET
SPECIFICATIONS ITEMS 3A & 3B - EVLI

I. DIESEL ENGINE:

MANUFACTURER - WARTSILA
MODEL - 6R32
HORSEPOWER/KW - 2869HP/2140KW (OUTPUT)
RPM

II. EMERGENCY GENERATOR

MANUFACTURER -
MODEL - 6V32
KVA-KW - 2675KVA; 2140KW
VOLTAGE - 480 VOLTS AC,30,60HZ
POWER FACTOR - 0.8 PF
RATED CURRENT - 3217.6 AMPS (@ 2675 KVA)
SYMMETRICAL SC - 65,000 AMPS

III. EMERGENCY GENERATOR CIRCUIT BREAKER

MANUFACTURER - TERASAKI
MODEL - AH-40C, 3 POLE, 4000 AF
OPERATION - ELECTRICAL (MOTOR CHARGE TYPE), 120V
MOUNTING - DRAWOUT
CIRCUIT VOLTAGE - 480VAC, 30, 3W, 60HZ
AMBIENT TEMP. - 50°C
CONTINUOUS = I_0 = 3218 AMPS, @115% = 3700A (5 TO 30 SEC.)
AL = LONG TIME = 3700 AMPS; 30 SEC.
AS = SHORT TIME = 9650 AMPS; 270 MLIIL SECS.
NI = NON ADJ INST = 50,000 AMPS; INST.
UNDERVOLTAGE TRIP = 120VAC,10,60HZ 500MS DELAY
TRIP INDICATION SWITCH

IV. VOLTAGE REGULATOR EQUIPMENT

BY STROMBERG (NON GE)
SEE STROMBERG DIAGRAM 3351 PA 3EL 001,REV.A. & 003 REV A DIMENSIONS
INPUT FROM GE: 120V,30,60HZ 50VA
RELAY F1 INPUT: 120V,10,60HZ 10VA.

V_{NL} = 42 VOLTS; I_{NL} = 2 AMPS
V_{FL} = 115 VOLTS; I_{FL} = 6 AMPS
} EXC. FIELD DATA. (MANUAL)

V. HYDRAULIC GOVERNOR EQUIPMENT

ASSUME BY WARTSILA-NON GE.

VI. CURRENT TRANSFORMER RATIOS

4000:5 CTA & CTC.

VII. POTENTIAL TRANSFORMER RATIOS

480:120 (4:1)

VIII. METER SCALES - MOVEMENT

VOLTMETER : 0-600V (RED LINE @ 480)
150 VOLT INPUT XFMR RATED.
AMMETER : 0-4200 AMPS
5.25A INPUT, (RED BAND @ 3218 AMPS.)
KILOWATT : 0-3000 KILOWATTS
IMA MOVEMENT INPUT
120V,10,60HZ,POWER INPUT.
RED BAND FROM 2175KW TO 3000KW
DB-40 TYPE SWITCHBOARD METER.

IX. "TIE" CIRCUIT BREAKER

MANUFACTURER : TERASAKI
MODEL : AH-30C,3 POLE, 3000AF
OPERATION : ELECTRICAL (MOTOR CHG.)
120V,10,60 HZ
MOUNTING : DRAWOUT
CIRCUIT VOLTAGE : 480V,30,60HZ
AMBIENT TEMP : 50°C
CONTINUOUS I_0 = 3000A @ 100% = 3000 AMPS
AL (LONG TIME) = 3000 A; 20 SECS.
AS (SHORT TIME) = 9000 A; 270 MILLISECS.
NI (NON.ADJ.INST) = 40,000 A; INST.
UNDERVOLTAGE = 120V,10,60HZ,500MS - TIME DELAY

X. AUTOMATIC TRANSFER CONTROL RELAYS

BASLER ELECTRIC CO. TYPE BE1-27/59
CAT.NO. BE1-27/59-A3N-C5L-AOSIP
RELAY POWER = 24VDC
RELAY SENSING VOLTAGE = 55-160VAC
TC=MEDIUM INVERSE
INST. BASLER PUBLICATION NO.
9 1706 00 990 JAN. 1986

XII. PREFERENTIAL TRIP UNITS AT1/AT2

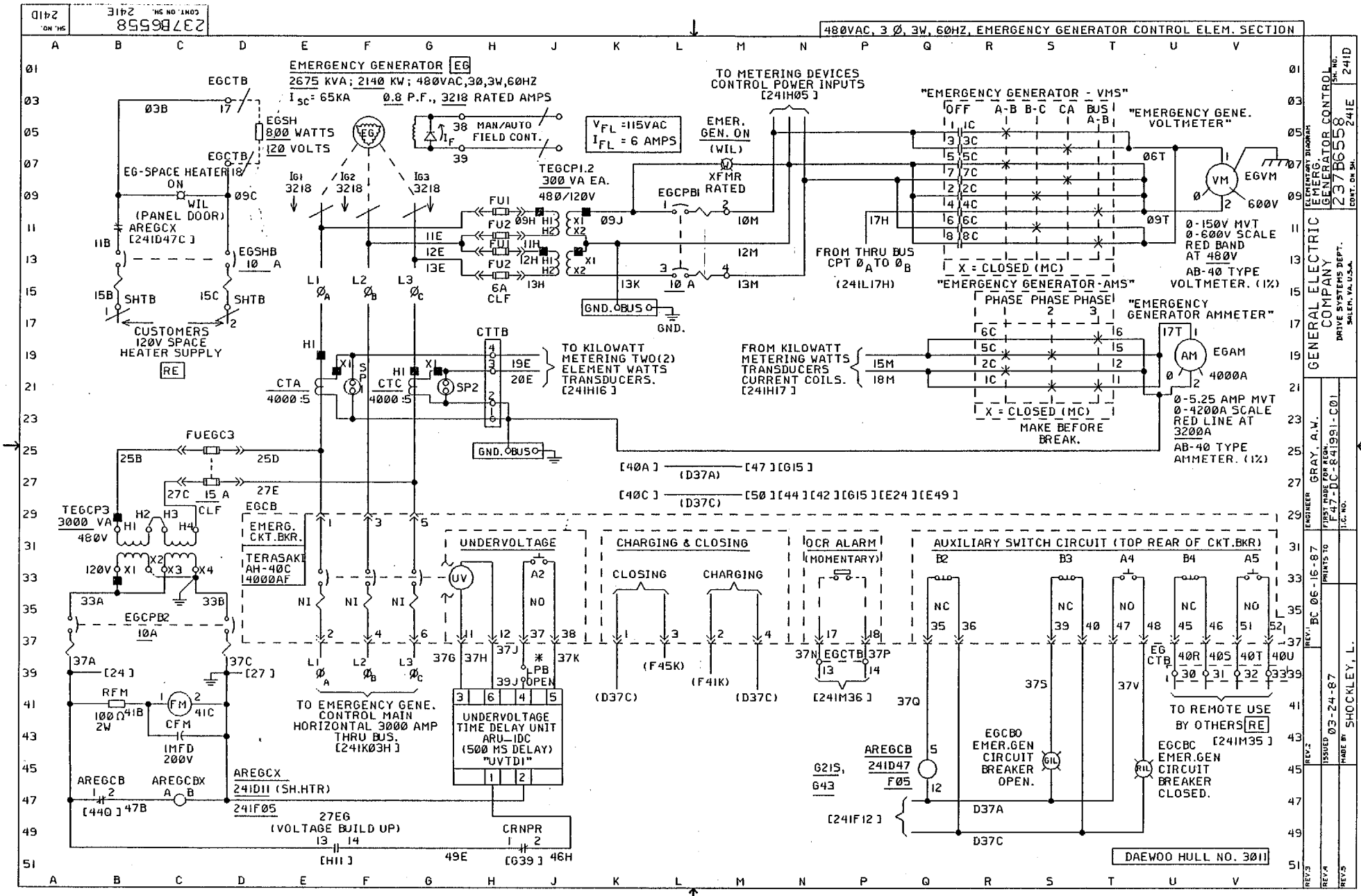
ROCHESTER INSTRUMENT SYSTEMS(RIS)
TYPE ET-12-19 DUAL TRIP CURRENT/ VOLTAGE ALARM. 0-1MA INPUT
CURRENT SENSING, 120VAC VOLTAGE POWER.
PUBLICATION NO. 1035-143(2-1-83)

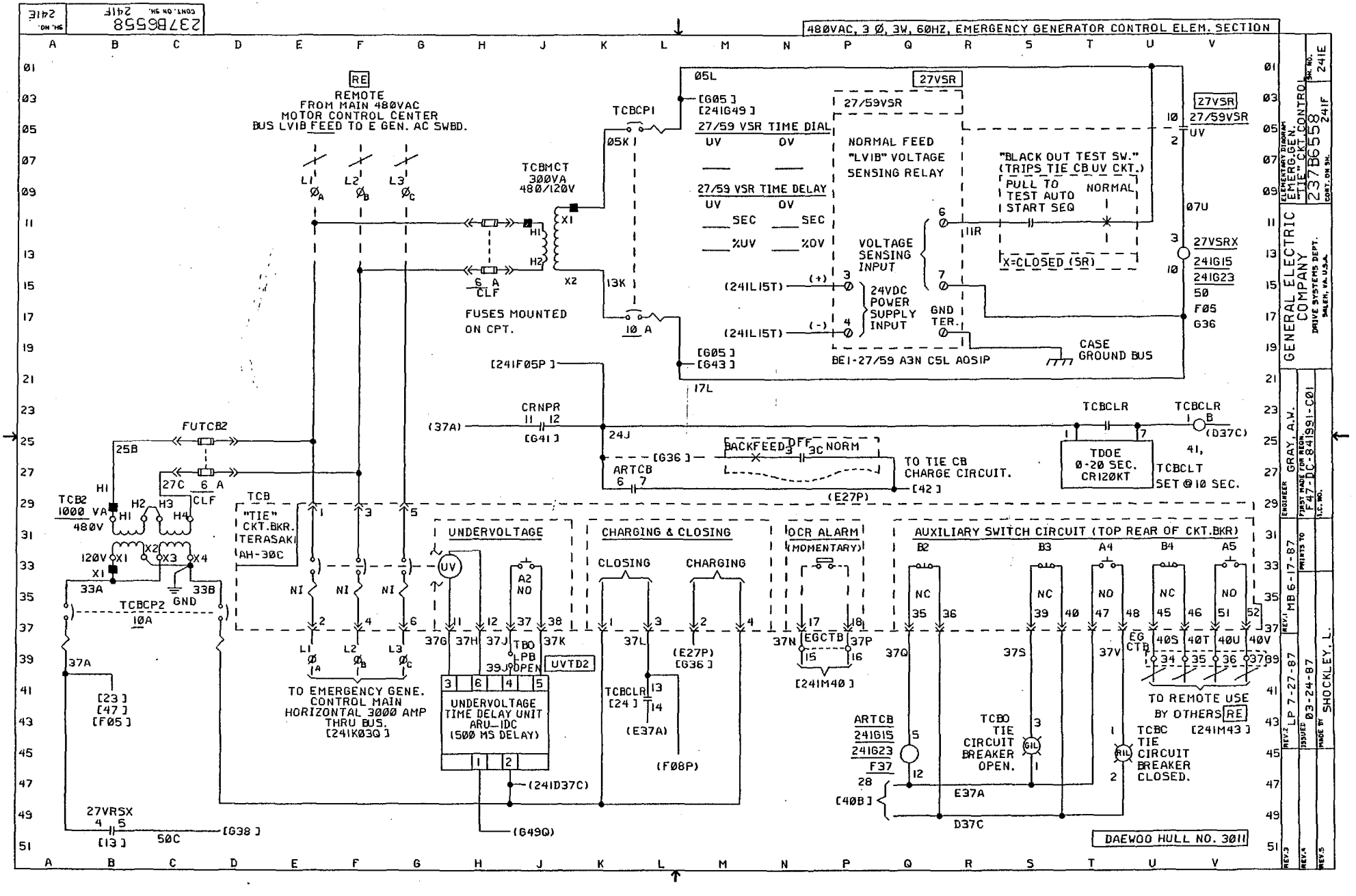
XIII.

DAEWOO HULL NO. 3011

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA, U.S.A.
ELEMENTARY DIAGRAM EMERG.GEN CONTROL DATA SHEET
SH. NO. 241C
237B6558 241D
CONT. ON SH.
ENGINEER GRAY, A.W.
FIRST MADE FOR REQN. F47-DC-841991-C01
I.C. NO.
REV.1 BC 06-15-87
PRINTS TO
REV.2
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5

237B6558 241D
SH. NO. 241C


480VAC, 3 Ø, 3W, 60HZ, EMERGENCY GENERATOR CONTROL ELEM. SECTION
EMERGENCY GENERATOR EG
2675 KVA; 2140 KW; 480VAC,3Ø,3W,60HZ
I SC= 65KA 0.8 P.F., 3218 RATED AMPS
EGCTB
EGSH 800 WATTS 120 VOLTS
EG-SPACE HEATER ON
WIL (PANEL DOOR)
AREGCX [241D47C]
EGSHB 10 A
SHTB
CUSTOMERS 120V SPACE HEATER SUPPLY
RE
MAN/AUTO FIELD CONT.
VFL =115VAC IFL = 6 AMPS
EMER. GEN. ON (WIL)
TO METERING DEVICES CONTROL POWER INPUTS [241H05]
TEGCP1,2 300 VA EA. 480/120V
XFMR RATED
EGCPB1
FU1 FU2 6A CLF
GND. BUS
"EMERGENCY GENERATOR - VMS"
OFF A-B B-C CA BUS A-B
X = CLOSED (MC)
"EMERGENCY GENE. VOLTMETER"
EGVM 600V
0-150V MVT 0-600V SCALE RED BAND AT 480V
AB-40 TYPE VOLTMETER. (1%)
FROM THRU BUS CPT ØA TO ØB (241L17H)
"EMERGENCY GENERATOR-AMS"
PHASE PHASE PHASE
"EMERGENCY GENERATOR AMMETER"
EGAM 4000A
0-5.25 AMP MVT 0-4200A SCALE RED LINE AT 3200A
AB-40 TYPE AMMETER. (1%)
MAKE BEFORE BREAK.
CTTB
TO KILOWATT METERING TWO(2) ELEMENT WATTS TRANSDUCERS. [241H16]
FROM KILOWATT METERING WATTS TRANSDUCERS CURRENT COILS. [241H17]
CTA 4000:5
CTC 4000:5
SP1 SP2
FUEGC3
TEGCP3 3000 VA 480V
EGCB
EMERG. CKT.BKR.
TERASAKI AH-40C 4000AF
UNDERVOLTAGE
CHARGING & CLOSING
CLOSING CHARGING
OCR ALARM (MOMENTARY)
AUXILIARY SWITCH CIRCUIT (TOP REAR OF CKT.BKR)
EGCPB2 10A
RFM 100Ω 2W
CFM 1MFD 200V
AREGCB AREGCBX
AREGCX 241D11 (SH.HTR)
241F05
27EG (VOLTAGE BUILD UP)
CRNPR
UNDERVOLTAGE TIME DELAY UNIT ARU-IDC (500 MS DELAY) "UVTDI"
TO EMERGENCY GENE. CONTROL MAIN HORIZONTAL 3000 AMP THRU BUS. [241K03H]
AREGCB 241D47 F05
G21S, G43
EGCBO EMER.GEN CIRCUIT BREAKER OPEN.
EGCBC EMER.GEN CIRCUIT BREAKER CLOSED.
TO REMOTE USE BY OTHERS RE [241M35]
[241F12]
D37A
D37C
DAEWOO HULL NO. 3011
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM EMERG. GENERATOR CONTROL
237B6558
241E
241D
ENGINEER GRAY, A.W.
DRAWN BC 06-16-87
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
I.C. NO. 47-DC-84199-C01


480VAC, 3 Ø, 3W, 60HZ, EMERGENCY GENERATOR CONTROL ELEM. SECTION
237B6558
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM EMERG.GEN. "TIE" CKT CONTROL
241E
241F
241G
CONT. ON SH. 241E
CONT. ON SH. 241F
RE
REMOTE FROM MAIN 480VAC MOTOR CONTROL CENTER BUS LV1B FEED TO E GEN. AC SWBD.
L1 Ø_A L2 Ø_B L3 Ø_C
TCBMCT 300VA 480/120V
X1 X2 H1 H2
6 A CLF
FUSES MOUNTED ON CPT.
05L
[G05] [241G49]
TCBCPI
05K
10 A
13K
27/59 VSR TIME DIAL
UV OV
27/59 VSR TIME DELAY
UV OV
SEC SEC
%UV %OV
27VSR
27/59VSR
NORMAL FEED "LV1B" VOLTAGE SENSING RELAY
"BLACK OUT TEST SW." (TRIPS TIE CB UV CKT.)
PULL TO TEST AUTO START SEQ
NORMAL
X=CLOSED (SR)
VOLTAGE SENSING INPUT
(241L15T) (+) 3
(241L15T) (-) 4
24VDC POWER SUPPLY INPUT
GND TER.
7
6
11R
BE1-27/59 A3N C5L A0S1P
CASE GROUND BUS
10 27/59VSR UV 2
07U
3 27VSRX
241G15
241G23
50
F05
G36
[241F05P]
[G05] [G43]
17L
CRNPR 11 12 [G41]
(37A)
24J
[G36]
ARTCB 6 7
BACKFEED OFF 3C NORM
TO TIE CB CHARGE CIRCUIT.
[42]
(E27P)
TCBCLR 1 7
TCBCLR 1 B (D37C)
TDOE 0-20 SEC. CR120KT
41, TCBCLT SET @ 10 SEC.
FUTCB2
25B
27C 6 A CLF
TCB2 1000 VA 480V
HI H2 H3 H4
120V X1 X2 X3 X4
X1
33A 33B GND
TCBCP2 10A
37A
[23] [47] [F05]
TCB "TIE" CKT.BKR. TERASAKI AH-30C
1 3 5
NI NI NI
2 4 6
L1 Ø_A L2 Ø_B L3 Ø_C
TO EMERGENCY GENE. CONTROL MAIN HORIZONTAL 3000 AMP THRU BUS. [241K03Q]
UNDERVOLTAGE
UV
A2 NO
TBO LPB OPEN
UVTD2
37G 37H 37J 37K 39J
UNDERVOLTAGE TIME DELAY UNIT ARU-1DC (500 MS DELAY)
3 6 4 5
1 2
(241D37C)
(G49Q)
CHARGING & CLOSING
CLOSING CHARGING
1 3 2 4
37L
TCBCLR 13 14 [24]
(E37A)
(E27P) [G36]
(F08P)
OCR ALARM (MOMENTARY)
17 18
37N EGCTB 37P
15 16
[241M40]
AUXILIARY SWITCH CIRCUIT (TOP REAR OF CKT.BKR)
B2 B3 A4 B4 A5
NC NC NO NC NO
35 36 39 40 47 48 45 46 51 52
37Q 37S 37V EG CTB 40S 40T 40U 40V
TO REMOTE USE BY OTHERS RE [241M43]
ARTCB 241G15 241G23 F37
5 12 28
[40B]
TCBO TIE CIRCUIT BREAKER OPEN.
TCBC TIE CIRCUIT BREAKER CLOSED.
E37A
D37C
27VRSX 4 5 [13]
50C
[G38]
DAEWOO HULL NO. 3011
GRAY, A.W.
MB 6-17-87
LP 7-27-87
03-24-87
SHOCKLEY, L.
F47-DC-841991-G01

MANUAL MODE TIE CB CHARGING/CLOSING CIRCUIT

EGMSS

AREGCB (EG CB OPEN) 10 11 AREGCX
(241E37A) [D43] 05E [D47] 05F
AUTO OFF MANUAL
[G18]
05J [E13] 27VSRX 6 7 05K
TIE CB CHARGE TBCHPB
N.O. 05M
TIE CB CLOSING TBCLPB
N.C.
(241E24J)
N.C. 08M N.O.
(241E37L)
[G32]

BACKFEED OP.MODE
TIE CB CHARGING/CLOSING
POWER.
"MAN.OFF AUTO" SW. IN MANUAL
"BACKFEED-OFF-NORMAL" SW.IN
BACKFEED.

MANUAL VOLTAGE CONTROL EQUIPMENT AND SWITCH

VOLTAGE VARYING "VARIAC" AUTO TRANS.
9T92A37
(241D37A) [241D50]
241D37C
FRAME 85
29E
37C
FULL WAVE DIODE RECTIFIER BRIDGE
FWB
25AMP CAP.

MANUAL/AUTO VOLTAGE CONTROL SELECTOR SWITCH

MAVCSS
MANUAL MODE AUTO MODE
FROM "AUTO" VR EQUIP [241AR__]
25K 27K 29K 32K
EGCTB 38 25N 39 27N
TO CUSTOMERS BRUSHLESS GENERATOR FIELD EXCITER VIA EMERGENCY GEN. CUSTOMER CONTROL INERFACE TERMINAL BOARD [241M49]
F = CLOSED MC
MAKE BEFORE BREAK CONTROL SW.

BRUSHLESS EXCITER RATINGS	
• VOLTAGE NO LOAD VNL = 42 VOLTS	• CURRENT NO LOAD INL = 2 AMPS
• VOLTAGE FULL LOAD VFL = 115 VOLTS	• CURRENT FULL LOAD IFL = 6 AMPS

EMER.GEN.CB MAN/AUTO CHARGING & CLOSING CIRCUIT

[G17] (241G17G)
ARTCB (TCB OPEN) 3 4 [E43] 37D
27EG VSRX EG VOLTAGE BUILD UP 6 7 [H11] 37F

EMER GEN. CB CHARGE (MANUAL)
N.O.41H
EMER GEN. CB CLOSE (MANUAL)
N.C. 41K
[D42L]
TO EMERG. GEN. CB "CHARGING" CIRCUIT.

N.C.45H N.O. 45K
[D44K]
TO EMERG. GEN. CB "CLOSING" CIRCUIT.

AST AUX (INST.) 6 7
[G09] (G13G) [G09] 47D
ASTDR (TDOE) 1 2 [G15]
ASTDR (TDOE) 13 14 [G15]
AUTOMATIC CLOSING

DAEWOO HULL NO. 3011

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM
SH. NO. 241F
237B6558 241G
CONT. ON SH.
ENGINEER GRAY, A.W.
FIRST MADE FOR RECN. F47-DC-8419-1-C01
I.C. NO.
PRINTS TO
REV.1 BC 06-17-87
REV.2
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5

480VAC, 3 Ø, 3W, 60HZ, EMERGENCY GENERATOR CONTROL ELEM. SECTION

27VSR
[TIMED OVERVOLTAGE CONTACT TD ON CLOSE]

A270VR
[LVIB PRIMARY VOLTAGE RETURN TO RATED]

(E05L) 17 18 [E090] G05E 4 11 G36 (E17L)

CLOSES AFTER 30 SEC.. TIME DELAY.

ASTAUX (INST.) 4 11 F47 (241D37C)

[F47]

"EMERGENCY GENERATOR MODE SELECT." SWITCH [EGMSS] AUTO OFF MANUAL

EMERGENCY GENERATOR PANEL CONTROL SOURCE POWER. (241D37A) 1 1C 13G

4 4C 17G [F37]

AREGCB EGCB CLOSED 3 4 [D43] (F05E) 3C F05J

20D 2C 21G [36]

5 5C 23G 8 9 [241E44] 8 9 [241E13] 23J EGCTB 23K 2 25G

X = CLOSED MAINTAINED CONTACT

ARTCB (TCB OPEN) 1 2 15J

27VSRX (VOLTAGE LOSS) 1 2 15K

ASTDR (TDOE) 6 7 4 11 15M

ASTDR F47 F49

TASTD TDOE 30SEC CR120KT 7 (TIME DELAY UNIT) SET AT 10 SEC.

DIESEL ENGINE COOLDOWN TIMING RELAY CIRCUITS

(D37C) (E37A) CRNPR 6 7 [40] ESDR 6 7 [19] 19S ESDRTD 1 2 21S

ESDR 4 11 ESDR 32

ESDTD 13 14 4 11 1 7 ESDTDT TDOE 3-5 MIN

6 ESDTD [25] 7 26R

ESDRTD 13 14 4 11 1 7 31S TDOE 20 SEC.

"EG ENGINE START" [EGP] N.O. 27J 27H AEGCTB 7 8

"EG ENGINE STOP" AEGCTB 5 N.O. 28L 6

ESDR 13 14 [19V] 29L EGCTB 19 20

RE TO EMERGENCY GENERATOR DIESEL ENGINE-EMERGENCY START [241M12]

RE TO EMERGENCY GENERATOR PUSH BUTTON START-STOP CIRCUIT. [241M16]

RE DIESEL ENGINE COOL DOWN CUSTOMER CIRCUITS [241M11]

EMERGENCY GENERATOR "BACKFEED" [EGP] - EGBSS SELECTOR SWITCH

BACK FEED OFF NORMAL 1 1C

(G17G) (F05K)

(21G) 2 2C 36G [G05G] A270VR 6 7 36H

(E24J) 3 3C (E27P)

[E50] 4 4C 50C

EGCB CLOSED AREGB 8 9 [D43] 5 5C 43D

X = CLOSED KEY REMOVABLE BOTH POS.

TIE CIRCUIT BREAKER CONTROL SOURCE POWER. (241E05L)

TIE CB OPEN ARTCB 14 13 [E43] 49D

4 3 5 10 CRNPR NORM. PWR. RESTORE TRIP EGCB COOL DOWN DIESEL ENGINE START CYCLE. E25, D50, 40, G19

TO PREFERENTIAL TRIP DEVICE ATI CIRCUIT-KIB. [241H41]

43G

(241D37C)

[241H42G] [E50] 49Q

TO KW SENSING TRANSDUCER [241H32] [241H27]

32N 23E

KWMR EMERGENCY GENERATOR KILOWATTS

DB-40 SWITCHBOARD METER

SCALE - 0-3000KW

MOVEMENT 0-1MA.

1% ACCURACY FS

RED BAND FROM 2175KW TO 3000KW

IMA 1 1000 2000 EMER GEN. KILO-WATTS RED BAND 0 2 3000KW CASE GND

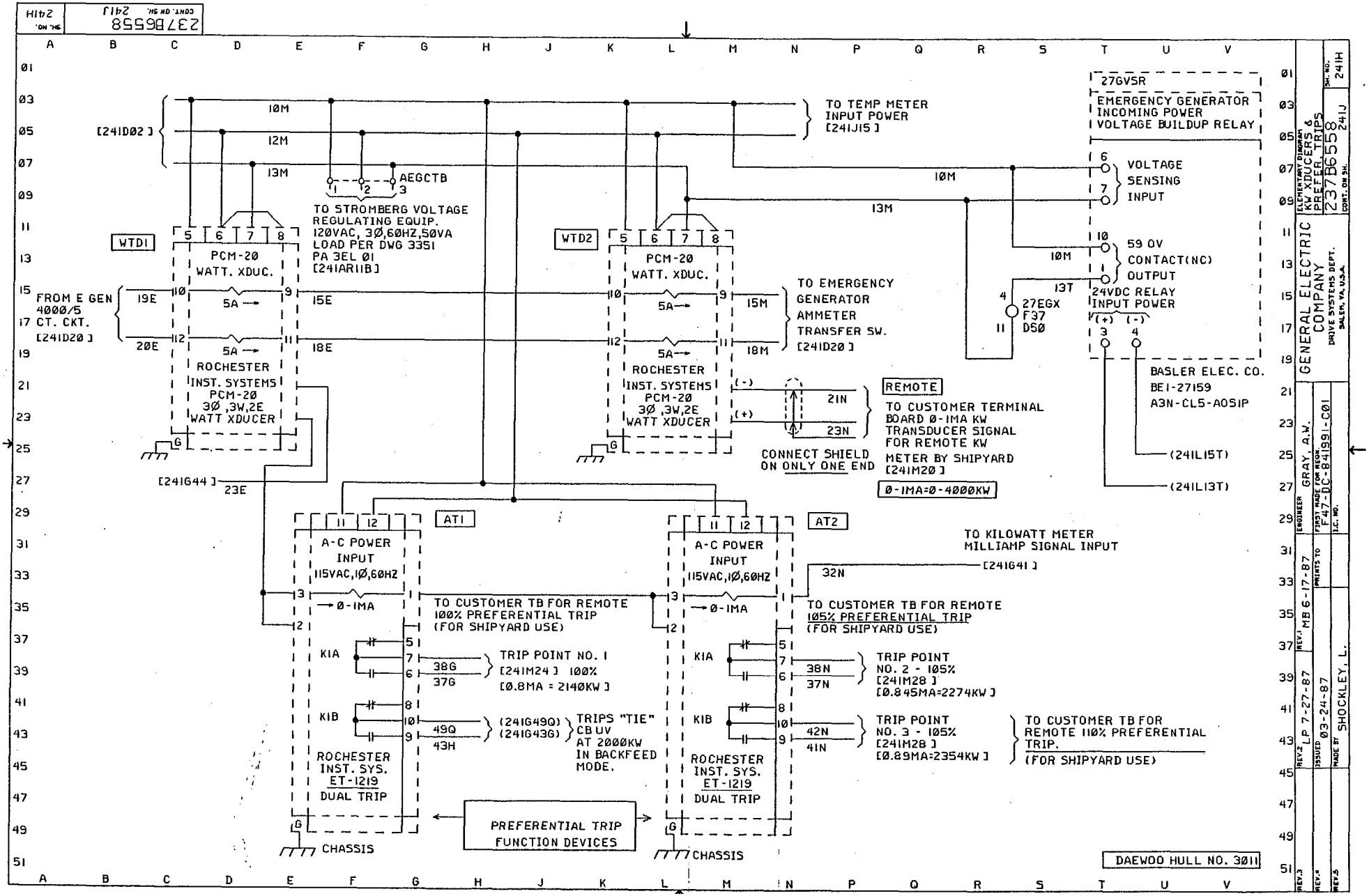
NOTE: KW METER INDICATES KW, WHILE PREFERENTIAL TRIP UNIT ATI (RELAY OUTPUT K1B) ACTS TO TRIP TIE CB UV IN BACKFEED MODE AT 2000KW. 93.4%

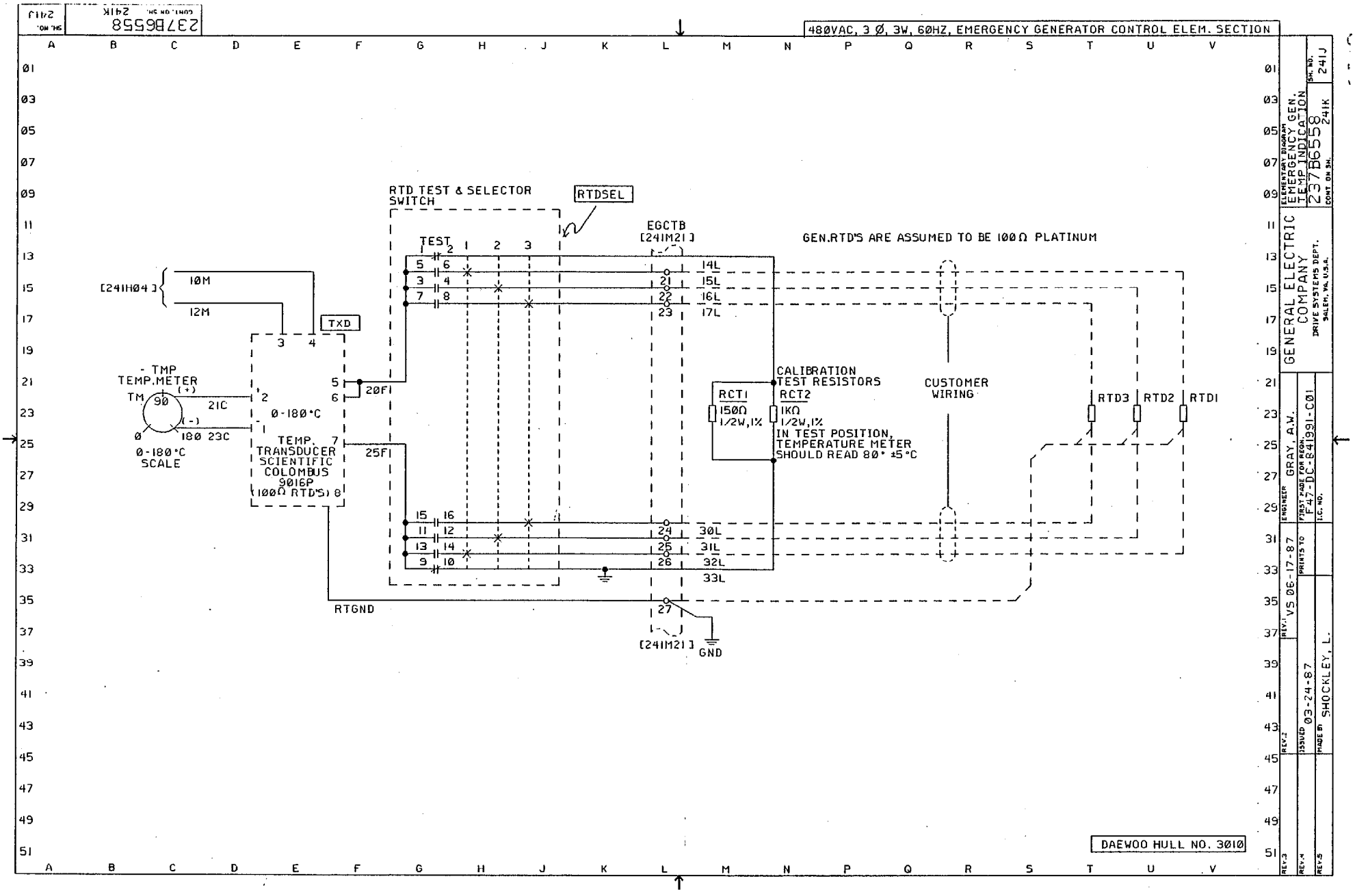
DAEWOO HULL NO 3011

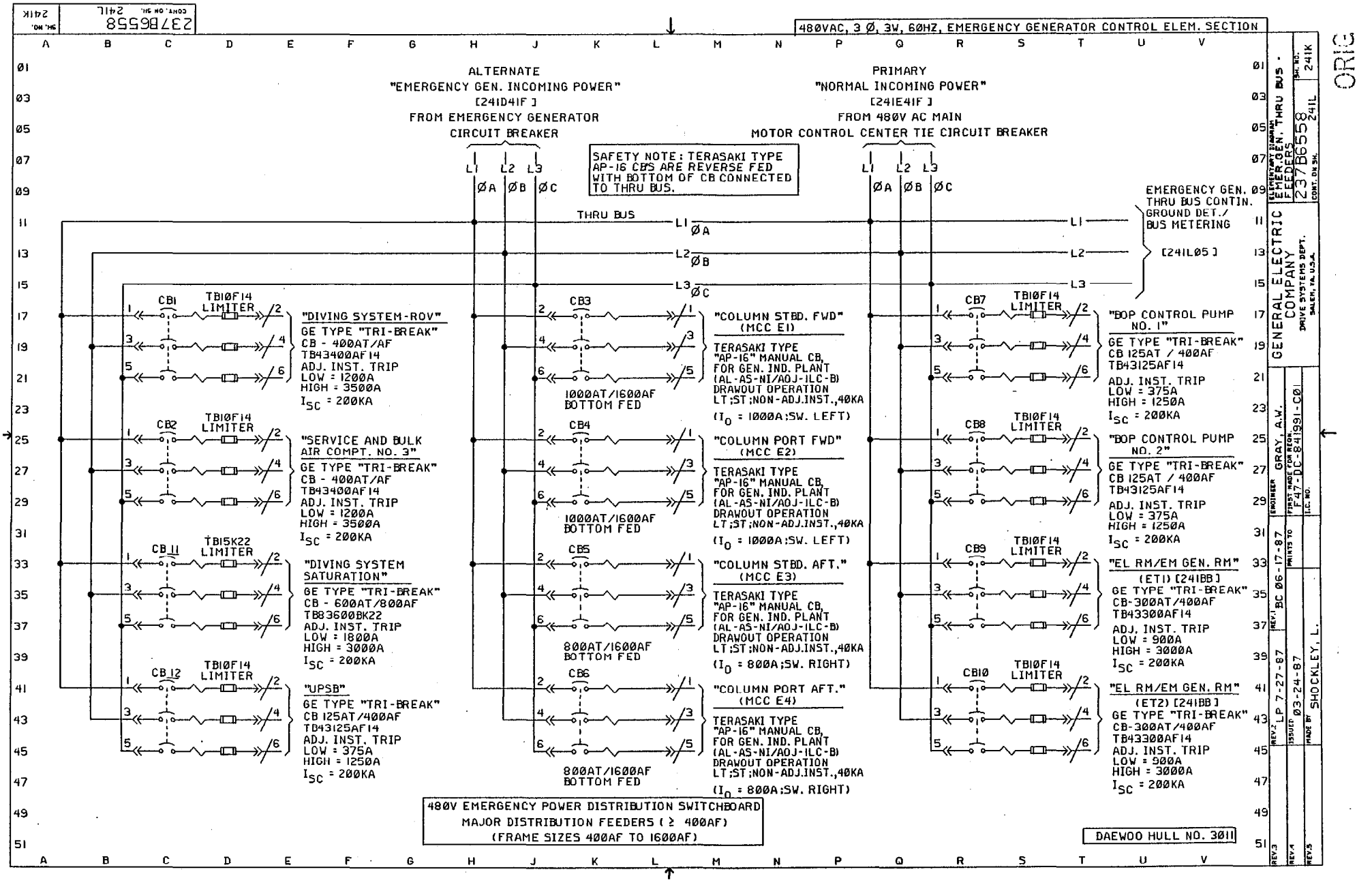
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.

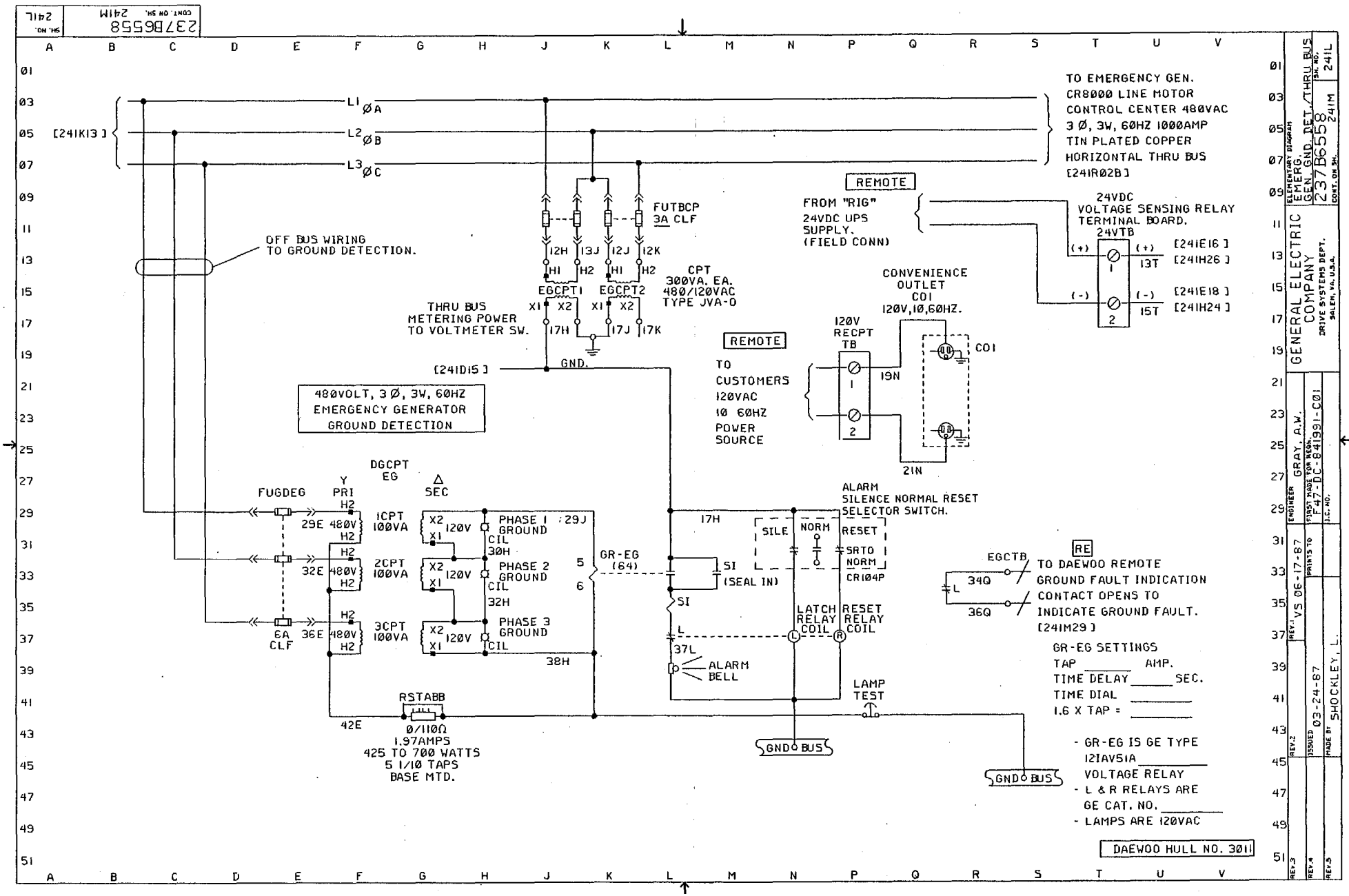
ELEMENTARY DIAGRAM
E. GEN. CONTROL CIRCUIT
237B6558
SH. NO. 2416
241H
CONT. ON SH.

ENGINEER GRAY, A.W.
REV.1 MB 6-17-87
REV.2
REV.3
REV.4
REV.5
FIRST MADE FOR REGN. E 47-DC-84199-C01
I.C. NO.
PRINTS TO
ISSUED 03-24-87
MADE BY SHOCKLEY, L.


SH. NO. 241H
CONT. ON SH. 241J
237B6558
[241D02]
10M
12M
13M
AEGCTB
1 2 3
TO STROMBERG VOLTAGE REGULATING EQUIP. 120VAC, 3Ø,60HZ,50VA LOAD PER DWG 3351 PA 3EL 01 [241AR11B]
TO TEMP METER INPUT POWER [241J15]
27GVSR
EMERGENCY GENERATOR INCOMING POWER VOLTAGE BUILDUP RELAY
6 VOLTAGE SENSING 7 INPUT
10M
13M
10 59 OV CONTACT(NC) OUTPUT
10M
13T
4 27EGX F37 11 D50
24VDC RELAY INPUT POWER
(+) (-)
3 4
BASLER ELEC. CO.
BE1-27159
A3N-CL5-A0S1P
(241L15T)
(241L13T)
WTD1
PCM-20 WATT. XDUC.
5 6 7 8
FROM E GEN 4000/5 CT. CKT. [241D20]
19E
20E
10 12 9 11
5A
5A
ROCHESTER INST. SYSTEMS PCM-20 3Ø ,3W,2E WATT XDUCER
15E
18E
WTD2
PCM-20 WATT. XDUC.
5 6 7 8
10 12 9 11
5A
5A
ROCHESTER INST. SYSTEMS PCM-20 3Ø ,3W,2E WATT XDUCER
TO EMERGENCY GENERATOR AMMETER TRANSFER SW. [241D20]
15M
18M
(-) (+)
21N
23N
REMOTE
TO CUSTOMER TERMINAL BOARD Ø-1MA KW TRANSDUCER SIGNAL FOR REMOTE KW METER BY SHIPYARD [241M20]
CONNECT SHIELD ON ONLY ONE END
Ø-1MA=Ø-4000KW
[241G44]
23E
AT1
A-C POWER INPUT 115VAC,1Ø,60HZ
11 12
3 1 2
→Ø-1MA
TO CUSTOMER TB FOR REMOTE 100% PREFERENTIAL TRIP (FOR SHIPYARD USE)
KIA
5 7 6
38G
37G
TRIP POINT NO. 1 [241M24] 100% [0.8MA = 2140KW]
KIB
8 10 9
49Q
43H
(241G49Q) (241G43G)
TRIPS "TIE" CB UV AT 2000KW IN BACKFEED MODE.
ROCHESTER INST. SYS. ET-1219 DUAL TRIP
G
CHASSIS
PREFERENTIAL TRIP FUNCTION DEVICES
AT2
A-C POWER INPUT 115VAC,1Ø,60HZ
11 12
3 1 2
→Ø-1MA
TO KILOWATT METER MILLIAMP SIGNAL INPUT
[241G41]
32N
TO CUSTOMER TB FOR REMOTE 105% PREFERENTIAL TRIP (FOR SHIPYARD USE)
KIA
5 7 6
38N
37N
TRIP POINT NO. 2 - 105% [241M28] [0.845MA=2274KW]
KIB
8 10 9
42N
41N
TRIP POINT NO. 3 - 105% [241M28] [0.89MA=2354KW]
TO CUSTOMER TB FOR REMOTE 110% PREFERENTIAL TRIP. (FOR SHIPYARD USE)
ROCHESTER INST. SYS. ET-1219 DUAL TRIP
G
CHASSIS
DAEWOO HULL NO. 3011
ELEMENTARY DIAGRAM KW XDUCERS & PREFER. TRIPS
237B6558
SH. NO. 241H
241J
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA, U.S.A.
ENGINEER GRAY, A.W.
PRINTS TO
I.C. NO. F47-DC-841991-C01
REV.1 MB 6-17-87
REV.2 LP 7-27-87
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5


480VAC, 3 Ø, 3W, 60HZ, EMERGENCY GENERATOR CONTROL ELEM. SECTION
RTD TEST & SELECTOR SWITCH
RTDSEL
EGCTB [241M21]
GEN.RTD'S ARE ASSUMED TO BE 100Ω PLATINUM
[241H04]
10M
12M
TXD
- TMP
TEMP.METER
TM
90
(+)
(-)
0
180
21C
23C
0-180°C SCALE
0-180°C
TEMP. TRANSDUCER SCIENTIFIC COLOMBUS 9016P (100Ω RTD'S)
20F
25F
RTGND
TEST
14L
15L
16L
17L
30L
31L
32L
33L
CALIBRATION TEST RESISTORS
RCT1
150Ω
1/2W,1%
RCT2
1KΩ
1/2W,1%
IN TEST POSITION, TEMPERATURE METER SHOULD READ 80° ±5°C
CUSTOMER WIRING
RTD3
RTD2
RTD1
[241M21]
GND
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA, U.S.A.
ELEMENTARY DIAGRAM
EMERGENCY GEN. TEMP INDICATION
237B6558
241K
SH. NO. 241J
CONT. ON SH.
ENGINEER GRAY, A.W.
FIRST MADE FOR REGN.
F47-DC-841991-C01
I.C. NO.
PRINTS TO
REV.1 VS 06-17-87
REV.2
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5
DAEWOO HULL NO. 3010


480VAC, 3 Ø, 3W, 60HZ, EMERGENCY GENERATOR CONTROL ELEM. SECTION
ALTERNATE
"EMERGENCY GEN. INCOMING POWER"
[241D41F]
FROM EMERGENCY GENERATOR
CIRCUIT BREAKER
PRIMARY
"NORMAL INCOMING POWER"
[241E41F]
FROM 480V AC MAIN
MOTOR CONTROL CENTER TIE CIRCUIT BREAKER
SAFETY NOTE: TERASAKI TYPE AP-16 CBS ARE REVERSE FED WITH BOTTOM OF CB CONNECTED TO THRU BUS.
THRU BUS
L1 ØA
L2 ØB
L3 ØC
EMERGENCY GEN. THRU BUS CONTIN. GROUND DET./ BUS METERING
[241L05]
CB1 TBI0F14 LIMITER
"DIVING SYSTEM-ROV"
GE TYPE "TRI-BREAK"
CB - 400AT/AF
TB43400AF14
ADJ. INST. TRIP
LOW = 1200A
HIGH = 3500A
ISC = 200KA
CB2 TBI0F14 LIMITER
"SERVICE AND BULK AIR COMPT. NO. 3"
GE TYPE "TRI-BREAK"
CB - 400AT/AF
TB43400AF14
ADJ. INST. TRIP
LOW = 1200A
HIGH = 3500A
ISC = 200KA
CB11 TBI5K22 LIMITER
"DIVING SYSTEM SATURATION"
GE TYPE "TRI-BREAK"
CB - 600AT/800AF
TB83600BK22
ADJ. INST. TRIP
LOW = 1800A
HIGH = 3000A
ISC = 200KA
CB12 TBI0F14 LIMITER
"UPSB"
GE TYPE "TRI-BREAK"
CB 125AT/400AF
TB43125AF14
ADJ. INST. TRIP
LOW = 375A
HIGH = 1250A
ISC = 200KA
CB3
1000AT/1600AF BOTTOM FED
"COLUMN STBD. FWD" (MCC E1)
TERASAKI TYPE "AP-16" MANUAL CB, FOR GEN. IND. PLANT (AL-AS-NI/AOJ-ILC-B) DRAWOUT OPERATION LT;ST;NON-ADJ.INST.,40KA
(IO = 1000A;SW. LEFT)
CB4
1000AT/1600AF BOTTOM FED
"COLUMN PORT FWD" (MCC E2)
TERASAKI TYPE "AP-16" MANUAL CB, FOR GEN. IND. PLANT (AL-AS-NI/AOJ-ILC-B) DRAWOUT OPERATION LT;ST;NON-ADJ.INST.,40KA
(IO = 1000A;SW. LEFT)
CB5
800AT/1600AF BOTTOM FED
"COLUMN STBD. AFT." (MCC E3)
TERASAKI TYPE "AP-16" MANUAL CB, FOR GEN. IND. PLANT (AL-AS-NI/AOJ-ILC-B) DRAWOUT OPERATION LT;ST;NON-ADJ.INST.,40KA
(IO = 800A;SW. RIGHT)
CB6
800AT/1600AF BOTTOM FED
"COLUMN PORT AFT." (MCC E4)
TERASAKI TYPE "AP-16" MANUAL CB, FOR GEN. IND. PLANT (AL-AS-NI/AOJ-ILC-B) DRAWOUT OPERATION LT;ST;NON-ADJ.INST.,40KA
(IO = 800A;SW. RIGHT)
CB7 TBI0F14 LIMITER
"BOP CONTROL PUMP NO. 1"
GE TYPE "TRI-BREAK"
CB 125AT / 400AF
TB43125AF14
ADJ. INST. TRIP
LOW = 375A
HIGH = 1250A
ISC = 200KA
CB8 TBI0F14 LIMITER
"BOP CONTROL PUMP NO. 2"
GE TYPE "TRI-BREAK"
CB 125AT / 400AF
TB43125AF14
ADJ. INST. TRIP
LOW = 375A
HIGH = 1250A
ISC = 200KA
CB9 TBI0F14 LIMITER
"EL RM/EM GEN. RM" (ET1) [241BB]
GE TYPE "TRI-BREAK"
CB-300AT/400AF
TB43300AF14
ADJ. INST. TRIP
LOW = 900A
HIGH = 3000A
ISC = 200KA
CBI0 TBI0F14 LIMITER
"EL RM/EM GEN. RM" (ET2) [241BB]
GE TYPE "TRI-BREAK"
CB-300AT/400AF
TB43300AF14
ADJ. INST. TRIP
LOW = 900A
HIGH = 3000A
ISC = 200KA
480V EMERGENCY POWER DISTRIBUTION SWITCHBOARD
MAJOR DISTRIBUTION FEEDERS (≥ 400AF)
(FRAME SIZES 400AF TO 1600AF)
DAEWOO HULL NO. 3011
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM EMER.GEN. THRU BUS - FEEDERS
237B6558
SHT. NO. 241K
241L
CONT. ON SH.
ENGINEER GRAY, A.W.
FIRST MADE FOR RIGA F47-DC-84191991-C01
I.C. NO.
PRINTS TO
REV.1 BC 06-17-87
REV.2 LP 7-27-87
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5


TO EMERGENCY GEN.
CR8000 LINE MOTOR
CONTROL CENTER 480VAC
3 Ø, 3W, 60HZ 1000AMP
TIN PLATED COPPER
HORIZONTAL THRU BUS
[241R02B]
[241K13]
L1 ØA
L2 ØB
L3 ØC
OFF BUS WIRING
TO GROUND DETECTION.
FUTBCP
3A CLF
12H 13J 12J 12K
H1 H2 H1 H2
EGCPT1
EGCPT2
CPT
300VA. EA.
480/120VAC
TYPE JVA-0
THRU BUS
METERING POWER
TO VOLTMETER SW.
X1 X2 X1 X2
17H 17J 17K
GND.
[241D15]
REMOTE
FROM "RIG"
24VDC UPS
SUPPLY.
(FIELD CONN)
24VDC
VOLTAGE SENSING RELAY
TERMINAL BOARD.
24VTB
(+)
(-)
[241E16]
[241H26]
[241E18]
[241H24]
13T
15T
CONVENIENCE
OUTLET
CO1
120V,1Ø,60HZ.
120V
RECPT
TB
19N
21N
CO1
REMOTE
TO
CUSTOMERS
120VAC
1Ø 60HZ
POWER
SOURCE
480VOLT, 3 Ø, 3W, 60HZ
EMERGENCY GENERATOR
GROUND DETECTION
DGCPT
EG
Y
PRI
Δ
SEC
FUGDEG
29E 480V
32E 480V
36E 480V
6A
CLF
1CPT
100VA
2CPT
100VA
3CPT
100VA
X2 X1 120V
PHASE 1 GROUND
CIL
30H
PHASE 2 GROUND
CIL
32H
PHASE 3 GROUND
CIL
38H
29J
5
6
GR-EG
(64)
RSTABB
Ø/110Ω
1.97AMPS
425 TO 700 WATTS
5 1/10 TAPS
BASE MTD.
42E
17H
SI
(SEAL IN)
SILE
NORM
RESET
SRTO
NORM
CR104P
ALARM
SILENCE NORMAL RESET
SELECTOR SWITCH.
SI
L
37L
ALARM
BELL
LATCH
RELAY
COIL
RESET
RELAY
COIL
LAMP
TEST
GND BUS
GND BUS
EGCTB
34Q
36Q
RE
TO DAEWOO REMOTE
GROUND FAULT INDICATION
CONTACT OPENS TO
INDICATE GROUND FAULT.
[241M29]
GR-EG SETTINGS
TAP _____ AMP.
TIME DELAY _____ SEC.
TIME DIAL _____
1.6 X TAP = _____
- GR-EG IS GE TYPE
12IAV51A _____
VOLTAGE RELAY
- L & R RELAYS ARE
GE CAT. NO. _____
- LAMPS ARE 120VAC
DAEWOO HULL NO. 3011
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
ENGINEER GRAY, A.W.
FIRST MADE FOR REQ. F47-DC-841991-C01
I.C. NO.
PRINTS TO
REV. VS 06-17-87
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
ELEMENTARY DIAGRAM
EMERG. GEN. GND. DET. /THRU BUS
SH. NO. 241L
241M
237B6558
CONT. ON SH.
SH. NO. 241L
237B6558
CONT. ON SH. 241M

480VAC, 3 Ø, 3W, 60HZ, EMERGENCY GENERATOR CONTROL ELEM. SECTION

EMERGENCY GENERATOR
TO REMOTE CUSTOMER DEVICES
CONTROL INTERFACE TERMINAL BOARD

EGCTB

[241G24] (241G23K) 1 — REMOTE — TO REMOTE GENERATOR DIESEL ENGINE-EMERGENCY START.
(241G25G) 2

3
4

[241H22] (241H2IN) 5 — REMOTE — CUSTOMER Ø-1MA KILOWATT TRANSDUCER SIGNAL FOR USE BY SHIPYARD
(241H23N) 6

[241H37G] (241H37G) 7 — REMOTE — CUSTOMER 100% PREFERENTIAL TRIP / LOAD SHED CIRCUIT.
(241H38G) 8

[241H37N] (241H37N) 9 — REMOTE — CUSTOMER 105% PREFERENTIAL TRIP / LOAD SHED CIRCUIT.
(241H38N) 10

[241H41N] (241H41N) 11 — REMOTE — CUSTOMER 110% PREFERENTIAL TRIP / LOAD SHED CIRCUIT.
(241H42N) 12

EG CB [241D40] (241D37N) 13 — REMOTE — "EMER. GEN." TRIP ON OVERCURRENT (OCR) MOMENTARY.
14

TIE CB [241E40] (241D37P) — REMOTE — "TIE CB" TRIP ON OVERCURRENT (OCR) MOMENTARY.
15
16

REMOTE — CUSTOMER'S SPACE HTR. POWER

EMERG. GEN. SPACE HEATER TB
SH TB
SPACE HTR. CONT. & SWITCH [241D13]
2
17
18 [09] — REMOTE — TO EMERGENCY GENERATOR 800W, 120VAC SPACE HEATER.

[45]

DIESEL ENGINE COOLDOWN [241G33]
32H 19 — REMOTE — CUSTOMER DIESEL ENGINE COOLDOWN CIRCUIT.
34H 20

FROM EMER. GEN. RTD SELECTOR SWITCH [241J25]
15L 21
16L 22
17L 23 — REMOTE — TO EMERGENCY GENERATOR 100Ω PLATINUM RTD'S
30L 24
31L 25
32L 26
GND 27

[241L34] 241L34Q 28 — REMOTE — TO DAEWOO REMOTE EMERGENCY GENERATOR GROUND FAULT INDICATION ALARM CIRCUIT. N.C. CONTACT OPENS TO ALARM
241L36Q 29

[241D41U] 30
31
32
33 — REMOTE — EMERGENCY GENERATOR CIRCUIT BREAKER NORMALLY CLOSED & NORMALLY OPEN INTERLOCKS FOR CUSTOMER USE.

[241E41U] 34
35
36
37 — REMOTE — "TIE" CIRCUIT BREAKER NORMALLY CLOSED & NORMALLY OPEN INTERLOCKS FOR CUSTOMER USE.

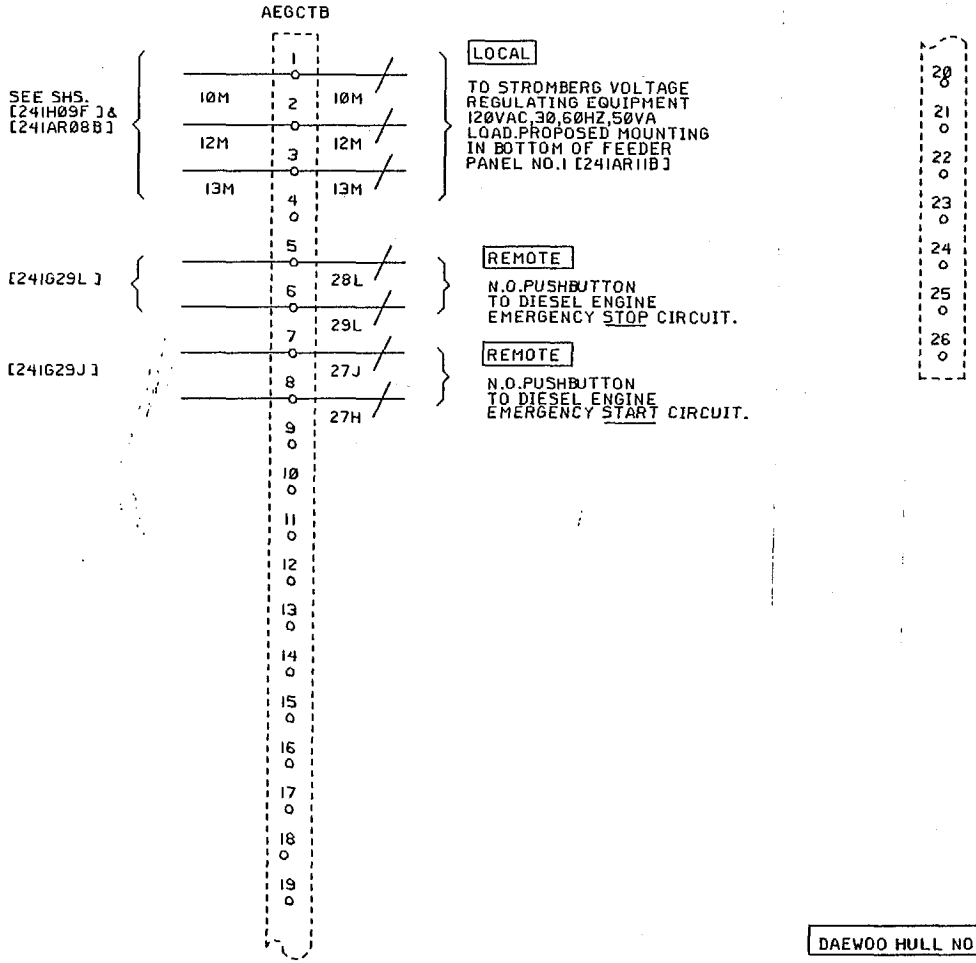
[241F26] 241F25N 38
241F27N 39 — REMOTE — TO EMERGENCY GENERATOR BRUSHLESS GENERATOR FIELD EXCITER.

DAEWOO HULL NO. 3011

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA, U.S.A.
ELEMENTARY DIAGRAM
EMERG. GEN. INTERFACE TB
237B6558
SH. NO. 241M
CONT. ON SH. 241NA

ENGINEER GRAY, A.W.
FIRST MADE FOR REQN. F47-DC-841991-C01
J.C. NO.
PRINTS TO
REV.1 VS 06-17-87
REV.2
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5

EMERGENCY GENERATOR
TO REMOTE CUSTOMER DEVICES
AUXILIARY CONTROL INTERFACE TB.



LOCAL

TO STROMBERG VOLTAGE REGULATING EQUIPMENT 120VAC,30,60HZ,50VA LOAD.PROPOSED MOUNTING IN BOTTOM OF FEEDER PANEL NO.1 [241AR11B]

REMOTE

N.O.PUSHBUTTON TO DIESEL ENGINE EMERGENCY STOP CIRCUIT.

REMOTE

N.O.PUSHBUTTON TO DIESEL ENGINE EMERGENCY START CIRCUIT.

DAEWOO HULL NO. 3011

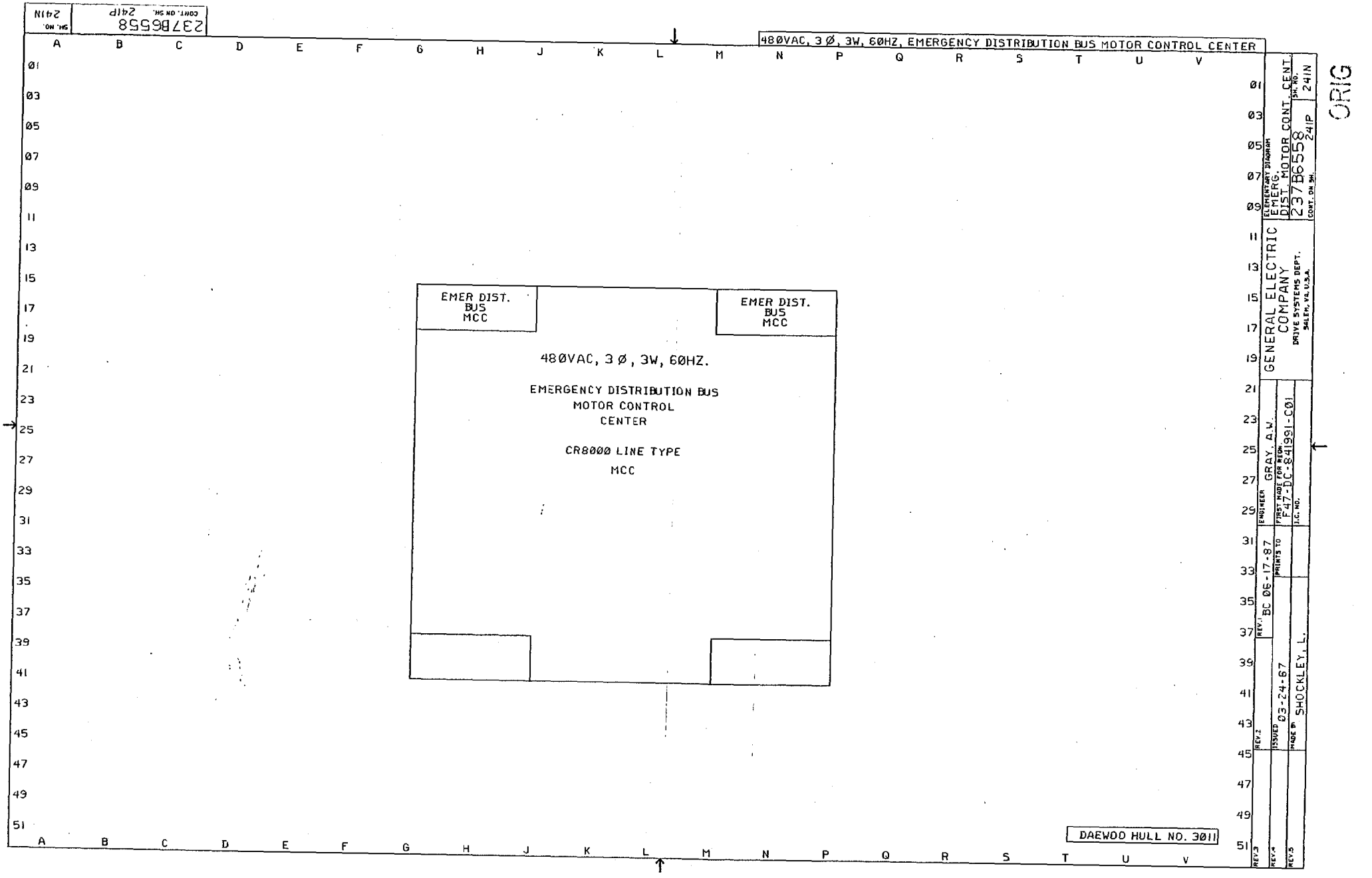
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
AUX. TB
EMER.GEN INTERFACE
237B6558
SH. NO. 241NA
CONT. ON SH. 241N

ENGINEER GRAY, A.W.
PRINTS TO
PRINTS MADE FOR REGN. F47-DC-84199I-C01
I.C. NO. DS4876

REV.1 LP 7-27-87
REV.2
REV.3
REV.4
REV.5

ISSUED 06-17-87
MADE BY SHOCKLEY, L.


480VAC, 3 Ø, 3W, 60HZ, EMERGENCY DISTRIBUTION BUS MOTOR CONTROL CENTER
EMER DIST. BUS MCC
EMER DIST. BUS MCC
480VAC, 3 Ø, 3W, 60HZ.
EMERGENCY DISTRIBUTION BUS MOTOR CONTROL CENTER
CR8000 LINE TYPE
MCC
DAEWOO HULL NO. 3011
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM
EMERG. DIST. MOTOR CONT. CENT.
237B6558 241P
SH. NO. 241N
CONT. ON SH.
ENGINEER GRAY, A.W.
FIRST MADE FOR REV. F47-DC-841991-C01
J.C. NO.
REV.1 BC 06-17-87
PRINTS TO
REV.2
ISSUED 03-24-67
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5
237B6558 241P
SH. NO. 241N

ORIG

480VAC, 3 Ø, 3W, 60HZ, EMERGENCY GENERATOR CONTROL ELEM. SECTION

UL	UL	UL	UL	UL	UL
EIC RADAR X-BAND XFMR.	DIC ELEV. NO.2 PORT AFT.	CIC HYD. PWR PK. 2A STBD	BIC STEAM GENER.	AIM	AØM
EID RADAR ARPA XFMR.	DID HYD. PWR PK. 1A STBD	CID HYD. PWR. PK. 3A PORT	BID ELEV. NO.1 STBD FWD.		
EIE ACOU. POS. IND. RE. PT.	DIM	CIE HYD. PWR PK. 4A PORT	BIE SPACE		
EIF ACOU. POS. IND. RE. ST.	MUD MIX PUMP NO.2 NEMA SIZE FIVE(5) FVNR 126HP	CIN LOW TEMP. FRESH WATER COOLING PUMP NO.1 NEMA SIZE FOUR(4) FVNR	BIG SPARE STARTER NO.1 NEMA 1 FVNR 10HP	"SEA WATER COOLING PUMP NO.3" NEMA SIZE 6 REDUCED VOLTAGE NON-REVERSING (RVNR) AUTO TRANSFORMER CLOSED TRANSITION STARTER	
EIG SPARE FDR NO.3			BIN		
EIH SPARE FDR NO.1			EMERG. GEN. SEA WATER COOLING PUMP NO.1 NEMA SIZE 4 FVNR 56 HP		
EII SPARE FDR NO.2					
EIN EMERG-ENCY GENERATOR ROOM SUPPLY FAN NEMA 2 251WVT 20HP					
20"	20"	20"	20"	48"	
300A VERT. BUS	600A VERT. BUS	600A VERT. BUS.	300A VERT. BUS.	NO. VERT. BUS	

ELVI
EMERGENCY
MCC 1000 AMP
BUS CONNECTS
TO EMERGENCY SWITCHBOARD
ELVI 3000 AMP BUS VIA QTY.2
646 MCM CABLES PER PHASE.

ELVI PANEL SPACE HEATERS.

AØM TB 1,3 — 2,4

E06 SH. 120V,1Ø 250W | E05 SH. 120V,1Ø 250W | E04 SH. 120V,1Ø 250W | E03 SH. 120V,1Ø 250W | E02 SH. 120V,1Ø 250W | E01 SH. 120V,1Ø 250W

ELVI MCC SPACE HEATERS

SH.AØ 62.5W | SH.AI 62.5W | SH.BI 62.5W | SH.CI 62.5W | SH.DI 62.5W | SH.EI 62.5W

SPACE HEATER THERMOSTAT TOP OF SECT. "C"

TT 3 — 1 — 33V

2 — 37V

RE — FROM CUSTOMER REMOTE 120V, 1 Ø, 60HZ ELVI MCC SPACE HEATER SUPPLY

DAEWOO HULL NO. 3011

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
EMERG. GEN. MCC STACKS
237B6558
SH. NO. 241P
241R
CONT. ON SH.

ENGINEER GRAY, A.W.
FIRST MADE FOR HIGH F47-DC-8419911-C01
I.C. NO.
PRINTS TO
REV.1 BC 06-17-87
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.2
REV.3
REV.4
REV.5

ORIG

480VAC, 3 Ø, 3W, 60HZ, EMERGENCY GENERATOR CONTROL ELEM. SECTION

[241L05]

480VAC, 3Ø, 3W, 60HZ 1000 AMP TIN PLATED COPPER HORIZONTAL THRU BUS.

[241505]

MCC UNIT BIC
CB TEDL 36060 LIMITER
"STEAM GENERATOR"
GE TYPE TEDL CB
50 AT/100 AF
TEDL60.
FIXED TRIP
I_{SC} = 200KA

MCC UNIT BID
CB TEDL 36100 LIMITER
"ELEVATOR NO.1 STBD. FORARD"
GE TYPE TEDL CB
100 AT/100 AF
TEDL100.
FIXED TRIP
I_{SC} = 200KA

MCC UNIT DIC
CB TEDL 36100 LIMITER
"ELEVATOR NO.2 PORT AFT."
GE TYPE TEDL CB
100 AT/100 AF
TEDL15.
FIXED TRIP
I_{SC} = 200KA

MCC UNIT DID
CB TEDL 36015 LIMITER
"HYD. POWER PACK VALVE 1A STBD. FWD." (SVCR)
GE TYPE TEDL CB
15 AT/100 AF
TEDL15.
FIXED TRIP
I_{SC} = 200KA

MCC UNIT CIC
CB TEDL 36015 LIMITER
"HYD. POWER PACK VALVE 2A PORT AFT" (SVC2)
GE TYPE TEDL CB
15 AT/100 AF
TEDL15.
FIXED TRIP
I_{SC} = 200KA

MCC UNIT CID
CB TEDL 36015 LIMITER
"HYD. POWER PACK VALVE 3A PORT AFT" (SVC3)
GE TYPE TEDL CB
15 AT/100 AF
TEDL15.
FIXED TRIP
I_{SC} = 200KA

MCC UNIT CIE
CB TEDL 36015 LIMITER
"HYD. POWER PACK VALVE 4A PORT AFT" (SVC4)
GE TYPE TEDL CB
15 AT/100 AF
TEDL15.
FIXED TRIP
I_{SC} = 200KA

MCC 300 AMP, (TP-CU) VERTICAL BUS SECTION "B".

MCC VERTICAL BUS "B" CONTINUATION TO STARTERS AND BLANK SPACE UNIT.
[241X13D]
[241AC13D]

MCC 600 AMP, (TP-CU) VERTICAL BUS SECTION "D".

MCC VERTICAL BUS "D" CONTINUATION TO STARTER.
[241T13D]

MCC 600 AMP, (TP-CU) VERTICAL BUS SECTION "C".

MCC VERTICAL BUS "C" CONTINUATION TO STARTER.
[241]

DAEWOO HULL NO. 3011

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
EMERG. GEN. MCC UNIT FDRS
237B6558 SH. NO. 241R
2415 CONT. ON SH.

ENGINEER GRAY, A.W.
FIRST MADE FOR REF. 7-DC-841991-C01
J.C. NO.

REV.1 BC 06-17-87 PRINTS TO
REV.2
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5

480VAC, 3 Ø, 3W, 60HZ, EMERGENCY GENERATOR CONTROL ELEM. SECTION

480VAC, 3 Ø, 3W, 60HZ 1000 AMP TIN PLATED COPPER HORIZONTAL THRU BUS.

MCC THRU BUS CONTINUATION. [241R05]

MCC THRU BUS CONTINUATION [241T03]

L1 ØA
L2 ØB
L3 ØC

MOTOR CONTROL CENTER
480VAC, 3Ø, 3W, 60HZ, 1000 AMP
MAIN HORIZONTAL THRY BUS.
TIN PLATED COPPER, BRACED
FOR 65KA SYM. SHORT CKT.

MOTOR CONTROL CENTER
480 VAC, 3Ø, 3W, 60HZ
300 AMP VERTICAL BUS
"E". (MCC SECTION "E")
TIN PLATED COPPER.

MCC UNIT EIC
CB TEDL 36015 LIMITER
"RADAR X-BAND TRANSFORMER"
GE TYPE TEDL CB
15 AT/100 AF
TEDL15.
FIXED TRIP
I_{SC} = 200KA

MCC UNIT EID
CB TEDL 36015 LIMITER
"RADAR ARPA TRANSFORMER"
GE TYPE TEDL CB
15 AT/100 AF
TEDL15.

MCC UNIT EIE
CB TEDL 36015 LIMITER
"ACOUSTIC POSITION INDICATION RELAY PORT."
GE TYPE TEDL CB
15 AT/100 AF
TEDL15.
FIXED TRIP
I_{SC} = 200KA

MCC UNIT EIF
CB TEDL 36015 LIMITER
"ACOUSTIC POSITION INDICATION RELAY STBD."
GE TYPE TEDL CB
15 AT/100 AF
TEDL15.
FIXED TRIP
I_{SC} = 200KA

MCC 300 AMP (TP-CU)
VERTICAL BUS "E"
CONTUNUATION TO
BLANK SPACES.

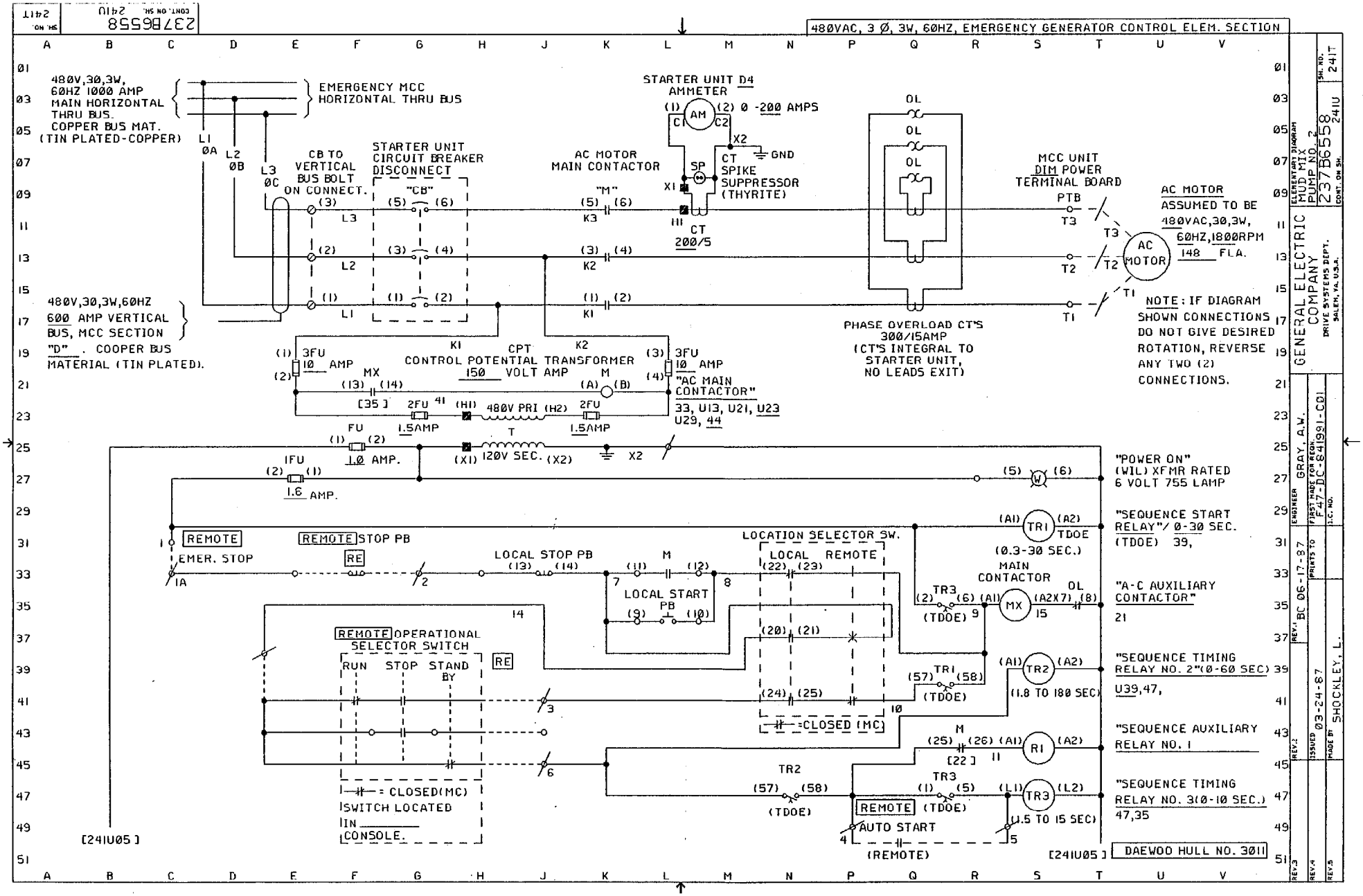
MCC UNIT EIG
CB TEDL 36015 LIMITER
"SPARE FEEDER NO.3"
GE TYPE TEDL CB
15 AT/100 AF
TEDL15.
FIXED TRIP
I_{SC} = 200KA

MCC UNIT EIH
CB TEDL 36060 LIMITER
"SPARE FEEDER NO.1"
GE TYPE TEDL CB
50 AT/100 AF
TEDL50.
FIXED TRIP
I_{SC} = 200KA

MCC UNIT EII
CB TEDL 36060 LIMITER
"SPARE FEEDER NO.2"
GE TYPE TEDL CB
30 AT/100 AF
TEDL30.
FIXED TRIP
I_{SC} = 200KA

DAEWOO HULL NO. 3011

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM
EMERG. GEN. MCC UNIT FDRS
237B6558 241T
SH. NO. 2415
ENGINEER GRAY, A.W.
FIRST MADE FOR REON F47-DC-841991-C01
I.C. NO.
PRINTS TO
REV.1 BC 06-17-87
REV.2
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5


480VAC, 3 Ø, 3W, 60HZ, EMERGENCY GENERATOR CONTROL ELEM. SECTION
480V,30,3W, 60HZ 1000 AMP MAIN HORIZONTAL THRU BUS. COPPER BUS MAT. (TIN PLATED-COPPER)
EMERGENCY MCC HORIZONTAL THRU BUS
L1 ØA
L2 ØB
L3 ØC
CB TO VERTICAL BUS BOLT ON CONNECT.
STARTER UNIT CIRCUIT BREAKER DISCONNECT
"CB"
STARTER UNIT D4
AMMETER
AM
0 -200 AMPS
AC MOTOR MAIN CONTACTOR
"M"
K1
K2
K3
SP
CT SPIKE SUPPRESSOR (THYRITE)
GND
X1
X2
CT 200/5
OL
PHASE OVERLOAD CT'S 300/15AMP (CT'S INTEGRAL TO STARTER UNIT, NO LEADS EXIT)
MCC UNIT DIM POWER TERMINAL BOARD
PTB
T1
T2
T3
AC MOTOR
AC MOTOR ASSUMED TO BE 480VAC,30,3W, 60HZ,1800RPM 148 FLA.
NOTE: IF DIAGRAM SHOWN CONNECTIONS DO NOT GIVE DESIRED ROTATION, REVERSE ANY TWO (2) CONNECTIONS.
480V,30,3W,60HZ 600 AMP VERTICAL BUS, MCC SECTION "D". COOPER BUS MATERIAL (TIN PLATED).
3FU 10 AMP
CPT CONTROL POTENTIAL TRANSFORMER 150 VOLT AMP
MX
M
"AC MAIN CONTACTOR" 33, U13, U21, U23 U29, 44
2FU 1.5AMP
480V PRI (H1) (H2)
T
120V SEC. (X1) (X2)
FU 1.0 AMP.
IFU 1.6 AMP.
"POWER ON" (WIL) XFMR RATED 6 VOLT 755 LAMP
REMOTE
EMER. STOP
REMOTE STOP PB
RE
LOCAL STOP PB
LOCAL START PB
LOCATION SELECTOR SW.
LOCAL REMOTE
TR1 TDOE (0.3-30 SEC.)
"SEQUENCE START RELAY"/ 0-30 SEC. (TDOE) 39,
MAIN CONTACTOR
"A-C AUXILIARY CONTACTOR" 21
REMOTE OPERATIONAL SELECTOR SWITCH
RUN STOP STAND BY
=CLOSED (MC)
TR2 (TDOE) (1.8 TO 180 SEC)
"SEQUENCE TIMING RELAY NO. 2"(0-60 SEC) U39,47,
"SEQUENCE AUXILIARY RELAY NO. 1
TR3 (TDOE) (1.5 TO 15 SEC)
"SEQUENCE TIMING RELAY NO. 3(0-10 SEC.) 47,35
AUTO START (REMOTE)
=CLOSED(MC) SWITCH LOCATED IN CONSOLE.
[241U05]
DAEWOO HULL NO. 3011
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM
MUD MIX PUMP NO. 2
237B6558
241U
241T
ENGINEER GRAY, A.W.
FIRST MADE FOR REQN. P47-DC-841991-C01
REV.1 BC 06-17-87
ISSUED 03-24-87
MADE BY SHOCKLEY, L.

480VAC, 3 Ø, 3W, 60HZ, EMERGENCY GENERATOR CONTROL ELEM. SECTION

[241U49] 120V, 1 Ø, 60HZ FROM SHEET 241T LINE 50 [241U49] X2

M (23) (24) [241T22] 12 13 GIL (1) (2) "RUN" GIL, XFMR. RATED 6V 755 LAMP

[49]

M (15) (16) 21Q 21R [241T22] 50 51 RE REMOTE FOR CUSTOMER USE.

M (13) (14) 25Q 25R [241T22] 60 61 RE REMOTE "RUN" NORMALLY OPEN (NO)

M (21) (22) 29Q 29R [241T22] 52 53 RE REMOTE "STAND BY" NORMALLY OPEN

RI (13) (14) 33R [241T43] 54

TB JUMPER

RI SPARE

TR2 (21) (22) 39Q 39R [241T39] 62 63 RE REMOTE ALARM

TR2 (66) (65) TO SPARE

DIM MCC STARTER UNIT TERMINAL BOARD

5 4 3 2 1A 1 X2 T3 T2 T1 TO AC MOTOR

[19]

STARTER UNIT LOCATION	STARTER NAMEPLATE DESIGNATION	MOTOR HORSEPOWER	STARTER NEMA SIZE	STARTER UNIT CIRCUIT BREAKER	STARTER OVERLOAD RELAY HEATERS
DIM	MUD MIX PUMP NO. 2	126	FIVE(5)	TBC4	C7.78A

DAEWOO HULL NO. 3011

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM MUD MIX PUMP NO. 2

237B6558 241V CONT. ON SH.

SH. NO. 241U

ENGINEER GRAY, A.W.

FIRST MADE FOR REQN. F47-DC-841981-C01

J.O. NO.

PRINTS TO

REV.1 DC 06-17-87

REV.2

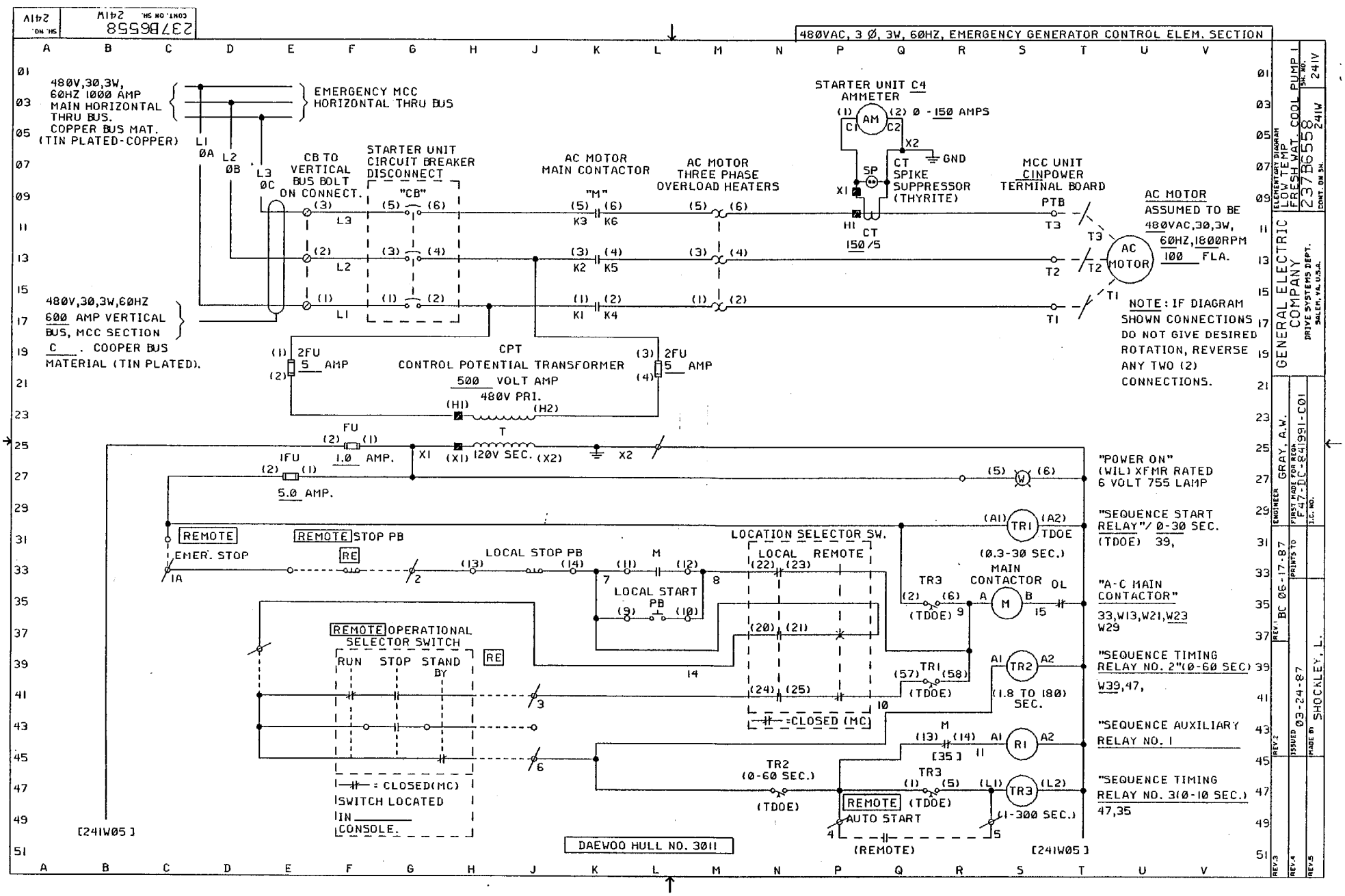
ISSUED 03-24-87

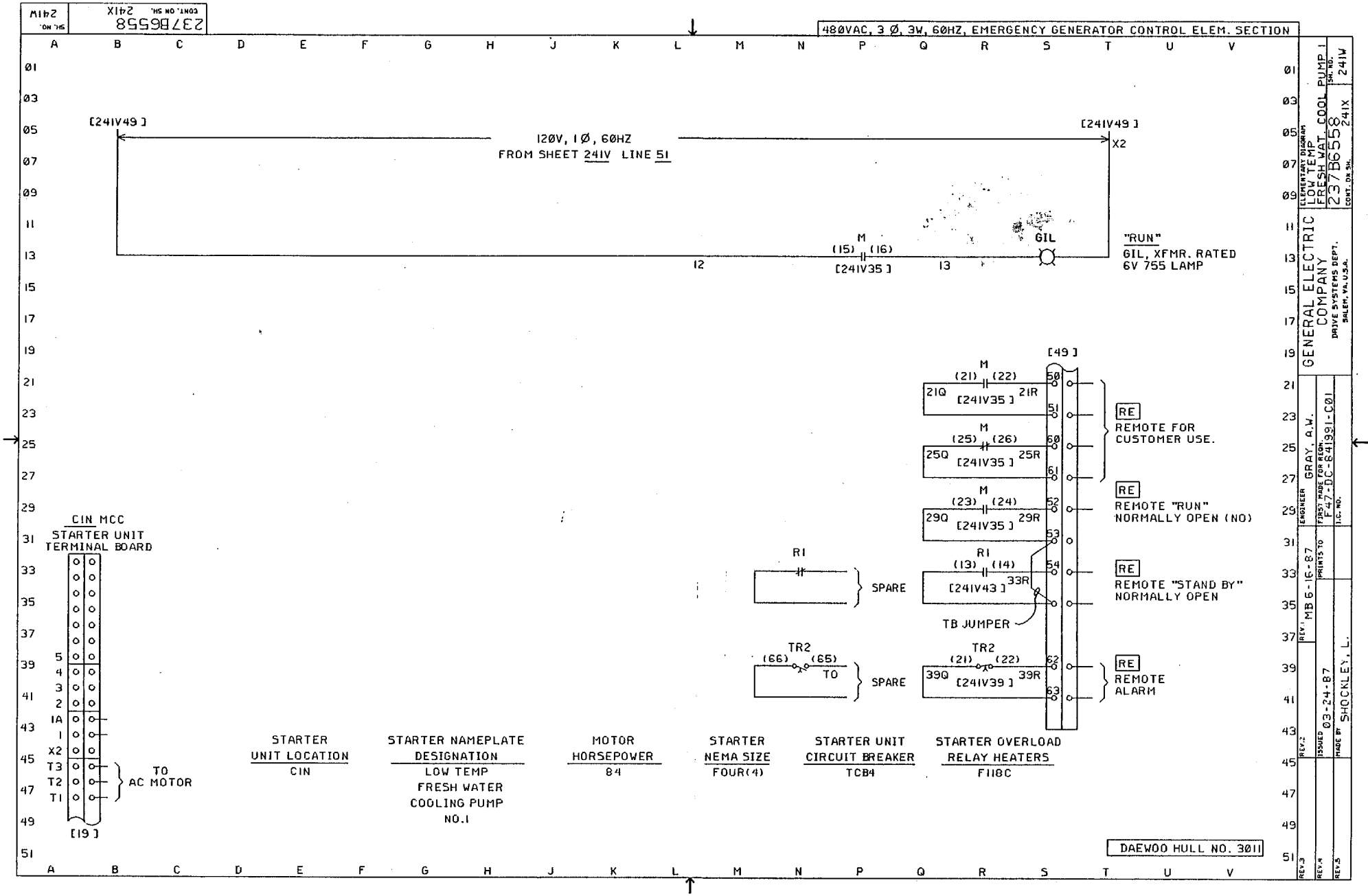
MADE BY SHOCKLEY, L.

REV.3

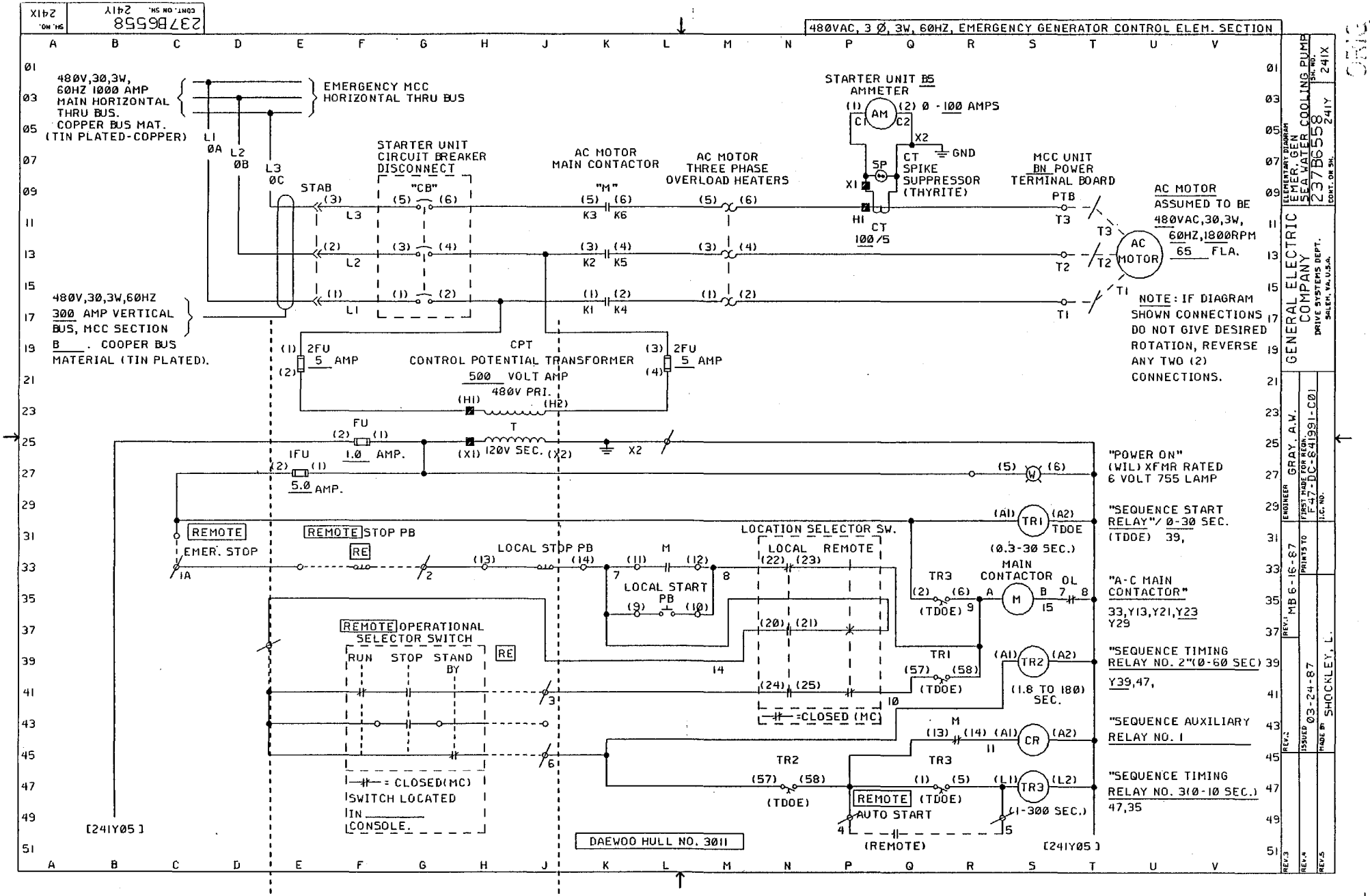
REV.4

REV.5


480VAC, 3 Ø, 3W, 60HZ, EMERGENCY GENERATOR CONTROL ELEM. SECTION
480V,30,3W, 60HZ 1000 AMP MAIN HORIZONTAL THRU BUS. COPPER BUS MAT. (TIN PLATED-COPPER)
EMERGENCY MCC HORIZONTAL THRU BUS
L1 ØA
L2 ØB
L3 ØC
CB TO VERTICAL BUS BOLT ON CONNECT.
STARTER UNIT CIRCUIT BREAKER DISCONNECT
"CB"
AC MOTOR MAIN CONTACTOR
"M"
AC MOTOR THREE PHASE OVERLOAD HEATERS
STARTER UNIT C4
AMMETER
AM
0 - 150 AMPS
CT SPIKE SUPPRESSOR (THYRITE)
GND
SP
CT 150/5
MCC UNIT CINPOWER TERMINAL BOARD
PTB
AC MOTOR
ASSUMED TO BE 480VAC,30,3W, 60HZ,1800RPM 100 FLA.
NOTE: IF DIAGRAM SHOWN CONNECTIONS DO NOT GIVE DESIRED ROTATION, REVERSE ANY TWO (2) CONNECTIONS.
480V,30,3W,60HZ 600 AMP VERTICAL BUS, MCC SECTION C. COOPER BUS MATERIAL (TIN PLATED).
2FU 5 AMP
CPT CONTROL POTENTIAL TRANSFORMER 500 VOLT AMP 480V PRI.
FU 1.0 AMP.
IFU 5.0 AMP.
T 120V SEC.
"POWER ON" (WIL) XFMR RATED 6 VOLT 755 LAMP
REMOTE EMER. STOP
REMOTE STOP PB
RE
LOCAL STOP PB
LOCAL START PB
LOCATION SELECTOR SW. LOCAL REMOTE
TR1 TDOE (0.3-30 SEC.)
"SEQUENCE START RELAY"/ 0-30 SEC. (TDOE) 39,
TR3 MAIN CONTACTOR
"A-C MAIN CONTACTOR" 33,W13,W21,W23 W29
REMOTE OPERATIONAL SELECTOR SWITCH
RUN STOP STAND BY
=CLOSED (MC)
SWITCH LOCATED IN CONSOLE.
TR2 (0-60 SEC.)
"SEQUENCE TIMING RELAY NO. 2"(0-60 SEC) W39,47,
(1.8 TO 180) SEC.
"SEQUENCE AUXILIARY RELAY NO. 1
R1
TR3 (0-10 SEC.)
"SEQUENCE TIMING RELAY NO. 3(0-10 SEC.) 47,35
REMOTE AUTO START
[241W05]
DAEWOO HULL NO. 3011
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM LOW TEMP FRESH WAT COOL PUMP 1
237B6558
241V
241W
ENGINEER GRAY, A.W.
FIRST MADE FOR REQ. F47-DC-841991-C01
D.C. NO.
PRINTS TO
REV.1 BC 06-17-87
REV.2
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5


480VAC, 3 Ø, 3W, 60HZ, EMERGENCY GENERATOR CONTROL ELEM. SECTION
237B6558 241X
CONT. ON SH. 241W
SH. NO.
[241V49]
120V, 1 Ø, 60HZ
FROM SHEET 241V LINE 51
[241V49]
X2
M
(15) (16)
[241V35]
12
13
GIL
"RUN"
GIL, XFMR. RATED
6V 755 LAMP
[49]
M
(21) (22)
21Q [241V35] 21R
RE
REMOTE FOR
CUSTOMER USE.
M
(25) (26)
25Q [241V35] 25R
M
(23) (24)
29Q [241V35] 29R
RE
REMOTE "RUN"
NORMALLY OPEN (NO)
R1
(13) (14)
[241V43] 33R
RE
REMOTE "STAND BY"
NORMALLY OPEN
TB JUMPER
R1
SPARE
TR2
(66) (65)
TO
TR2
(21) (22)
39Q [241V39] 39R
RE
REMOTE
ALARM
SPARE
CIN MCC
STARTER UNIT
TERMINAL BOARD
5
4
3
2
1A
1
X2
T3
T2
T1
[19]
TO
AC MOTOR
STARTER
UNIT LOCATION
CIN
STARTER NAMEPLATE
DESIGNATION
LOW TEMP
FRESH WATER
COOLING PUMP
NO.1
MOTOR
HORSEPOWER
84
STARTER
NEMA SIZE
FOUR(4)
STARTER UNIT
CIRCUIT BREAKER
TCB4
STARTER OVERLOAD
RELAY HEATERS
F118C
DAEWOO HULL NO. 3011
ELEMENTARY DIAGRAM
LOW TEMP
FRESH WAT. COOL PUMP 1
237B6558
241X
241W
GENERAL ELECTRIC
COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA, U.S.A.
ENGINEER GRAY, A.W.
I.C. NO. 47-DC-841991-C01
REV. MB 6-16-87
PRINTS TO
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.2
REV.3
REV.4
REV.5

ORIG


480VAC, 3 Ø, 3W, 60HZ, EMERGENCY GENERATOR CONTROL ELEM. SECTION
480V,30,3W, 60HZ 1000 AMP MAIN HORIZONTAL THRU BUS. COPPER BUS MAT. (TIN PLATED-COPPER)
EMERGENCY MCC HORIZONTAL THRU BUS
L1 ØA
L2 ØB
L3 ØC
480V,30,3W,60HZ 300 AMP VERTICAL BUS, MCC SECTION B . COOPER BUS MATERIAL (TIN PLATED).
STARTER UNIT CIRCUIT BREAKER DISCONNECT
"CB"
STAB
AC MOTOR MAIN CONTACTOR
"M"
AC MOTOR THREE PHASE OVERLOAD HEATERS
STARTER UNIT B5 AMMETER
AM
0 - 100 AMPS
CT 100/5
SP
CT SPIKE SUPPRESSOR (THYRITE)
GND
MCC UNIT BN POWER TERMINAL BOARD
PTB
AC MOTOR ASSUMED TO BE 480VAC,30,3W, 60HZ,1800RPM 65 FLA.
AC MOTOR
NOTE: IF DIAGRAM SHOWN CONNECTIONS DO NOT GIVE DESIRED ROTATION, REVERSE ANY TWO (2) CONNECTIONS.
2FU 5 AMP
CPT CONTROL POTENTIAL TRANSFORMER 500 VOLT AMP 480V PRI.
FU 1.0 AMP.
IFU 5.0 AMP.
T 120V SEC.
"POWER ON" (WIL) XFMR RATED 6 VOLT 755 LAMP
REMOTE EMER. STOP
REMOTE STOP PB
RE
LOCAL STOP PB
LOCAL START PB
LOCATION SELECTOR SW.
LOCAL REMOTE
TR1 TDOE (0.3-30 SEC.)
"SEQUENCE START RELAY"/ 0-30 SEC. (TDOE) 39,
MAIN CONTACTOR
"A-C MAIN CONTACTOR" 33,Y13,Y21,Y23 Y29
REMOTE OPERATIONAL SELECTOR SWITCH
RUN STOP STAND BY
= CLOSED(MC) SWITCH LOCATED IN CONSOLE.
= CLOSED (MC)
TR2 (1.8 TO 180) SEC.
"SEQUENCE TIMING RELAY NO. 2"(0-60 SEC) Y39,47,
"SEQUENCE AUXILIARY RELAY NO. 1
CR
TR3 (1-300 SEC.)
"SEQUENCE TIMING RELAY NO. 3(0-10 SEC.) 47,35
REMOTE AUTO START (REMOTE)
[241Y05]
DAEWOO HULL NO. 3011
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA, U.S.A.
ELEMENTARY DIAGRAM EMER. GEN SEA WATER COOLING PUMP
237B6558
ENGINEER GRAY, A.W.
FIRST MADE FOR RGN. F47-DC-841991-C01
I.C. NO.
REV.1 MB 6-16-87
PRINTS TO
REV.2 ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5

480VAC, 3 Ø, 3W, 60HZ, EMERGENCY GENERATOR CONTROL ELEM. SECTION

[241X49] 120V, 1Ø, 60HZ FROM SHEET 241X LINE 50 [241X49] X2

12 M (15) (16) [241X35] 13 GIL (1) (2)

"RUN"
GIL, XFMR. RATED
6V 755 LAMP

B5 MCC
STARTER UNIT
TERMINAL BOARD

5
4
3
2
1A
1
X2
T3
T2
T1

TO AC MOTOR

[19]

CR (13) (14) SPARE

TR2 (66) (65) (TDOE) SPARE

[49]

M (21) (22) 21Q [241X35] 21R 50
51
RE REMOTE FOR CUSTOMER USE.

M (25) (26) 25Q [241X35] 25R 60
61
RE REMOTE FOR CUSTOMER USE.

M (23) (24) 29Q [241X35] 29R 52
53
RE REMOTE "RUN" NORMALLY OPEN (NO)

CR (13) (14) [241X43] 33R 54
RE REMOTE "STAND BY" NORMALLY OPEN

TB JUMPER

TR2 (21) (22) 39Q [241X39] 39R 62
63
RE REMOTE ALARM

STARTER UNIT LOCATION	STARTER NAMEPLATE DESIGNATION	MOTOR HORSEPOWER	STARTER NEMA SIZE	STARTER UNIT CIRCUIT BREAKER	STARTER OVERLOAD RELAY HEATERS
BIN	EMERGENCY GENERATOR SEA WATER COOLING PUMP.	56	FOUR(4)	TECL 109	C7.72B

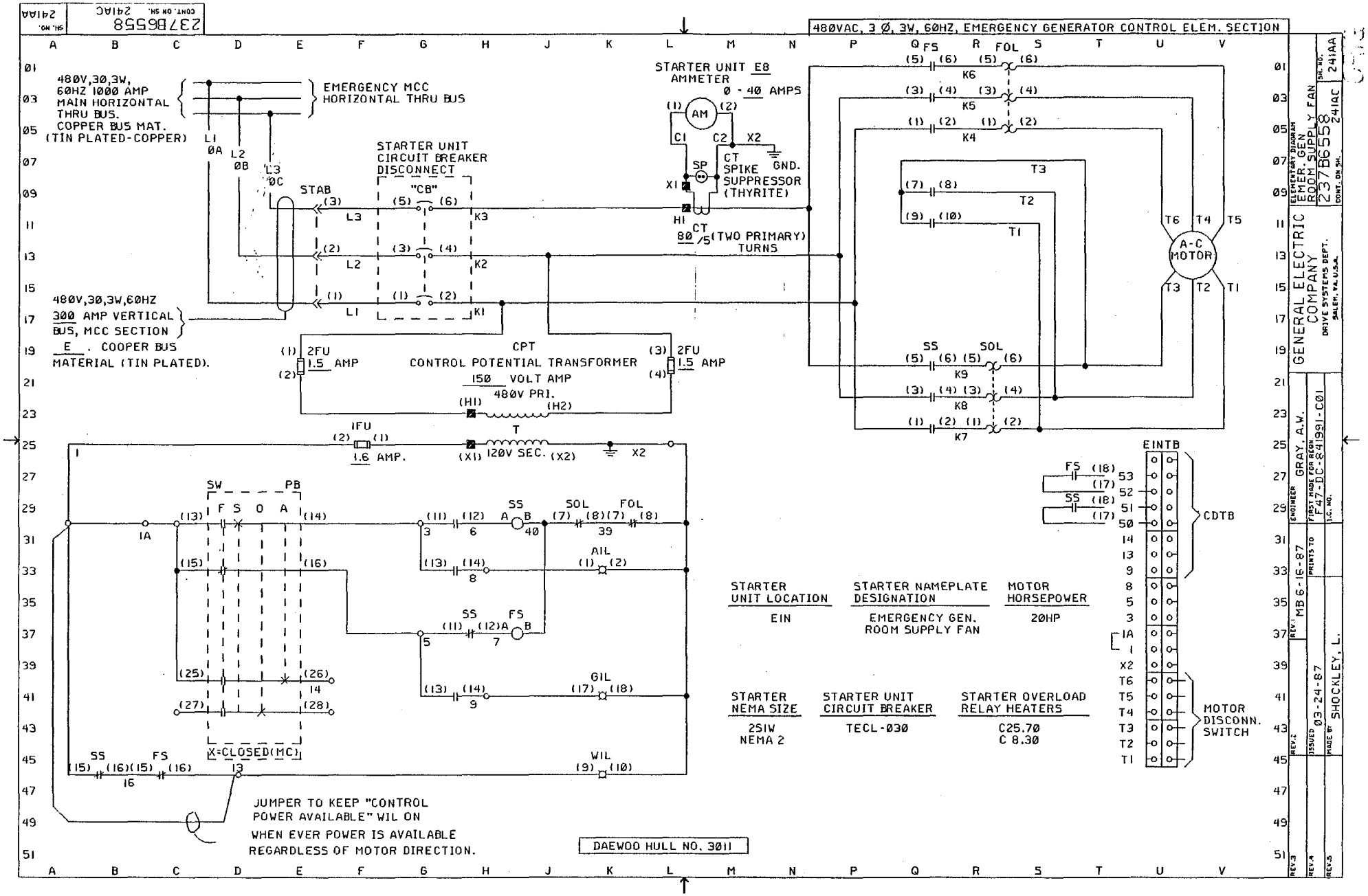
DAEWOO HULL NO. 3010

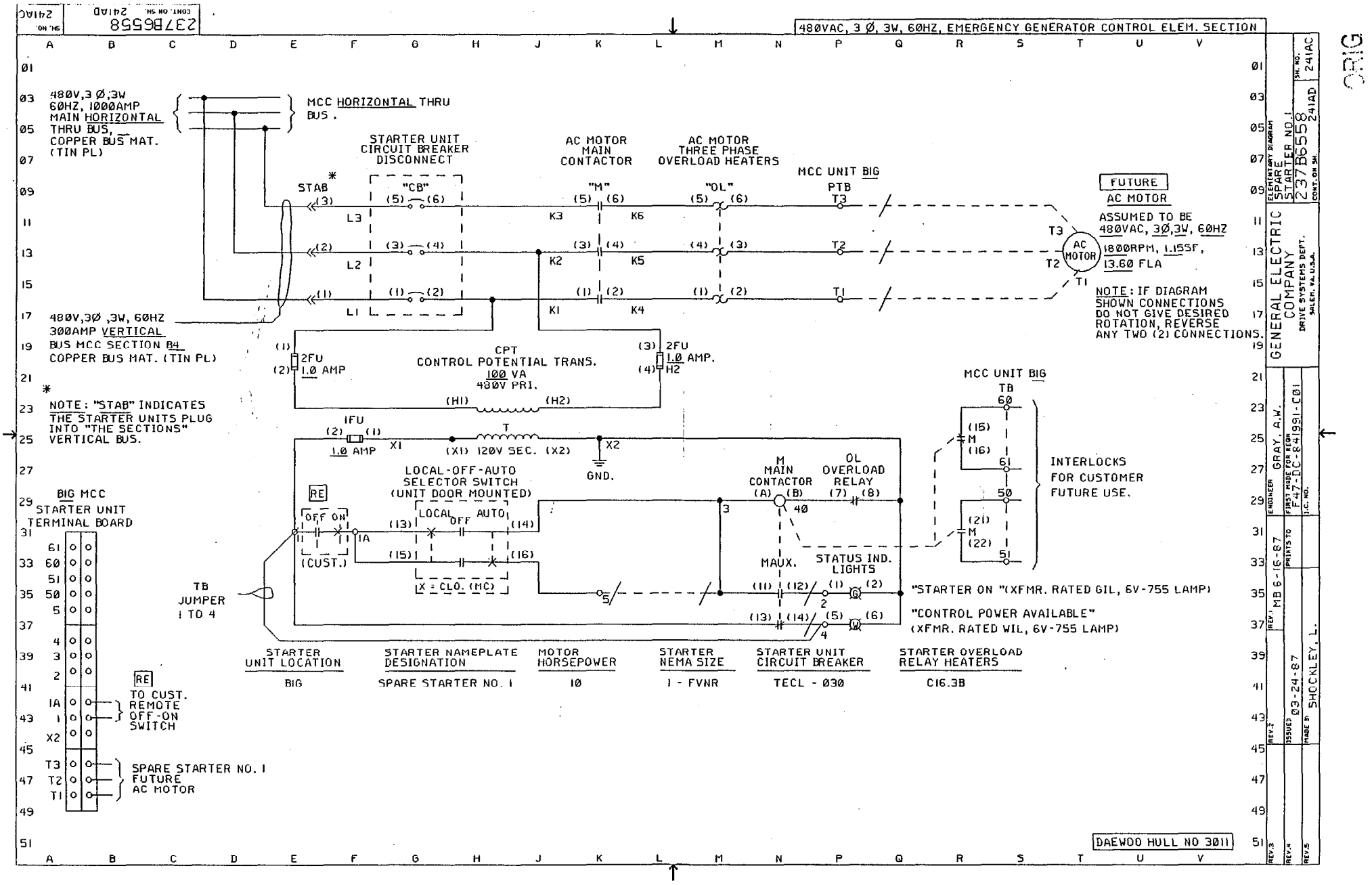
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.

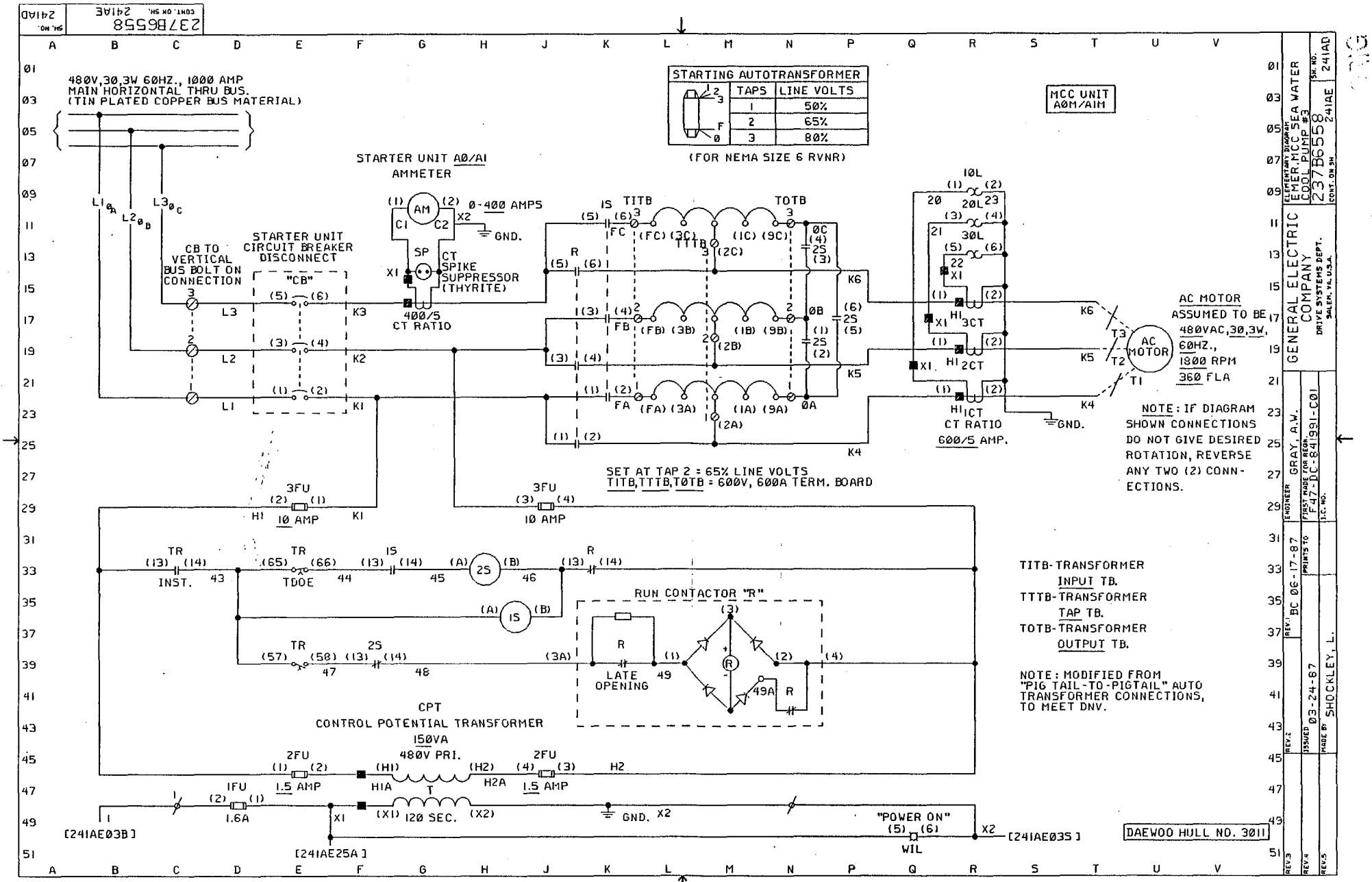
ELEMENTARY DIAGRAM
EMER. GEN. SEA WATER COOLING PUMP
237B6558
SH. NO. 241Y
CONT. ON SH. 241AA

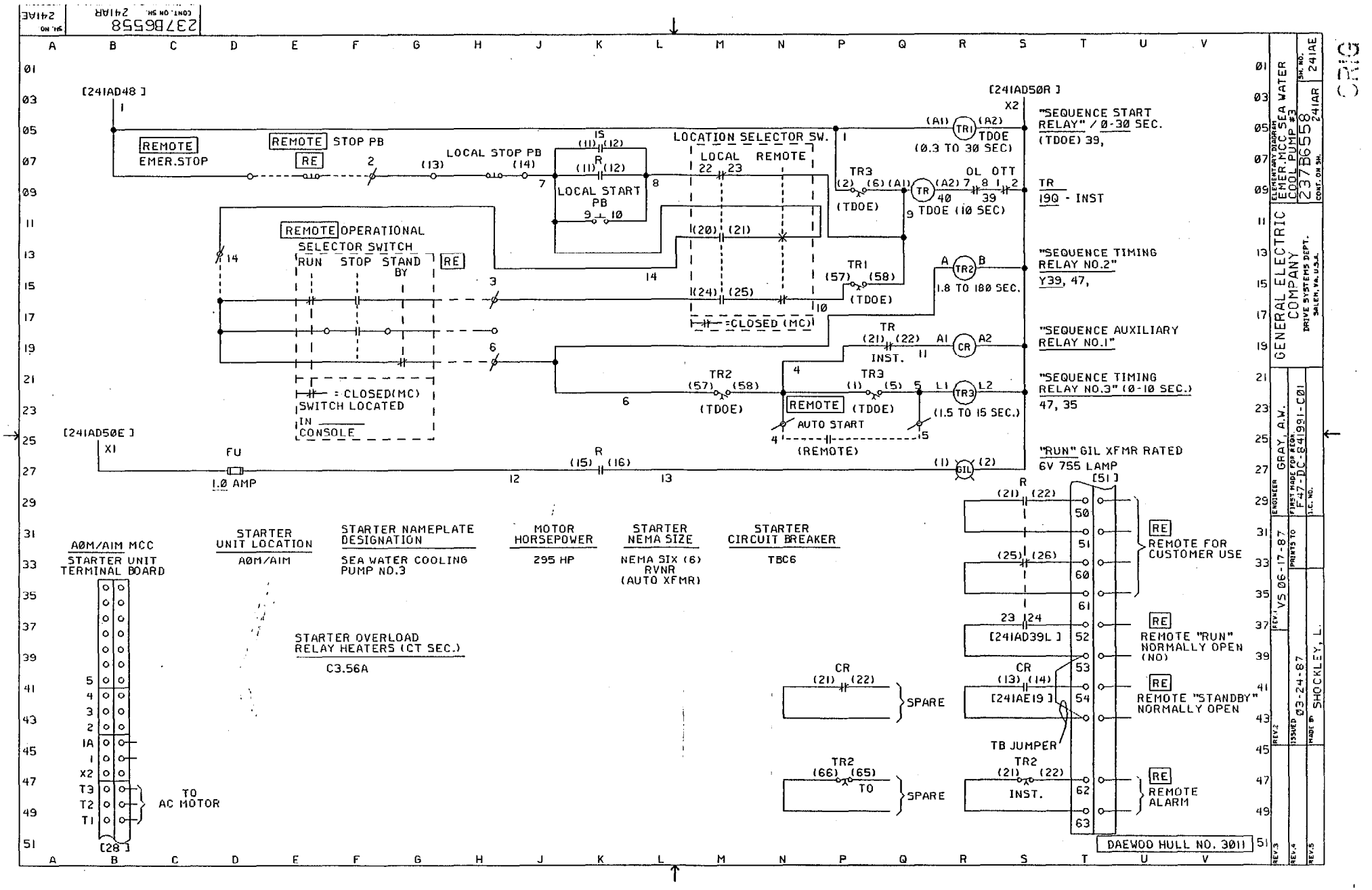
ENGINEER GRAY, A.W.
PRINTS TO
I.C.C. NO.
FIRST MADE FOR REQN. F47-DC-841991-C01

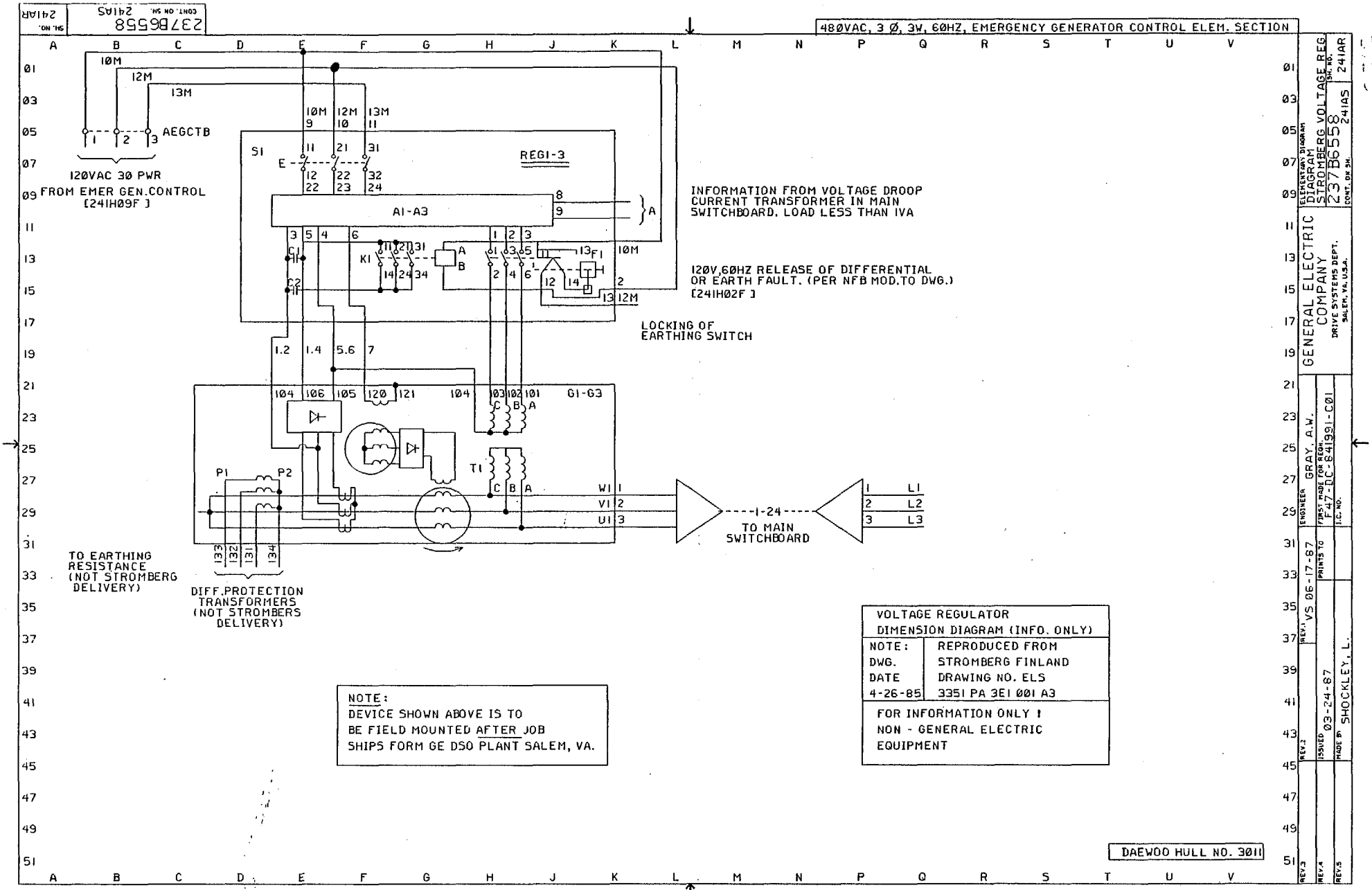
REV.1 MB 6-16-87
REV.2
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5


480VAC, 3 Ø, 3W, 60HZ, EMERGENCY GENERATOR CONTROL ELEM. SECTION
480V,30,3W, 60HZ 1000 AMP MAIN HORIZONTAL THRU BUS. COPPER BUS MAT. (TIN PLATED-COPPER)
EMERGENCY MCC HORIZONTAL THRU BUS
L1 ØA
L2 ØB
L3 ØC
STARTER UNIT CIRCUIT BREAKER DISCONNECT
"CB"
STAB
STARTER UNIT E8
AMMETER
Ø - 40 AMPS
AM
C1
C2
X2
CT SPIKE SUPPRESSOR (THYRITE)
GND.
SP
X1
HI
CT 80/5 (TWO PRIMARY) TURNS
480V,30,3W,60HZ 300 AMP VERTICAL BUS, MCC SECTION E. COOPER BUS MATERIAL (TIN PLATED).
2FU 1.5 AMP
CPT
CONTROL POTENTIAL TRANSFORMER
150 VOLT AMP
480V PRI.
(H1)
(H2)
1FU
1.6 AMP.
T
(X1) 120V SEC. (X2)
X2
SW
PB
F S O A
IA
AIL
SS
FS
SOL
FOL
GIL
WIL
X=CLOSED(MC)
JUMPER TO KEEP "CONTROL POWER AVAILABLE" WIL ON WHEN EVER POWER IS AVAILABLE REGARDLESS OF MOTOR DIRECTION.
DAEWOO HULL NO. 3011
K1
K2
K3
K4
K5
K6
K7
K8
K9
T1
T2
T3
A-C MOTOR
T4
T5
T6
EINTB
CDTB
MOTOR DISCONN. SWITCH
STARTER UNIT LOCATION
EIN
STARTER NAMEPLATE DESIGNATION
EMERGENCY GEN. ROOM SUPPLY FAN
MOTOR HORSEPOWER
20HP
STARTER NEMA SIZE
2SIW NEMA 2
STARTER UNIT CIRCUIT BREAKER
TECL-030
STARTER OVERLOAD RELAY HEATERS
C25.70 C 8.30
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM EMER. GEN ROOM SUPPLY FAN
237B6558 241AC
CONT. ON SH.
SH. NO. 241AA
ENGINEER GRAY, A.W.
FIRST MADE FOR RGDN F47-DC-8419911-C01
I.C. NO.
PRINTS TO
REV.1 MB 6-16-87
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.2
REV.3
REV.4
REV.5
237B6558 241AC
CONT. ON SH. 241AA
SH. NO.

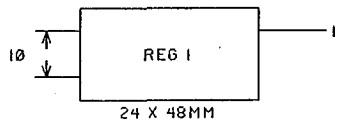

480VAC, 3 Ø, 3W, 60HZ, EMERGENCY GENERATOR CONTROL ELEM. SECTION
480V,3 Ø,3W
60HZ, 1000AMP
MAIN HORIZONTAL
THRU BUS,
COPPER BUS MAT.
(TIN PL)
MCC HORIZONTAL THRU BUS .
480V,3Ø ,3W, 60HZ
300AMP VERTICAL
BUS MCC SECTION B4
COPPER BUS MAT. (TIN PL)
NOTE: "STAB" INDICATES THE STARTER UNITS PLUG INTO "THE SECTIONS" VERTICAL BUS.
STARTER UNIT CIRCUIT BREAKER DISCONNECT
AC MOTOR MAIN CONTACTOR
AC MOTOR THREE PHASE OVERLOAD HEATERS
MCC UNIT BIG PTB
STAB
"CB"
"M"
"OL"
L3
L2
L1
K3
K6
K2
K5
K1
K4
FUTURE AC MOTOR
ASSUMED TO BE 480VAC, 3Ø,3W, 60HZ
1800RPM, 1.15SF,
13.60 FLA
AC MOTOR
NOTE: IF DIAGRAM SHOWN CONNECTIONS DO NOT GIVE DESIRED ROTATION, REVERSE ANY TWO (2) CONNECTIONS.
2FU
1.0 AMP
CPT
CONTROL POTENTIAL TRANS.
100 VA
480V PRI.
(H1)
(H2)
2FU
1.0 AMP.
H2
IFU
1.0 AMP
X1
T
(X1) 120V SEC. (X2)
X2
GND.
MCC UNIT BIG TB
INTERLOCKS FOR CUSTOMER FUTURE USE.
LOCAL-OFF-AUTO SELECTOR SWITCH (UNIT DOOR MOUNTED)
RE
OFF ON
LOCAL OFF AUTO
(CUST.)
X = CLO. (HC)
M MAIN CONTACTOR
OL OVERLOAD RELAY
MAUX.
STATUS IND. LIGHTS
"STARTER ON "(XFMR. RATED GIL, 6V-755 LAMP)
"CONTROL POWER AVAILABLE" (XFMR. RATED WIL, 6V-755 LAMP)
TB JUMPER 1 TO 4
BIG MCC STARTER UNIT TERMINAL BOARD
TO CUST. REMOTE OFF-ON SWITCH
SPARE STARTER NO. 1 FUTURE AC MOTOR
STARTER UNIT LOCATION
STARTER NAMEPLATE DESIGNATION
MOTOR HORSEPOWER
STARTER NEMA SIZE
STARTER UNIT CIRCUIT BREAKER
STARTER OVERLOAD RELAY HEATERS
BIG
SPARE STARTER NO. 1
10
1 - FVNR
TECL - 030
C16.3B
DAEWOO HULL NO 3011
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
ENGINEER GRAY, A.W.
DRAWING NO. 237B6558
SH. NO. 241AC
241AD
CONT. ON SH.
REV.1 MB 6-16-87
REV.2
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5

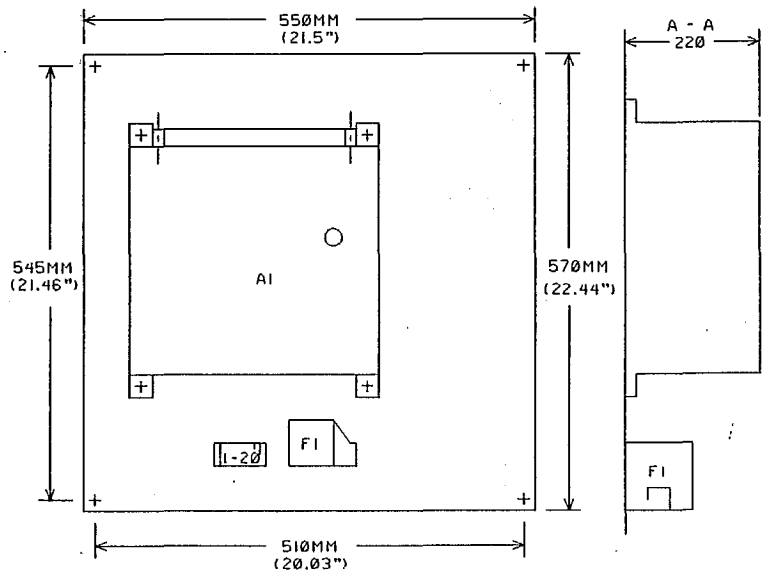

480V,30,3W 60HZ., 1000 AMP
MAIN HORIZONTAL THRU BUS.
(TIN PLATED COPPER BUS MATERIAL)
STARTING AUTOTRANSFORMER
TAPS | LINE VOLTS
1 | 50%
2 | 65%
3 | 80%
(FOR NEMA SIZE 6 RVNR)
MCC UNIT
A0M/A1M
STARTER UNIT A0/A1
AMMETER
0-400 AMPS
GND.
SP
CT
SPIKE
SUPPRESSOR
(THYRITE)
400/5
CT RATIO
CB TO
VERTICAL
BUS BOLT ON
CONNECTION
STARTER UNIT
CIRCUIT BREAKER
DISCONNECT
"CB"
L1
L2
L3
K1
K2
K3
K4
K5
K6
3FU
10 AMP
SET AT TAP 2 = 65% LINE VOLTS
TITB,TTTB,T0TB = 600V, 600A TERM. BOARD
AC MOTOR
ASSUMED TO BE
480VAC,30,3W,
60HZ.,
1800 RPM
360 FLA
CT RATIO
600/5 AMP.
NOTE: IF DIAGRAM
SHOWN CONNECTIONS
DO NOT GIVE DESIRED
ROTATION, REVERSE
ANY TWO (2) CONN-
ECTIONS.
TITB-TRANSFORMER
INPUT TB.
TTTB-TRANSFORMER
TAP TB.
TOTB-TRANSFORMER
OUTPUT TB.
NOTE: MODIFIED FROM
"PIG TAIL-TO-PIGTAIL" AUTO
TRANSFORMER CONNECTIONS,
TO MEET DNV.
RUN CONTACTOR "R"
LATE
OPENING
CPT
CONTROL POTENTIAL TRANSFORMER
150VA
480V PRI.
1.5 AMP
1.6A
120 SEC.
"POWER ON"
WIL
[241AE03B]
[241AE25A]
[241AE03S]
DAEWOO HULL NO. 3011
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM
EMER.MCC SEA WATER
COOL PUMP #3
237B6558
241AD
241AE
ENGINEER GRAY, A.W.
FIRST MADE FOR REGN. F47-DC-841991-C01
PRINTS TO
I.C. NO.
REV.1 BC 06-17-87
REV.2
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5

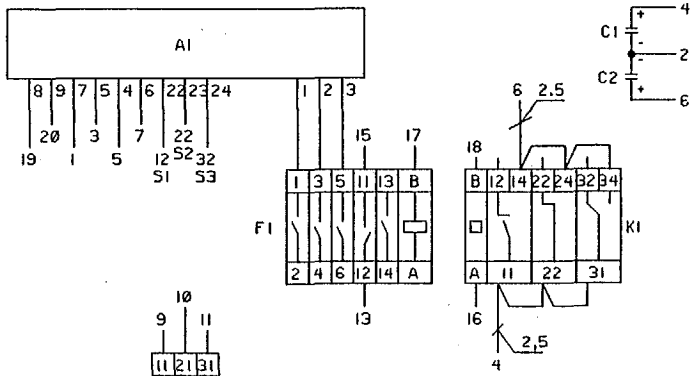

237B6558
[241AD48]
[241AD50R]
[241AD50E]
REMOTE
EMER.STOP
REMOTE
RE
REMOTE STOP PB
LOCAL STOP PB
LOCAL START PB
LOCATION SELECTOR SW.
LOCAL
REMOTE
REMOTE OPERATIONAL SELECTOR SWITCH
RUN
STOP
STAND BY
RE
= CLOSED(MC) SWITCH LOCATED IN CONSOLE
= CLOSED (MC)
TR3
TDOE
TDOE (10 SEC)
OL OTT
TR1
TDOE
(0.3 TO 30 SEC)
"SEQUENCE START RELAY" / 0-30 SEC.
(TDOE) 39,
TR
190 - INST
"SEQUENCE TIMING RELAY NO.2"
Y39, 47,
1.8 TO 180 SEC.
TR2
TDOE
"SEQUENCE AUXILIARY RELAY NO.1"
INST.
"SEQUENCE TIMING RELAY NO.3" (0-10 SEC.)
47, 35
(1.5 TO 15 SEC.)
REMOTE
AUTO START
(REMOTE)
FU
1.0 AMP
"RUN" GIL XFMR RATED
6V 755 LAMP
GIL
[51]
RE
REMOTE FOR CUSTOMER USE
REMOTE "RUN" NORMALLY OPEN (NO)
REMOTE "STANDBY" NORMALLY OPEN
TB JUMPER
REMOTE ALARM
[241AD39L]
[241AE19]
SPARE
SPARE
TO AC MOTOR
[28]
A0M/AIM MCC
STARTER UNIT TERMINAL BOARD
STARTER UNIT LOCATION
A0M/AIM
STARTER NAMEPLATE DESIGNATION
SEA WATER COOLING PUMP NO.3
MOTOR HORSEPOWER
295 HP
STARTER NEMA SIZE
NEMA SIX (6) RVNR (AUTO XFMR)
STARTER CIRCUIT BREAKER
TBC6
STARTER OVERLOAD RELAY HEATERS (CT SEC.)
C3.56A
DAEWOO HULL NO. 3011
ELEMENTARY DIAGRAM
EMER MCC SEA WATER COOL PUMP #3
237B6558
241AE
241AR
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
ENGINEER GRAY, A.W.
J.C. NO. F47-DC-841991-C01
REV.1 VS 06-17-87
REV.2
REV.3
REV.4
REV.5
ISSUED 03-24-87
MADE BY SHOCKLEY, L.

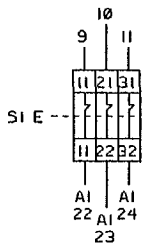

480VAC, 3 Ø, 3W, 60HZ, EMERGENCY GENERATOR CONTROL ELEM. SECTION
10M
12M
13M
AEGCTB
120VAC 30 PWR
FROM EMER GEN.CONTROL
[241H09F]
S1
REG1-3
A1-A3
INFORMATION FROM VOLTAGE DROOP
CURRENT TRANSFORMER IN MAIN
SWITCHBOARD. LOAD LESS THAN 1VA
120V,60HZ RELEASE OF DIFFERENTIAL
OR EARTH FAULT. (PER NFB MOD.TO DWG.)
[241H02F]
LOCKING OF
EARTHING SWITCH
K1
F1
G1-G3
T1
P1
P2
TO EARTHING
RESISTANCE
(NOT STROMBERG
DELIVERY)
DIFF.PROTECTION
TRANSFORMERS
(NOT STROMBERS
DELIVERY)
W1
V1
U1
1-24
TO MAIN
SWITCHBOARD
L1
L2
L3
NOTE:
DEVICE SHOWN ABOVE IS TO
BE FIELD MOUNTED AFTER JOB
SHIPS FORM GE DSO PLANT SALEM, VA.
VOLTAGE REGULATOR
DIMENSION DIAGRAM (INFO. ONLY)
NOTE: REPRODUCED FROM
DWG. STROMBERG FINLAND
DATE DRAWING NO. ELS
4-26-85 3351 PA 3E1 001 A3
FOR INFORMATION ONLY !
NON - GENERAL ELECTRIC
EQUIPMENT
DAEWOO HULL NO. 3011
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM
DIAGRAM
STROMBERG VOLTAGE REG
SH. NO. 241AR
237B6558
241AS
CONT. ON SH.
ENGINEER GRAY, A.W.
FIRST MADE FOR REGN. F47-DC-841991-C01
I.C. NO.
REV.1 VS 06-17-87
PRINTS TO
REV.2
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5
237B6558 241AS
SH. NO. 241AR
CONT. ON SH. 241AS

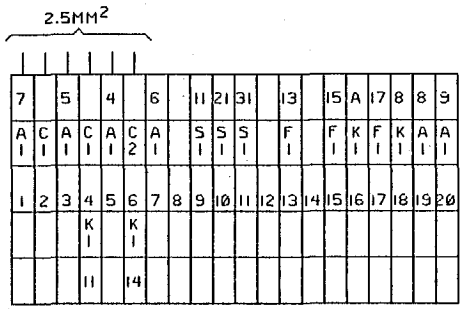
480VAC, 3 Ø, 3W, 60HZ, EMERGENCY GENERATOR CONTROL ELEM. SECTION











"PROPOSED" MOUNTING IN BOTTOM PORTION OF FEEDER PANEL NO.1 (E01).

VOLTAGE REGULATOR DIMENSION DIAGRAM (INFO. ONLY)	
NOTE:	REPRODUCED FROM
DWG.	STROMBERG FINLAND
DATE	DRAWING NO. ELS
4-26-85	3351 PA 3EL 004-B

FOR INFORMATION ONLY !
NON - GENERAL ELECTRIC EQUIPMENT

DAEWOO HULL NO. 3011

ELEMENTARY DIAGRAM
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
STROMBERG VOLTAGE REG.
237B6558
SH. NO. 241BA
CONT. ON SH.
ENGINEER GRAY, A.W.
FIRST MADE FOR REQN. F47-DC-841991-C01
J.C. NO.
REV.1 VS 06-17-87
PRINTS TO
REV.2
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5

CONT. ON SH. 241BA
237B6558
SH. NO. 241AS

ORIG

SH. NO. 241BA | CONT. ON SH. 241BB | 237B6558

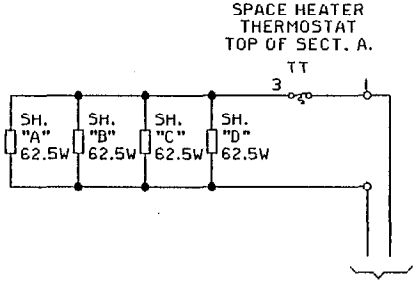
ELV2 MCC | DAEWOO FOR SONAT DRILLING INC. GVA 4500 CLASS SEMI. | ELV2 MCC

120 VOLT EMERGENCY
POWER DISTRIBUTION (ELV2)
MOTOR CONTROL CENTER.
SPECFICATION ITEM 3C
ELEMENTARY SECTION
CR8000 TYPE MCC.

CLASS: NEMA II; WIRING: TYPE "B"
ENCLOSURE: NEMA I (GASKETED/DRIPSHIELDS/HANDRAILS)
RATING: 120 VAC, 3 PHASE, 3 WIRE 60HZ.
GPC MEBANE PO# : A60IC134Y28.
BUILT PER: UL 845 & 508; IP23 ENCLOSURE.
INST PUBLICATIONS: GET-6728A; GEH-4961

- 120 VOLT EMERGENCY POWER DISTRIBUTION ELV2 ONE LINE DIAGRAM
- MCC FRONT (ELEVATION) VIEW
- KEY INTERLOCKED INCOMING CIRCUIT BREAKERS.
- 120 VOLT, 3Ø , 3W, 60HZ., GROUND DETECTION WITH REMOTE ALARM. (UNIT BIN)
- 120 VOLT, 3Ø , 3W, 60HZ., ELV2 FEEDER CIRCUIT BREAKERS.

ELV2 MCC
SPACE HEATERS



DAEWOO HULL NO.3011

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM 3Ø, 3W, 60HZ ELV2 120VAC.
237B6558 SH. NO. 241BA CONT. ON SH. 241BB

ENGINEER GRAY, A.W.
FIRST MADE FOR REQN. F47-EC-84119911-C01
I.C. NO.

REV.1 VS 06-17-87 PRINTS TO
REV.2 ISSUED 03-24-87 MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5

120VAC, 3Ø, 3W, 60HZ EMERGENCY POWER DISTRIBUTION ELV2 ONE LINE DIAGRAM

FROM 480V, 3Ø, 3W, 60HZ. EMERGENCY GENERATOR ELV1, FEEDER PANEL NO.2 "EL RM/ EM.GEN. RM." (ET1) [241K35]

FROM 480V, 3Ø, 3W, 60HZ. EMERGENCY GENERATOR ELV1, FEEDER PANEL NO.2 "EL RM/ EM.GEN. RM." (ET2) [241K43]

EMERGENCY LIGHTING TRANSFORMER ET1 — Δ 480V — ET1 225KVA — Δ 120V — REMOTE BY OTHERS.

EMERGENCY LIGHTING TRANSFORMER ET2 — Δ 480V — ET2 225KVA — Δ 120V — REMOTE BY OTHERS.

1082 AMP CAPABILITY AT 225KVA AND 120VAC.

QTY.5-4/0 CABLES PER PHASE.

1082 AMP CAPABILITY AT 225KVA AND 120VAC.

QTY.5-4/0 CABLES PER PHASE.

EMERGENCY TRANSFORMER NO.1 INCOMING CIRCUIT BREAKER. MCC UNIT - "B1" 1200AT/1200AF TYPE TKM-1200CB.

KEY INTERLOCKED TO PERMIT ONLY ONE CB TO CLOSE AT A TIME.

EMERGENCY TRANSFORMER NO.2 INCOMING CIRCUIT BREAKER. MCC UNIT - "C1" 1200AT/1200AF TYPE TKM-1200CB.

1000 AMP UL/1200 AMP NEMA TIN PLATED COPPER MAIN HORIZONTAL THRU BUS. 42KA-Isc.

1FU 3A

120V, 3Ø, 3W 60HZ GROUND DET. & REMOTE ALARM.

MCC UNIT - "B4" ELEM. SH. 241BG.

600 AMP TIN PLATED COPPER VERTICAL BUS "A".

600 AMP TIN PLATED COPPER VERTICAL BUS "B".

600 AMP TIN PLATED COPPER VERTICAL BUS "C".

600 AMP TIN PLATED COPPER VERTICAL BUS "D".

VERTICAL BUS "A" CONTINUATION [241BC03B]

VERTICAL BUS "B" CONTINUATION [241BC03G]

VERTICAL BUS "C" CONTINUATION [241BC03M]

VERTICAL BUS "D" CONTINUATION [241BC03R]

DAEWOO HULL NO.3010

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM 3Ø, 3W, 60HZ-ELV2 120VAC, CONT. ON SH.

SH. NO. 241BB

237B6558 241BC

ENGINEER GRAY, A.W.

FIRST MADE FOR REQ. P.O. 47-DC-841991-C01

I.C. NO.

REV.1 VS 06-17-87 PRINTS TO

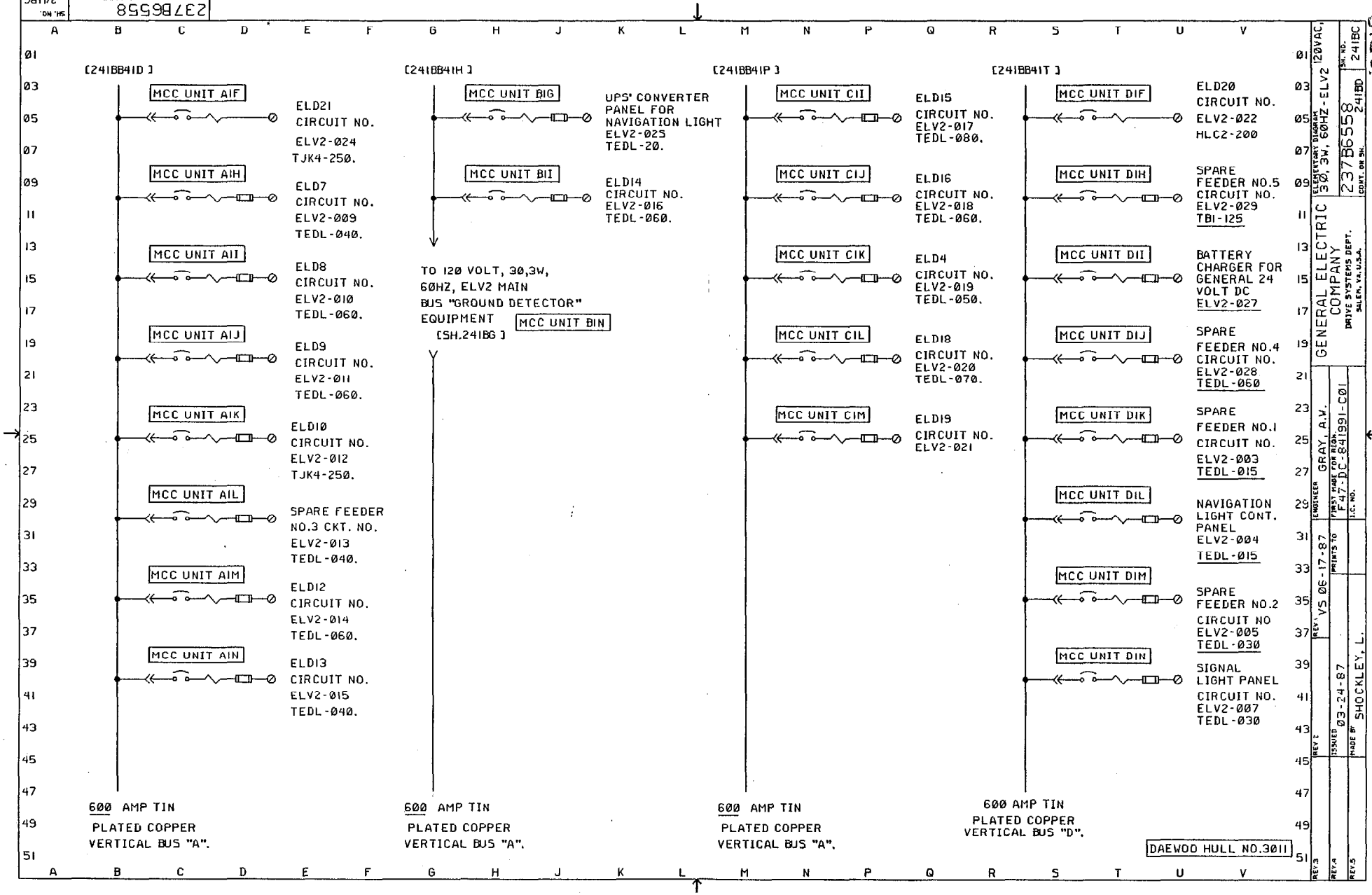
REV.2 ISSUED 03-24-87

MADE BY SHOCKLEY, L.

REV.3

REV.4

REV.5

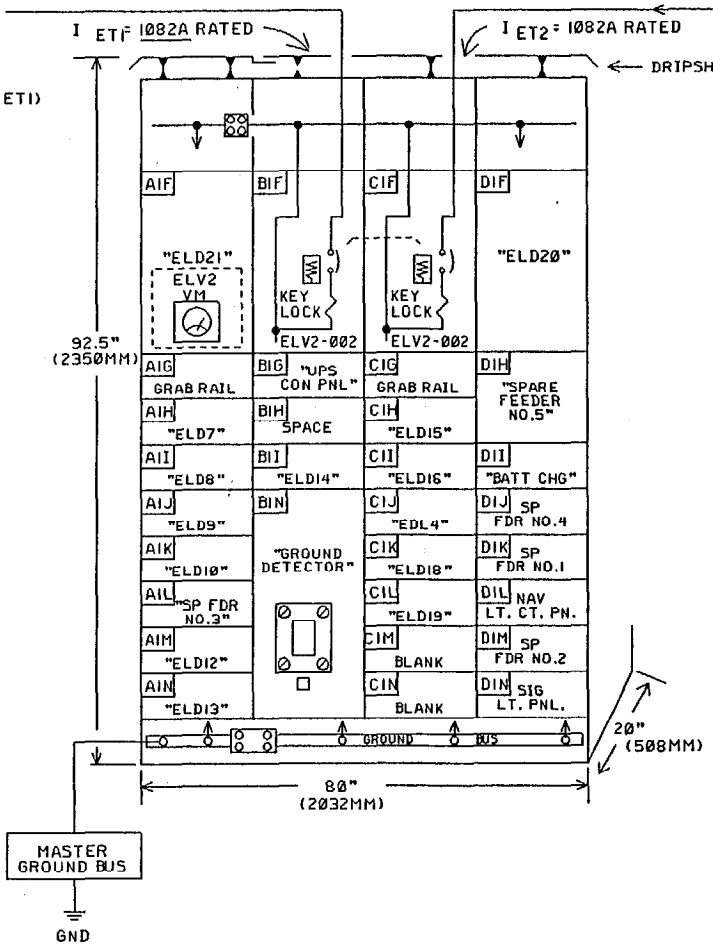

[241BB41D]
MCC UNIT AIF
ELD21 CIRCUIT NO. ELV2-024 TJK4-250.
MCC UNIT AIH
ELD7 CIRCUIT NO. ELV2-009 TEDL-040.
MCC UNIT AII
ELD8 CIRCUIT NO. ELV2-010 TEDL-060.
MCC UNIT AIJ
ELD9 CIRCUIT NO. ELV2-011 TEDL-060.
MCC UNIT AIK
ELD10 CIRCUIT NO. ELV2-012 TJK4-250.
MCC UNIT AIL
SPARE FEEDER NO.3 CKT. NO. ELV2-013 TEDL-040.
MCC UNIT AIM
ELD12 CIRCUIT NO. ELV2-014 TEDL-060.
MCC UNIT AIN
ELD13 CIRCUIT NO. ELV2-015 TEDL-040.
600 AMP TIN PLATED COPPER VERTICAL BUS "A".
[241BB41H]
MCC UNIT BIG
UPS' CONVERTER PANEL FOR NAVIGATION LIGHT ELV2-025 TEDL-20.
MCC UNIT BII
ELD14 CIRCUIT NO. ELV2-016 TEDL-060.
TO 120 VOLT, 30,3W, 60HZ, ELV2 MAIN BUS "GROUND DETECTOR" EQUIPMENT [SH.241BG]
MCC UNIT BIN
600 AMP TIN PLATED COPPER VERTICAL BUS "A".
[241BB41P]
MCC UNIT CII
ELD15 CIRCUIT NO. ELV2-017 TEDL-080.
MCC UNIT CIJ
ELD16 CIRCUIT NO. ELV2-018 TEDL-060.
MCC UNIT CIK
ELD4 CIRCUIT NO. ELV2-019 TEDL-050.
MCC UNIT CIL
ELD18 CIRCUIT NO. ELV2-020 TEDL-070.
MCC UNIT CIM
ELD19 CIRCUIT NO. ELV2-021
600 AMP TIN PLATED COPPER VERTICAL BUS "A".
[241BB41T]
MCC UNIT DIF
ELD20 CIRCUIT NO. ELV2-022 HLC2-200
MCC UNIT DIH
SPARE FEEDER NO.5 CIRCUIT NO. ELV2-029 TBI-125
MCC UNIT DII
BATTERY CHARGER FOR GENERAL 24 VOLT DC ELV2-027
MCC UNIT DIJ
SPARE FEEDER NO.4 CIRCUIT NO. ELV2-028 TEDL-060
MCC UNIT DIK
SPARE FEEDER NO.1 CIRCUIT NO. ELV2-003 TEDL-015
MCC UNIT DIL
NAVIGATION LIGHT CONT. PANEL ELV2-004 TEDL-015
MCC UNIT DIM
SPARE FEEDER NO.2 CIRCUIT NO ELV2-005 TEDL-030
MCC UNIT DIN
SIGNAL LIGHT PANEL CIRCUIT NO. ELV2-007 TEDL-030
600 AMP TIN PLATED COPPER VERTICAL BUS "D".
DAEWOO HULL NO.3011
ELEMENTARY DIAGRAM 120VAC, 30, 3W, 60HZ - ELV2
SH. NO. 241BC
237B6558 241BD
CONT. ON SH.
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
ENGINEER GRAY, A.W.
FIRST MADE FOR RIG NO. 7 - DC-841991-C01
I.C. NO.
REV.1 VS 06-17-87
PRINTS TO
REV.2 ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5

DAEWOO FOR SONAT OFFSHORE DRILLING, INC.
G.E. REQUISITION F47-DC-841990-C01
G.E. MATERIAL LIST 0504X0628W62
120V EMERGENCY POWER DISTRIBUTION ELV2
SPECIFICATION ITEM 3C

FROM SECONDARY OF
CUSTOMER SUPPLIED 225 KVA
480V TO 120V, 3 Ø, 3W,
60HZ, DELTA (Δ) - DELTA (Δ)
CONNECTED TRANSFORMER
ET1
SPECIFICATION ITEM NO. 2D-2(ET1)
PROVISIONS FOR QTY.5 - 4/0
CABLES PER PHASE.

DNV REQUIRED CHANGES:

1. 120VAC, 3Ø, 3W, 60HZ. "BUS VOLTMETER. 0-150V MOVEMENT, 0 -150V SCALE, RED LINE AT 120VAC.
2. ADDED GLASTIC LINE SHIELDS TO ALL CBS.
3. GROUND ALL DOORS TO VERTICAL SECTION.
4. ADD ONE SCREW PER WIRE PER GROUND BUS CONNECTION.
5. ADDED SCREWS TO ATTACH ALL NAMEPLATES.
6.
7.
8.
9.



FROM SECONDARY OF
CUSTOMER SUPPLIED 225 KVA
480V TO 120V, 3 Ø, 3W, 60HZ,
DELTA (Δ) - DELTA (Δ)
CONNECTED TRANSFORMER
ET2
SPECIFICATION ITEM NO. 2D-2 (ET2)
PROVISIONS FOR QTY.5-4/0
CABLES PER PHASE.

MAIN AC THRU BUS
DATA
.VOLTAGE: 120 VAC, 3 PHASES, 3 WIRE
60 HERTZ
.CURRENT CAPACITY: 1000HORIZ./600VER. AMPERES
.BUS MATERIAL: COPPER (TIN PLATED)
.AMBIENT RATING: 50°C
.BUS DIMENSIONS: 0.25" X 2.0" X 2/PHASE
.MCM N.A.
.SHORT CIRCUIT BRACING 42 KA

BUILT PER UL 845 & UL 508; IP23 ENCLOSURE
- VERTICAL GROUND BUS EACH SECTION, WITH PLUG IN UNITS ATTACHED VIA CLAMP AT LEFT REAR OF UNIT.
- WEIGHT: 500 LBS/SECTION(AVG) X 4 SECTIONS = 2000 LBS.(1361Kg).
- GRAB/HAND RAILS: SECTIONS

U.S. COAST GUARD REQUIRED INFORMATION
DOT - COAST GUARD - 46 CFR FED. REGISTER
.GE REQN. NO.: F47-DC-841991-C01
.GE MTL. LIST: 0504X0629W62
.SHIPYARD NAME: DAEWOO, OKPO KOREA
.SHIPYARD HULL NO.: DAEWOO HULL 3011
.VESSEL NAME: GEORGE RICHARDSON
.VESSEL OWNER: SONAT OFFSHORE DRILLING, INC.
.VESSEL CLASS: GVA 4500 SEMI-SUBMER.
.MODU MOBILE OFFSHORE DRILLING UNIT
.DATE: FEB. 14, 1987

DAEWOO HULL NO.3011

ELEMENTARY DIAGRAM 120VAC, 3Ø, 3W, 60HZ - ELV2
SH. NO. 241BD
237B6558
241BE
CONT. ON SH.
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
ENGINEER GRAY, A.W.
PRINTS TO
FIRST MADE FOR REQN. F47-DC-841991-C01
J.C. NO.
REV.1 VS 06-16-87
REV.2 LP 7-27-87
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5

DAEWOO FOR SONAT OFFSHORE DRILLINGS, INC.

120 VOLT EMERGENCY POWER DISTRIBUTION

ELEV2

REMOTE
FROM CUSTOMER SUPPLIED REMOTE TRANSFORMER. ET1
PROVISIONS AT INCOMING CB FOR QTY.5-4/0 CABLES PER PHASE.

L1-ØA
L2-ØB
L3-ØC

INCOMING MAIN POWER
120VAC, 3Ø , 3W, 60HZ., 1200 AMPS.

KEY INTERLOCKED SUCH THAT ONLY ONE CB AT A TIME CLOSES!

EMERGENCY TRANSFORMER NO. 1 - "ELV2 - 002"
MAIN KEY INTERLOCKED

17G 17H 17J

TKMA 1200 TYPE CB
1200 AT / 1200 AF
MCC UNIT BIF

L1-ØA
L2-ØB
L3-ØC

REMOTE
FROM CUSTOMER SUPPLIED REMOTE TRANSFORMER. ET2
PROVISIONS AT INCOMING CB FOR QTY.5-4/0 CABLES PER PHASE.

EMERGENCY TRANSFORMER NO. 2 - "ELV2 - 111"
MAIN KEY INTERLOCKED

17P 17Q 17R

TKMA 1200 TYPE CB
1200 AT / 1200 AF
MCC UNIT CIF

MAIN HORIZONTAL AC THRU BUS
VOLTAGE 120 VAC, 3 PHASES, 3 WIRE
60 HZ.
CURRENT CAPACITY 1000 AMPS UL / 1200 NEMA
BUS MATERIAL COPPER (TIN PLATED).
AMBIENT RATING 50° C
SHORT CKT. BRACING 42,000 AMPS.
0.25"TX2.0"H TWO(2) PER PHASE.

L1
L2
L3

600 AMP TIN PLATED COPPER VERTICAL BUS "A" EXTENSION.

600 AMP TIN PLATED COPPER VERTICAL BUS "B" EXTENSION.

FVM 3A CLF

VM

MCC UNIT AIF

"ELV2 VOLTS"
GE TYPE AB-40
0-150VOLT SCALE
0-150VOLT INPUT
RED LINE AT 120V.

600 AMP TIN PLATED COPPER VERTICAL BUS "C" EXTENSION.

600 AMP TIN PLATED COPPER VERTICAL BUS "D" EXTENSION.

L1 L2 L3

TO 600 AMP COPPER VERTICAL BUS "A" FEEDER LOADS.
[241BF04H]

TO 600 AMP COPPER VERTICAL BUS "B" FEEDER LOADS AND "GROUND DETECTOR" EQUIPMENT.
[241BG04H]

TO 600 AMP COPPER VERTICAL BUS "C" FEEDER LOADS.
[241BH04H]

TO 600 AMP COPPER VERTICAL BUS "D" FEEDER LOADS.
[241BJ04H]

DAEWOO HULL NO. 3011

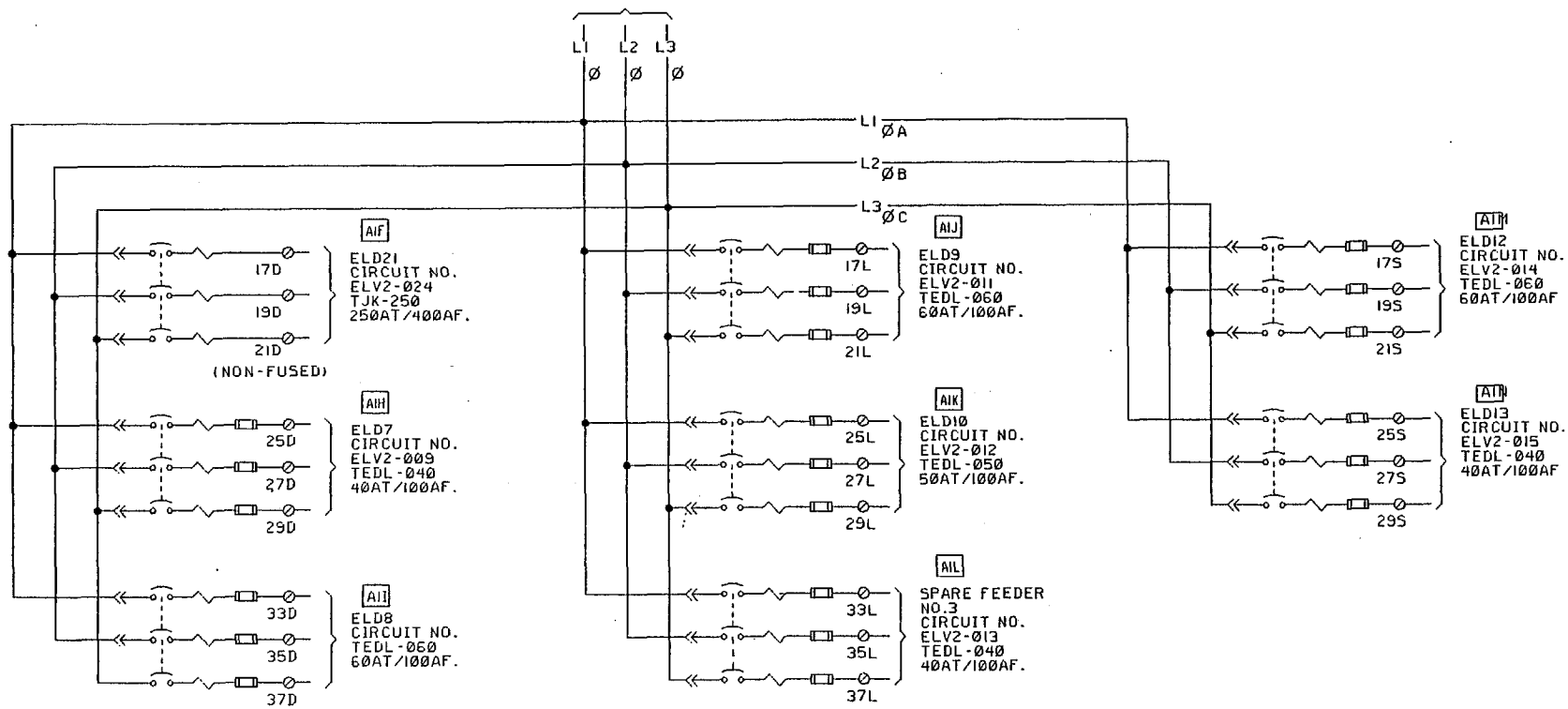
ELEMENTARY DIAGRAM
120VAC 3Ø, 3W, 60HZ.-ELV2
SH. NO. 241BE
2378658
CONT. ON SH. 241BF

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.

ENGINEER GRAY, A.W.
FIRST MADE FOR REQN. F47-DC-841991-C01
I.C. NO.

REV.1 BC 06-17-87
PRINTS TO
REV.2 LP 7-27-87
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5

600AMP COPPER VERTICAL
BUS "A" CONTINUATION & LOADS.
[241BE43C]



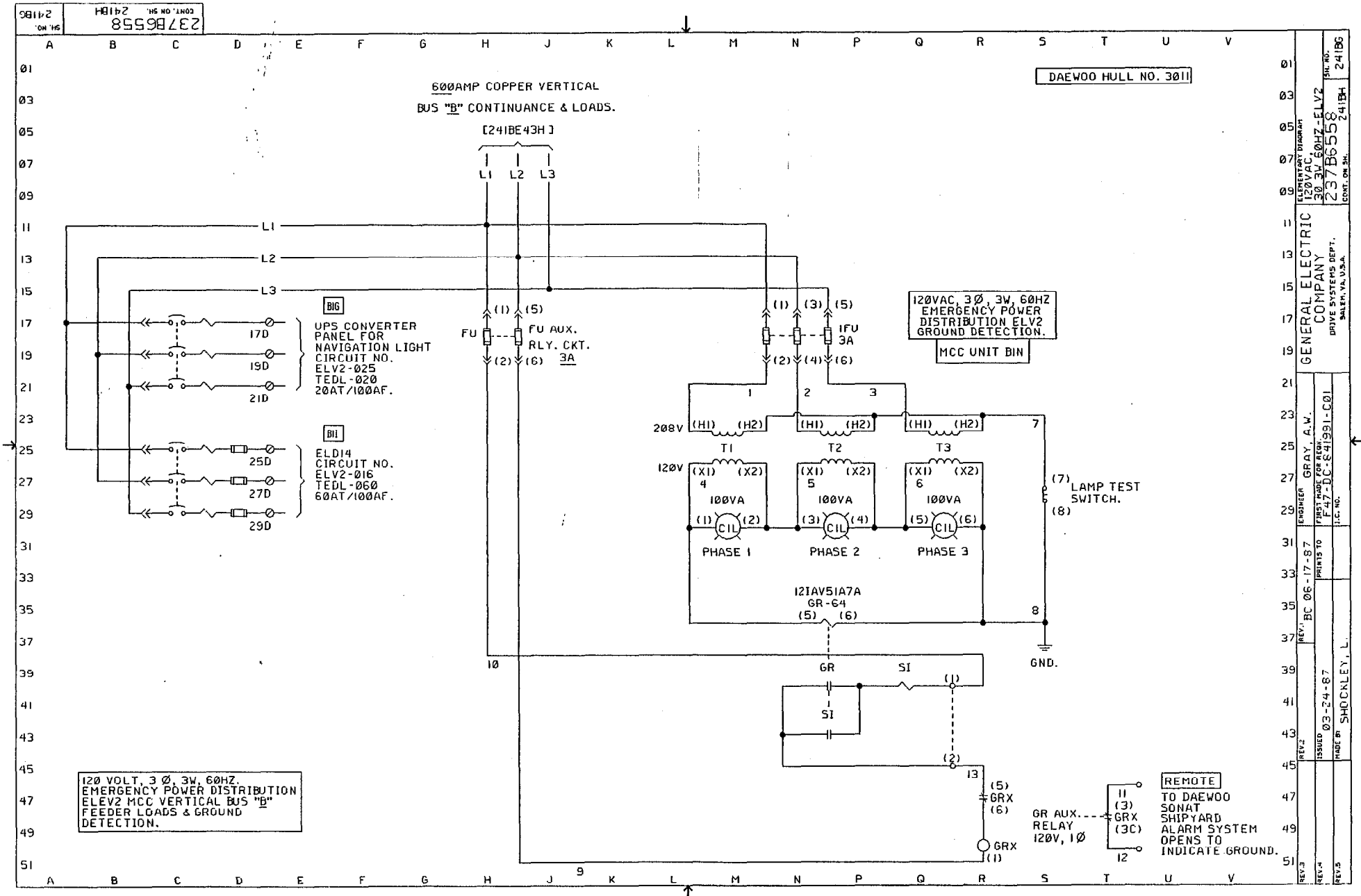
120VOLT,3Ø,3W,60HZ
EMERGENCY POWER DISTRIBUTION ELV2 MCC
VERTICAL BUS "A" FEEDER LOADS.

DAEWOO HULL NO. 3011

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.

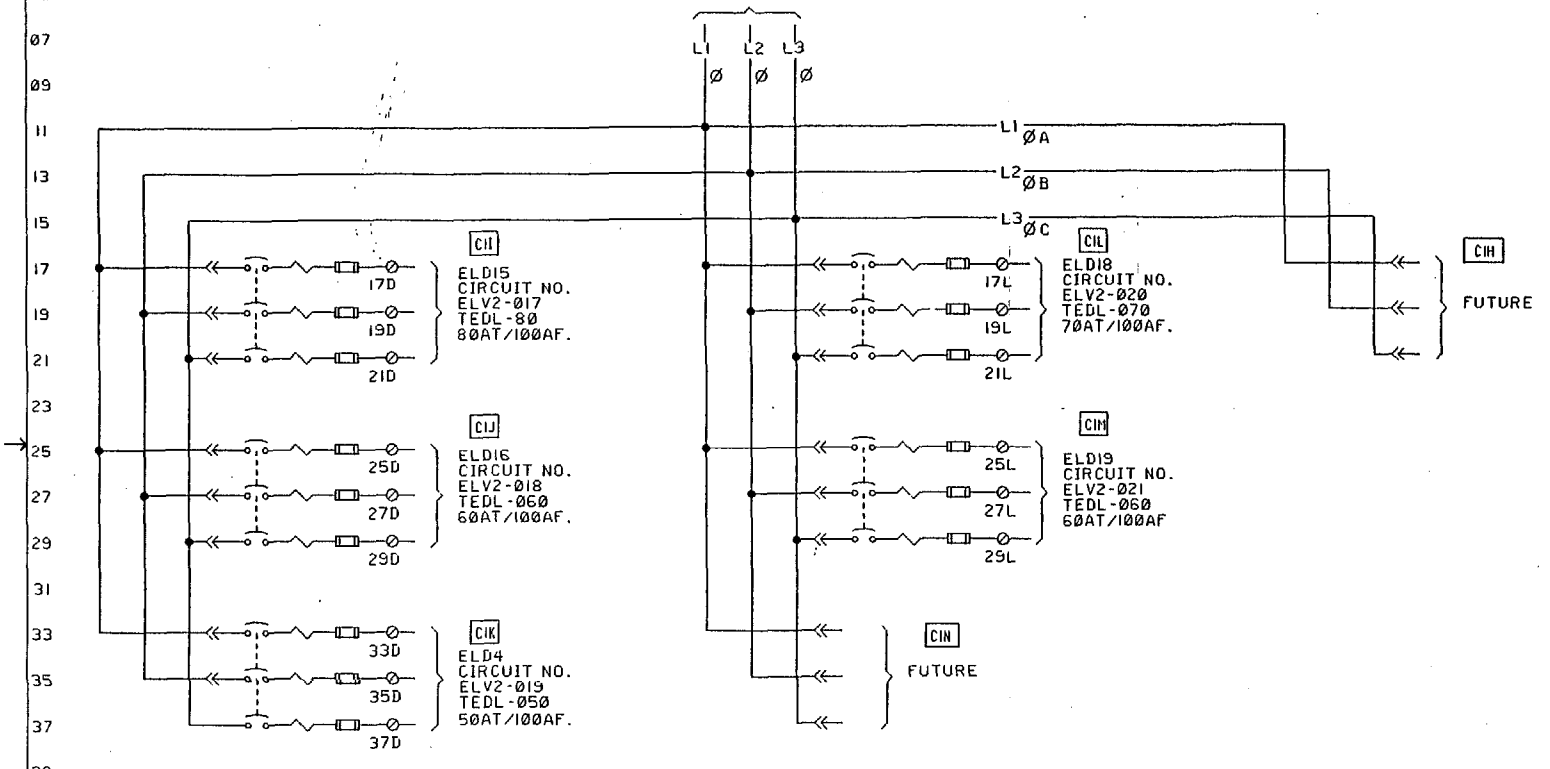
ELEMENTARY DIAGRAM
120VAC, 3Ø 3W 60HZ-ELV2
237B6558
CONT. ON SH. 241BG
SH. NO. 241BF

ENGINEER GRAY, A.W.
FIRST MADE FOR REGN F47-DC-841991-C01
J.C. NO.
PRINTS TO
REV.1 BC 06-17-87
REV.2
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5


600AMP COPPER VERTICAL
BUS "B" CONTINUANCE & LOADS.
[241BE43H]
DAEWOO HULL NO. 3011
L1 L2 L3
BIG
UPS CONVERTER PANEL FOR NAVIGATION LIGHT CIRCUIT NO. ELV2-025 TEDL-020 20AT/100AF.
17D
19D
21D
BII
ELDI4 CIRCUIT NO. ELV2-016 TEDL-060 60AT/100AF.
25D
27D
29D
FU
FU AUX. RLY. CKT. 3A
IFU 3A
120VAC, 3Ø, 3W, 60HZ EMERGENCY POWER DISTRIBUTION ELV2 GROUND DETECTION.
MCC UNIT BIN
208V
120V
T1
T2
T3
100VA
PHASE 1
PHASE 2
PHASE 3
(7) LAMP TEST SWITCH.
(8)
GND.
121AV51A7A
GR-64
GR
SI
GRX
GR AUX. RELAY 120V, 1Ø
REMOTE
TO DAEWOO SONAT SHIPYARD ALARM SYSTEM OPENS TO INDICATE GROUND.
120 VOLT, 3 Ø, 3W, 60HZ. EMERGENCY POWER DISTRIBUTION ELEV2 MCC VERTICAL BUS "B" FEEDER LOADS & GROUND DETECTION.
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM 120VAC 3Ø 3W 60HZ-ELV2
237B6558
241BH
241BG
ENGINEER GRAY, A.W.
FIRST MADE FOR REQ. PF-47-DC-&41991-C01
PRINTS TO
J.C. NO.
REV.1 BC 06-17-87
REV.2
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5

ORIG

600AMP COPPER VERTICAL
BUS "C" CONTINUATION & LOADS.
[241BE43]



120VOLT,3Ø,3W,60HZ
EMERGENCY POWER DISTRIBUTION ELV2 MCC
VERTICAL BUS "C" FEEDER LOADS.

DAEWOO HULL NO. 3011

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM
120VAC, 3Ø 3W 60HZ-ELV2
237B6558
CONT. ON SH. 241BJ
SH. NO. 241BH
ENGINEER GRAY, A.W.
FIRST MADE FOR REQN. F47-DC-E41991-C01
J.C. NO.
PRINTS TO
REV.1 BC 06-17-87
REV.2
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5

600AMP COPPER VERTICAL
BUS "B" CONTINUATION & LOADS.
[241BE43T]

L1 Ø
L2 Ø
L3 Ø

L1 ØA
L2 ØB
L3 ØC

DIF
17D
19D
21D
ELD20
CIRCUIT NO.
ELV2-022
"HLC2-200"
200AT/200AF.
NON-FUSED.

DIH
25D
27D
29D
SPARE FEEDER
NO.5 CIRCUIT NO.
ELV2-029
TBI-125
125AT/400AF

DII
33D
35D
37D
BATTERY CHARGER
FOR GENERAL 24
VOLT DC.
CIRCUIT NO.
ELV2-027
TEDL-15
15AT/100AF.

DIJ
17L
19L
21L
SPARE FEEDER
NO.4-CIRCUIT NO.
ELV2-028
TEDL-060
60AT/100AF.

DIK
25L
27L
29L
SPARE FEEDER
NO.1-CIRCUIT NO.
ELV2-003
TEDL-015

DIL
33L
35L
37L
NAVIGATION LIGHT
CONTROL PANEL
CIRCUIT NO.
ELV2-004
TEDL-015.
15AT/100AF

DIM
17S
19S
21S
SPARE FEEDER
NO.2-CIRCUIT NO.
ELV2-005
TEDL-030
30AT/100AF

DIN
25S
27S
29S
SIGNAL LIGHT
PANEL-CIRCUIT NO.
ELV2-007
TEDL-030
30AT/100AF

120VOLT,30,3W,60HZ
EMERGENCY POWER DISTRIBUTION ELV2 MCC
VERTICAL BUS "D" FEEDER LOADS.

DAEWOO HULL NO. 3011

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
120VAC, 30,3W 60HZ ELV2
237B6558
CONT. ON SH. 241B
SH. NO. 241BJ

ENGINEER GRAY, A.W.
FIRST MADE FOR REQN. F47-DC-841991-C01
I.C. NO.
PRINTS TO

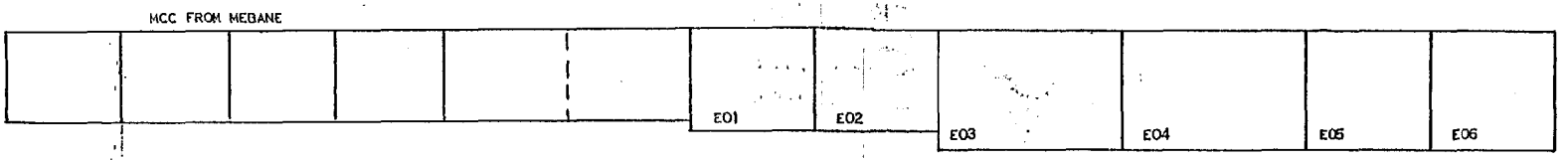
REV.1 BC 06-17-87
REV.2 LP 7-27-87
ISSUED 03-24-87
MADE BY SHOCKLEY, L.
REV.3
REV.4
REV.5

09 SH NO

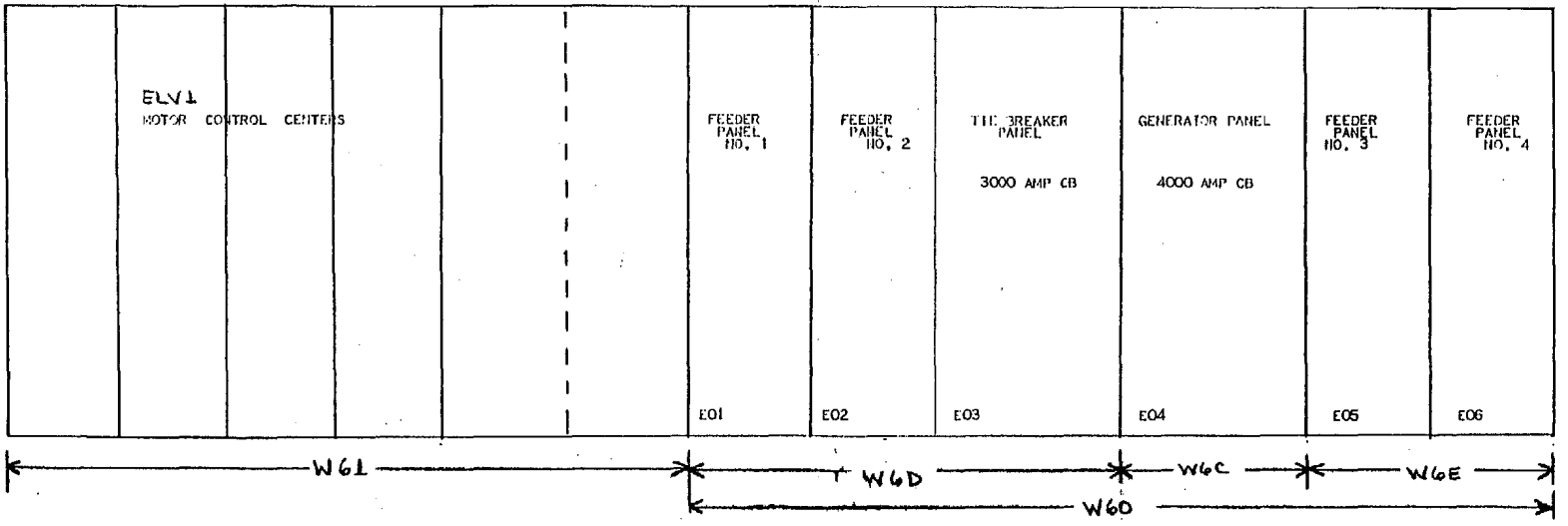
2378656 CONT ON SH 60A

EMERGENCY AC GENERATOR SWITCHBOARD - ELV1

TOP VIEW



FRONT VIEW



GENERAL ELECTRIC
D.S.D.
SALEM VA. U.S.A.

ASSEMBLY DIAGRAM
EMERG. SWITCHBOARD

2378656G6 CONT ON SH 60A

SH NO 60

REV. 1

APPROVALS

PRINTS TO

PURCHASE ORDER NO. 990 '-C01

FAC. NO. 4876

REV. 2

ISSUED 10-30-86

DRAWN BY L SHOCKLEY

REV. 3

REV. 4

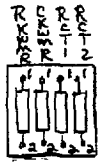
REV. 5

SH. NO. 60A

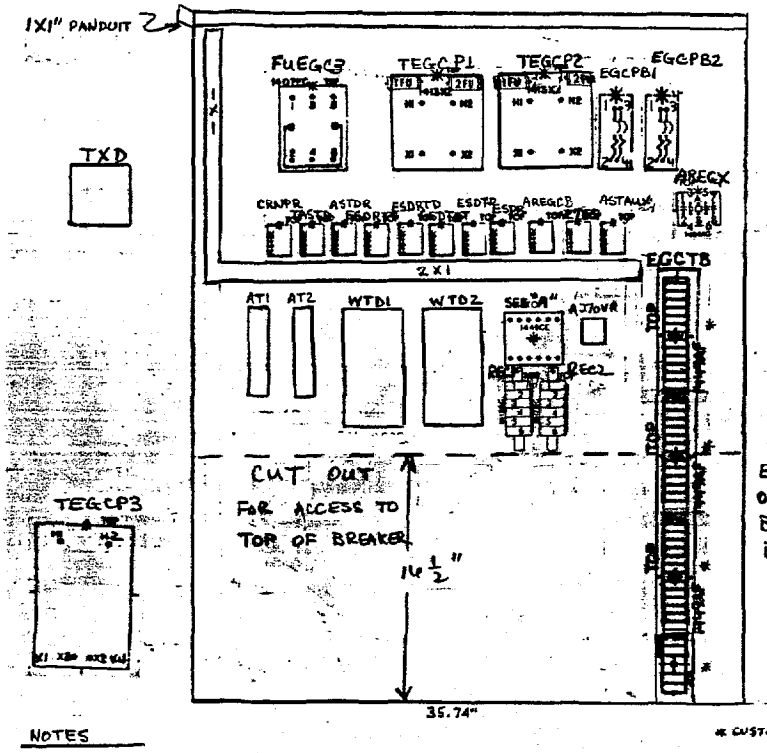
23786556 CONT. ON SH. 60B

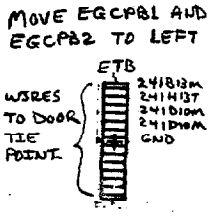
M.L. ITEM NO	DEVICE NOMENCLATURE
2	SHTB
3, 4A	FUEGCP3
26	TEGCP1,2
20, 21	EGCPB1,2
27	TEGCP3
103	AREGCB
104	AREGCX
147	TXD
145, 150, NPB	RCT1, 2
123	REC1, REC2
132	CRNPR
133	ESDR, ESDTD ESDRTD
134	ESDTDT
135	ESTRTDT
136	ASTDR, ASTNK
139	TASTD
140	27EGVSR
143	27EG
151	UVTD1, 2
152	AT1
153	AT2
160, 161	EGCTB

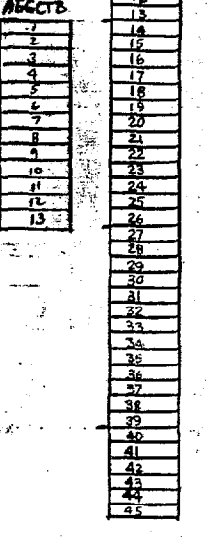
DETAIL "A"



NOTE-KEEP WIRES FROM THRU BUS TO FUEGCS AND TEGCP1,2 SEPERATE FROM OTHER WIRING. USE FIRE ZONE WIRE.







* CUSTOMER SIDE OF TB

NOTES

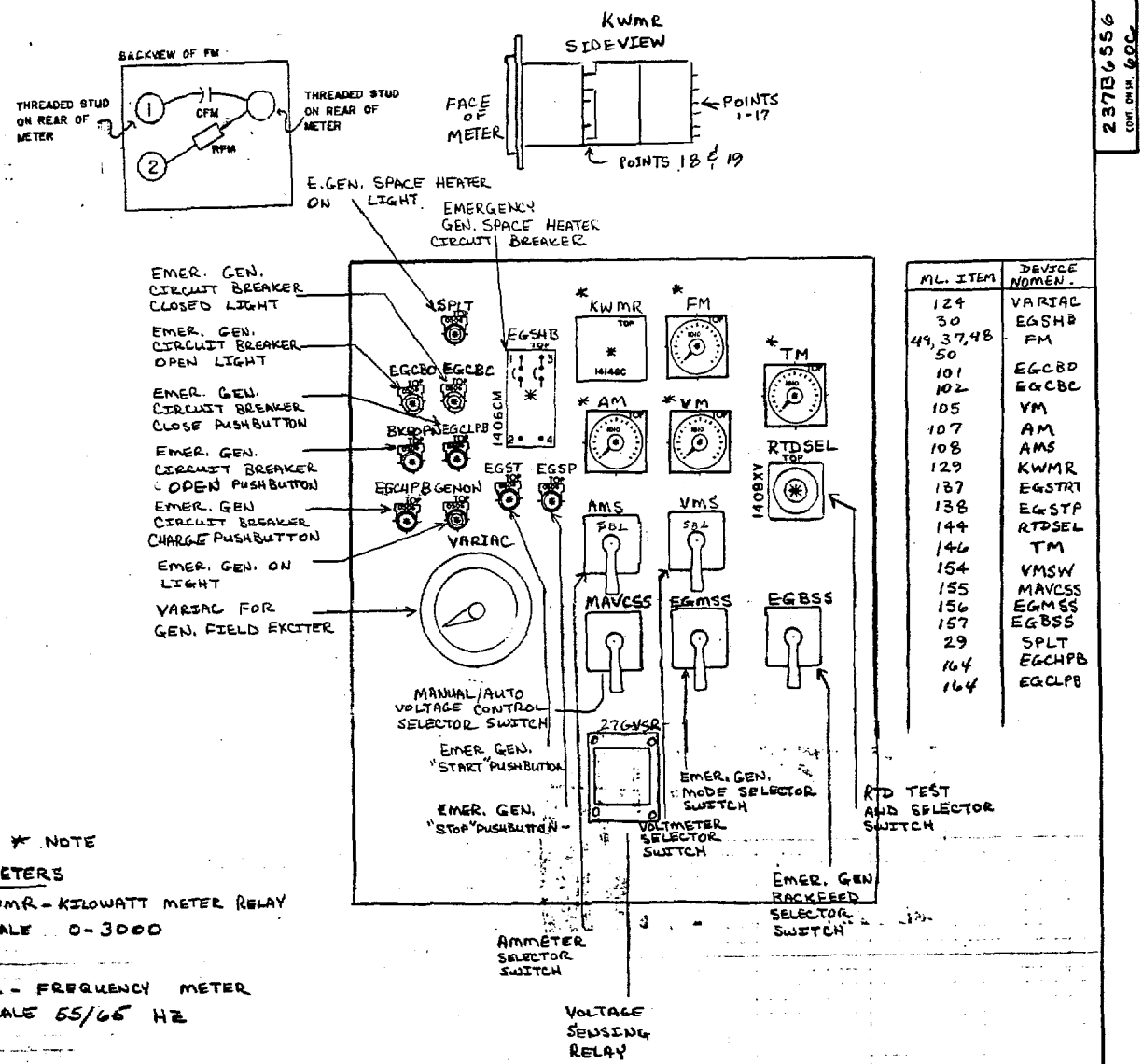
1) UVTD1 SUPPLIED BY VENDOR WITH BREAKER MOVE TO SIDE WALL
2) MOVE SHTB TO SIDE WALL
3) ADD ETB TO SPLICE WIRES GOING TO 27GVSR
4) TXD AND TEGCP3 MOVED TO LEFT SIDE WALL.

G.E. ITEM #	PANEL #	LAYER #	WIRE TABLE #	CAT # 0504X0628W6L E04
REV. 3	REV. 2	REV. 1 REV 1 [illegible]	APPROVALS [illegible]	GENERAL ELECTRIC D. S. D SALEM VA. U.S.A.
REV. 4	ISSUED OCT. 27, 1986	PRINTS TO	FIRST MADE FOR REQ. E47-DC-841990-C01	ASSEMBLY DIAGRAM EMER. GEN. PNL
REV. 5	MADE BY L. SHOCKLEY		P.C. NO. 4876	23786556 CONT. ON SH. 60B — SH. NO. 60A

ODRIVE PRESS

SH NO 60B

23786556 CONT ON SH. 60C



ML. ITEM	DEVICE NOMEN.
124	VARIAC
30	EGSHB
49, 37, 48	FM
50	
101	EGCBD
102	EGCBC
105	VM
107	AM
108	AMS
129	KWMR
137	EGSTRT
138	EGSTP
144	RTDSEL
146	TM
154	VMSW
155	MAVCSS
156	EGMSS
157	EGBSS
29	SPLT
164	EGCHPB
164	EGCLPB

* NOTE

METERS

KWMR - KILOWATT METER RELAY
SCALE 0-3000

FM - FREQUENCY METER
SCALE 55/65 HZ

TM - TEMPERATURE METER
SCALE 0-180°C

AM - AMMETER
SCALE 0-4200 AMPS

VM - VOLTMETER
SCALE 0-600 VOLTS

G.E. ITEM #	PANEL #	LAYER #	WIRE TABLE #	CAT # 0504X0628WGC D04

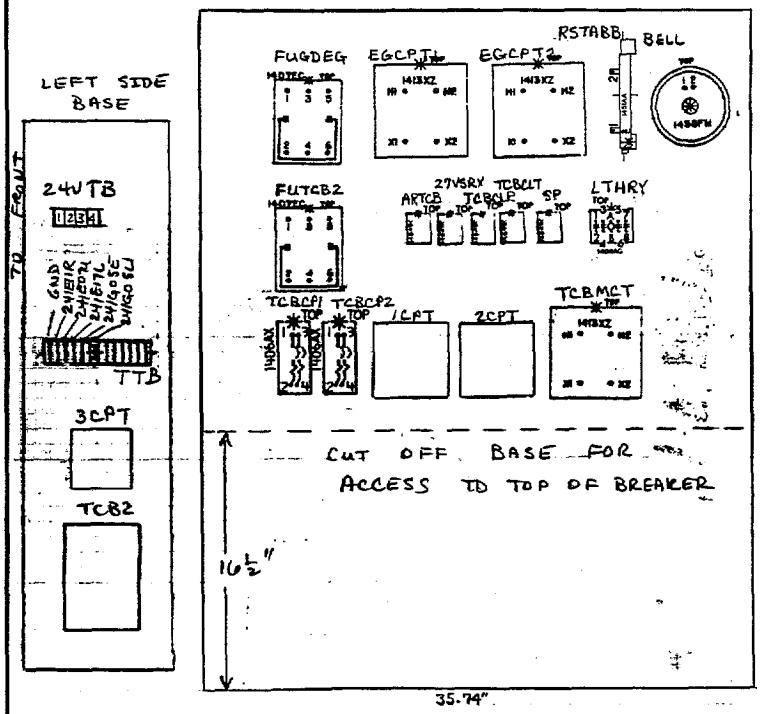
REV. 3 LS 4-27-87	REV. 2 LS 3-20-87	REV. 1 LS 2-25-87	APPROVALS LL	GENERAL ELECTRIC D.S.D. SALEM, VA. U.S.A.	ASSEMBLY DIAGRAM EMER. GEN. DOOR
REV. 4 RC 5/18/87	ISSUED OCT. 26, 1986	PRINTS TO	FIRST MADE FOR REQ. EW7-DC-841990-C01		23786556 CONT ON SH. 60C / SH NO. 60B
REV. 5	MADE BY L. SHOCKLEY		I.C. NO. 4876		

93559

SH NO 60C

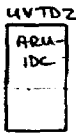
237B6556 CONT ON SH. 60D

ML. ITEM	DEVICE NOMEN.
154,155	FUGDEG
161	EGCPT1,2
168,158	RSTABB
162	BELL
112,113	FUTCB2
103	ARTCB
118	27VSRX
119	TCBCLR
120	TCBCLT
163	LTHRY
114,115	TCBCP1
114,115	TCBCP2
157	1CPT, 2CPT 3CPT
111	TCBMCT
116	TCB2
117	27VSR
172,173	CTB



UVTD2 SUPPLIED WITH TIE BREAKER

MOVED TO RIGHT WALL



NOTE* OVER HEAD GROUND BUS TO BE MOUNTED VERTICAL ON BRACKETS WITH 90° BEND.

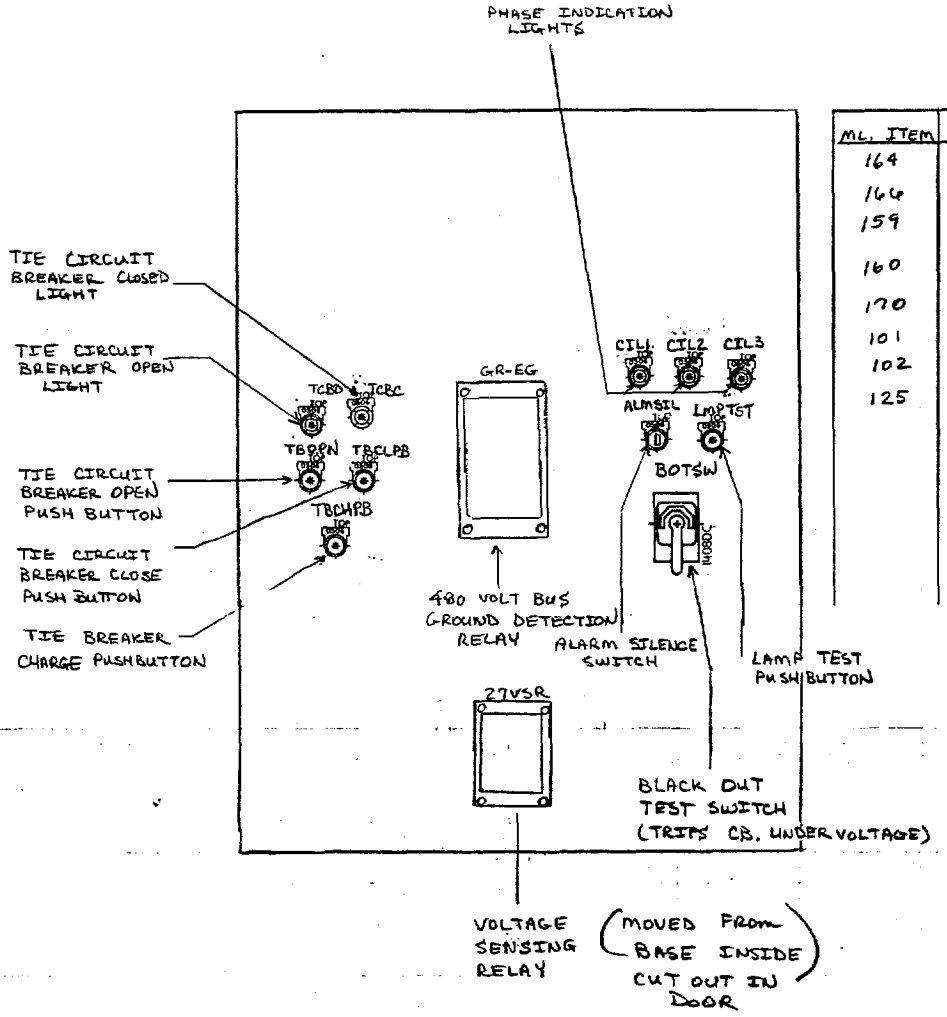
G.E. ITEM #	PANEL #	LAYER #	WIRE TABLE #	CAT # 0504X0628 W6D E03

REV. 3	REV. 2	REV. 1 L5 4-27-87	APPROVALS LL	GENERAL ELECTRIC D.S.D. SALEM VA. U.S.A.	ASSEMBLY DIAGRAM TIE BREAKER PNL	
REV. 4	ISSUED OCT. 28 1986	PRINTS TO	FIRST MADE FOR REQ. FU7-DC-841990-C01		237B6556 CONT ON SH. 60D	SH NO 60C
REV. 5	MADE BY L. SHOCKLEY		I.C. NO. 4876			

02559

SH NO 60D

23786556 CONT ON SH. 60E



ML. ITEM	DEVICE NOMEN.
164	ALMSIL
166	LMPTST
159	CIL1-3
160	GR-EG
170	BOTSW
101	TCBO
102	TCBC
125	TBC-PB

G.E. ITEM # | PANEL # | LAYER # | WIRE TABLE # | CAT # 0504X0628 W6D D03

REV. 3	REV. 2 LS 4-27-87	REV. 1 2-25-87 LS	APPROVALS LS	GENERAL ELECTRIC D.S.D SALEM VA. U.S.A.	ASSEMBLY DIAGRAM TIE BKR. DOOR
REV. 4	ISSUED OCT. 28 1986	PRINTS TO	FIRST MADE FOR REQ F47-DC-841990-C01		23786556 CONT ON SH. 60E / SH NO. 60D
REV. 5	MADE BY L. SHOCKLEY		I.C. NO. 4876		

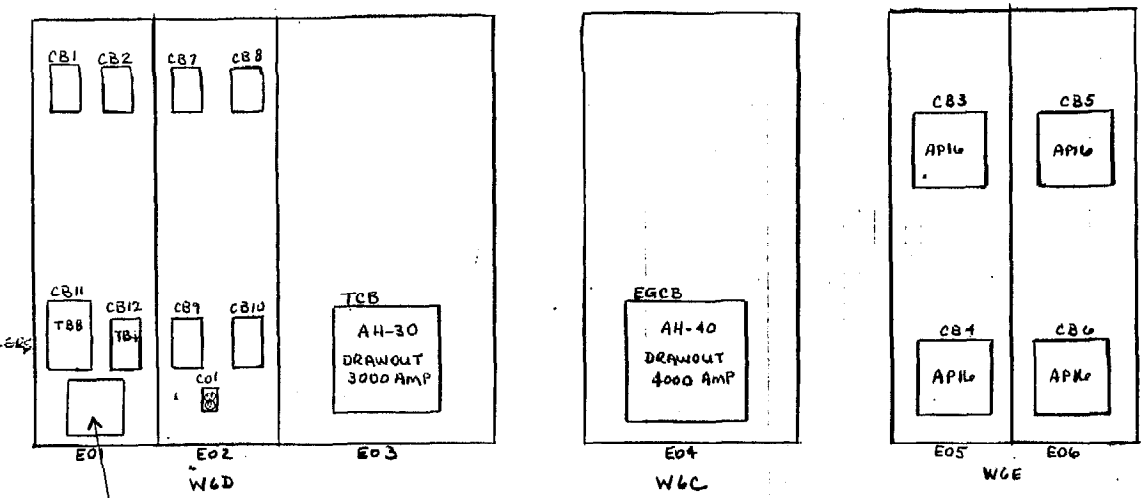
82559

309 SH NO

2378 6556 CONT ON SH FL

(W6D)

M.L. ITEM	DEVICE NOMEN.
	(E01)
120,121,126	CB1, CB2
122,123,127	CB11
120,121, 2	CB12
	(E02)
2,120,121	CB7, CB8
1,120,121	CB9, CB10
	(E03)
1	TCB
	W6C (E04)
1	EGCB
	W6E (E05)
1	CB3, CB4
	(E06)
1	CB5, CB6



W6D E01 CIRCUIT BREAKERS

CB1, CB2 - 400 AMP FRAME
400 AMP TRIP

CB11 - 800 AMP FRAME
600 AMP TRIP

CB12 - 400 AMP FRAME
125 AMP TRIP

W6D E02 CIRCUIT BREAKERS

CB7, CB8 - 400 AMP FRAME
125 AMP TRIP

CB9, CB10 - 400 AMP FRAME
300 AMP TRIP

FOR STROMBERG VOLTAGE REGULATOR

NOTE - DELETE CONVENIENCE OUTLET AND POWER SUPPLY IN E02. TO BE WIRED TO TB IN E03 UNIT.

W6E E05, E06 CIRCUIT BKRS
CB3, CB4, CB5, CB6

AP16 - DRAWOUT BREAKERS
1600 AMP FRAME
1000 AMP TRIP

CAT # 0504X0628 AC

GENERAL ELECTRIC
D, S, D
SALEM VA U.S.A.

ASSEMBLY DIAGRAM
BREAKERS
2378 6556
SH NO. 60E
CONT ON SH FL.

WIRE TABLE # 28
APPROVED
FIRST MADE FOR REQ. P47-0C-84190-001
T.C. NO. 4876

G.E. ITEM #
PANEL #
LAYER #
DRAWN BY L. SHOCKLEY
ISSUED 10-30-86
PRINTS TO
REV. 1 2-25-87 LS
REV. 2 4-27-87
REV. 3
REV. 4
REV. 5

6GGED



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Printed in U.S.A.